

Full-System Design, Implementation, and Analysis for Quantum Key Distribution on Mobile Platforms

Dissertation

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By

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Abstract

Recent advances in quantum computing threaten the future of information security in classical cryptography. However, parallel advances in quantum communication and quantum key distribution (QKD) in particular show promise as a new paradigm for information security without the same vulnerabilities. Because the security of QKD does not depend on the computational power of the adversary, it can be considered “future proof”. Nevertheless QKD brings its own set of technical challenges and security vulnerabilities that must be addressed to bring it forth as a practical option for secure information transfer.

One outstanding issue is the so-called “last mile” problem wherein connecting a communication network to the end users requires significant expense and infrastructure. Mobile and reconfigurable communication platforms present advantages in this area due to their ability to meet the needs of many end users with a smaller network of nodes. The implementation of a QKD system on mobile platforms such as drones and cars brings this advantage to a hypothetical future quantum communication network.

The security of QKD hinges on limiting an adversary to only performing measurements on the quantum state in the encoded degree-of-freedom of the wavepacket, at which point the attack can be detected and dealt with via privacy amplification. This involves preparing and measuring high fidelity quantum states with sufficient transmission volume to overcome the challenges of finite key effects. Additionally, it requires high levels of indistinguishability in the non-encoding degrees-of-freedom lest the information be leaked to the adversary via a side-channel attack.

In this thesis, I present the designs and experimental efforts undertaken to implement a QKD system on mobile platforms and operated in an outdoor environment. I also present the detailed analysis, modeling, and data processing used to determine the central figures-

of-merit for the performance of the system.

I explain the light emitting diode (LED) based photonic source driven by a field-programmable gate array, along with the apparatus with which we collect the photonic wavepackets. These custom LED-to-fiber couplers can achieve mean photon numbers on the order of unity, which is as high an intensity as is optimal for most QKD protocols. As measures against potential side-channel attacks, the transmitter optics bench incorporates a single-mode spatial filter fiber and a spectral filter while the field-programmable gate array dynamically controls the timing of the wavepackets. For the indistinguishability of transmitted states, I derive the maximum mutual information that can be leaked to the adversary through side-channel attacks. The spectral degree-of-freedom yields a leaked mutual information of 2.4×10^{-5} (with a 2.44×10^{-5} measurement bias and uncertainty of 1.20×10^{-5}). The temporal degree-of-freedom yields a leaked mutual information of 4.31×10^{-5} (with a 3.18×10^{-6} measurement bias and a 7.99×10^{-7} uncertainty).

I describe the design of a full optical train for state preparation and measurement, including polarization fidelity compensation to combat the birefringent effects of the single-mode spatial filter fiber as well as other optical components. These designs are informed by experience in implementing tabletop versions of the system as well as extensive modeling of the non-ideal polarization properties of the components. Additionally, I detail the procedures required to effectively deploy this system on mobile platforms for outdoor field testing. The polarization preparation and measurement system is able to achieve average quantum bit error rates as low as 2.21% in field demonstrations.

I introduce a suite of custom data analysis and processing software, which stitches together the asynchronous receiver data streams, synchronizes the data streams of the two users with a custom qubit-based Bayesian algorithm, generates sifted keys, and filters noise in the post-processing. The synchronization algorithm in particular allows the reconciliation of the two users' data streams without the need for external hardware solutions and without sacrificing any secure key. My simulations show that the algorithm can tolerate relative clock drift rates of up to 1 part in 4140.

I present our team's demonstrations of this system in various mobile platform configurations in the context of relevant state-of-the-art QKD demonstrations in the literature. The

primary figure-of-merit of the aggregate system is the finite secure key rate, and we achieve finite key rates of 7.98 kbits/s for the air-to-air demonstration, 1.52 kbits/s for the air-to-car demonstration, and 19.6 kbits/s for the car-to-car demonstration.

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Vita

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Publications and Presentations

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Chapter 1

Introduction

Quantum information science (QIS) is becoming an increasingly important field in the modern technology landscape. Recent decades have seen significant technological developments in both quantum computing and quantum communication, and in many ways developments in these two areas drive each other. One of the most anticipated applications of mature quantum computing is the ability to implement Shor’s algorithm [1], which can more efficiently factor very large numbers. Rivest–Shamir–Adleman (RSA) cryptography is the current standard for classical encryption, but its security rests on the computational difficulty of factoring large numbers, and thus that security is increasingly at risk.

This presents a potential void in information security, a cornerstone of modern society. There are two main approaches to filling that void: post-quantum public-key cryptography and quantum key distribution (QKD). Post-quantum cryptography aims to design new classical encryption techniques that are computationally hard to break for both classical and quantum computers. This strategy holds the advantage that it can be implemented on the current classical communication infrastructure; however, it is an open question whether classical algorithms exist that are computationally hard for both classical and quantum computation. Many post-quantum cryptography strategies have been proposed and evaluated to

some extent along the guidelines from the National Institute of Standards and Technology (NIST) [2], but none have yet been selected for widescale implementation. Additionally, NIST expects that full adoption of a new post-quantum algorithm may take up to 20 years [3].

Introduced in 1984 [4], QKD is a symmetric encryption protocol that promises unconditional information security founded on the fundamental laws of physics, rather than on the difficulty of computational problems. The fundamental information security in QKD comes from the observation that an eavesdropper cannot measure a quantum system without affecting it. The two cryptographic parties, Alice and Bob, can measure and quantify these effects and determine what fraction of their information could have been leaked to the eavesdropper. Once this fraction is known, they can employ classical privacy amplification techniques [5] to reduce the raw transmitted key to a shorter, fully secure one. This fully secure key can be used to classically encrypt information using one-time pads [6].

Bennett and Brassard established the first QKD protocol (BB84) [4], which used the polarization degree-of-freedom of single photons to transmit information. In the original BB84 protocol, Alice prepares single-photon pulses in two mutually unbiased polarization bases. In our system, these two bases consist of horizontal/vertical (H/V) polarization, which is the monitoring basis, and left circular/right circular (L/R) polarization, which is the signal basis. Alice prepares a series of many photonic wavepackets, choosing randomly one of these four states for each time step and recording her choices. Bob receives these photonic wavepackets and randomly chooses to measure in the H/V or L/R basis for each time step and records each result. In an ideal system, when Bob measures in the same basis as Alice prepares her state, the state he detects will match the state she prepares. For example, if Alice sends the H polarization state for a given wavepacket and Bob measures it in the H/V basis, he will register H as his detection event. However, when Alice and Bob's

bases are unmatched, the detected event will not be deterministic. For example, if Alice sends the H polarization state for a given wavepacket and Bob measures it in the R/L basis, there is a 50% chance that he will register L and a 50% chance that he will register R. Thus, for these detections with unmatched bases, Alice and Bob cannot share useful information.

After this measurement phase is complete, Alice and Bob publicly share which basis was used at each time step. By only keeping the states when the same bases are used, Alice and Bob reduce down to a matching string of random states in the sifting process. After further reduction for the error correction and privacy amplification [7], this becomes the secret classical key securely shared between Alice and Bob. However, if an eavesdropper attempts to measure the polarization of the single-photon wavepackets after they are sent by Alice, the eavesdropper will often affect the polarization state, leading to errors between the sifted keys of Alice and Bob. Alice and Bob can detect these errors by publishing some fraction of their data and performing a comparison. Even in the absence of an eavesdropper, qubits are lost to the environment via transmission loss and environmental radiation is detected due to stray light and thermal effects, so the system is formally considered open. However, the security of the system is still guaranteed using privacy amplification based on the quantum bit error rate (QBER). An important aspect to note is that the quantum users pessimistically assume that all quantum bit errors or other deviations from the ideal statistics are caused by the actions of the eavesdropper, and they perform sufficient privacy amplification to ensure that any potentially compromised information is eliminated.

Subsequently developed protocols have extended QKD to different types of systems [8, 9] and relaxed the requirement for a true single-photon source with the introduction of the decoy state protocol [10, 11]. The ability to use attenuated classical sources that can transmit much faster than current single-photon emitters has paved the way for practical implementations of quantum cryptography. Our system uses a pulsed stochastic photonic source with decoy

states [10], where the decoys are photonic wavepackets with a lower mean photon number. In this decoy-state protocol, Alice occasionally sends a lower intensity wavepacket or the vacuum quantum state. To further simplify the system and make it more efficient, we only transmit one state in the monitoring basis, which gives an equivalent secure key rate in comparison to transmitting both states in this basis [12, 13].

However, QKD systems are still subject to significant limitations. A key drawback is the difficulty in authenticating Alice and Bob. Sharing a secure, private key with another user is undermined by the inability to verify the identity of the other user. Additionally, QKD requires the development of extensive and technically challenging new infrastructure in order to add quantum capabilities to existing communication systems. Great challenges also remain in performing QKD over longer distances. Many QKD implementations employ fiber networks [14, 15, 16], but these are constrained over long distance connections due to signal attenuation in fiber. Unlike with classical fiber signals, quantum fiber signals cannot be simply amplified due to the no-cloning theorem; this, coupled with the immaturity of quantum repeater technology, has limited the maximum distance of current state-of-the-art fiber-based QKD to on the order of 1000 km [14].

Free-space optical links are subject to significantly less attenuation as a function of distance compared to fiber-based systems [17, 18]. QKD links on mobile platforms utilize these free-space optical links and provide additional benefits due to their reconfigurable nature. Reconfigurable communication nodes can help solve the so-called “last mile” problem [19], alleviating the need for expensive infrastructure used to connect a large number of different end users.

The technology developed to create a secure QKD connection between mobile platforms can also be easily applied to remote entanglement distribution. Distributing entangled photon pairs facilitates not only quantum communication but the creation of a quantum network

wherein remote quantum systems can be connected [20]. A full mobile quantum network can be used to connect remote quantum computing resources, an important building block for an eventual quantum internet. Another exciting application of quantum networks is the ability to create systems of entangled quantum sensors, the sensitivity of which scales with the number of nodes N (whereas the sensitivity of a classical network only scales with $\sqrt{N}$). Remote entanglement distribution is the immediate target for future work on this project.

A quantum network on mobile platforms presents a very different set of technical challenges compared to its terrestrial counterpart. Maintaining an optical link between mobile platforms requires an advanced pointing-and-tracking system. Additionally, because of limitations in the carrying capacity, space, and flight time of the mobile platform, all systems intended for a mobile quantum network must be designed with low size, weight, and power (SWaP) restrictions in mind. Many design choices in a quantum system already involve performance trade-offs. For example, a given single-photon detector may have high efficiency, but also come with high intrinsic detector noise. Choosing the optimal equipment is a matter of maximizing the figure-of-merit in the context of the overall system limitations and variables, and SWaP restrictions are simply another set of constraints for this optimization.

In QKD, the primary figure-of-merit is the secure key rate, which is the number of theoretically secure bits that can be transmitted per second. The asymptotic secure key rate is the secure key rate that can be achieved in the case of an infinite number of channel access attempts. However in a real QKD system, one can only access the channel finitely many times, and some data must be sacrificed to estimate the parameters that determine how much privacy amplification is necessary. Additionally, because finite data gives imperfect knowledge of these parameters (and the fraction of information leaked to the eavesdropper may be higher over the whole data set than what is measured in a particular subset therein), one must make pessimistic assumptions in one's privacy amplification to strengthen claims

on security. Due to the limited flight times for our system (and thus limited number of channel access attempts), these finite key effects [21], play a significant role in the secure key rate. This puts increased importance on the SWaP considerations for this system.

1.1 Outline and Summary of Contributions

In this thesis, I present the essential aspects of our designs and experimental procedures that make it possible to perform mobile QKD in a low SWaP setting amid outdoor environmental conditions. I present a particular focus on the optical train that allows the preparation and measurement of high fidelity photonic qubits while simultaneously dealing with common side-channel attack vulnerabilities. I also focus on the data management and post-processing techniques I develop, which are critical in outputting the sifted keys and other metrics needed to calculate the primary figure-of-merit: the finite secure key rate. The culmination of this work is the finite secure key rates presented for experiments in the air-to-air, air-to-car, and car-to-car mobile QKD configurations.

This project is the result of an extensive, multi-institutional collaboration between Prof. Daniel J. Gauthier’s group at the Ohio State University (OSU), and Prof. Paul Kwiat’s group at the University of Illinois (UIUC). Beginning in 2016, this project has involved both current and former graduate students, who are summarized in Table 1.1.

In Table 1.2, I summarize my particular contributions to project in each chapter of this thesis. The following section gives a brief summary of the key points of each chapter (including a representative figure from each), as well a more comprehensive explanation of the contributions of each project member to that part of the work.

	Name	Institution
Previous Team Members	Alex Hill	UIUC
	Joseph Szabo	OSU
	Akash Gupta	OSU
Current Team Members	Roderick Cochran	OSU
	Andrew Conrad	UIUC
	Samantha Issac	UIUC
	Timur Javid	UIUC
	Daniel Sanchez-Rosales	OSU

Table 1.1: Current and former members of the Drone QKD project.

Chapter 2: QKD Source Design

A QKD transmitter consists of three main subsystems: the control system, the photonic source, and the state preparation. The control system drives the photonic source, which sends the photonic wavepackets to the state preparation. From there, the encoded photonic qubits exit the transmitter and are sent to the receiver for measurement. In Chapter 2, I focus on the photonic source design and properties such as the mean photon number and the spectral, spatial, and temporal indistinguishability. I also discuss the systems that connect the photonic source to the control system and the state preparation, and the options in photonic sources along with their advantages and drawbacks in the context of this QKD system. Figure 1.1 shows an annotated photograph of our QKD source and connecting systems.

The basic design choices for these systems including the photonic source were made by former project members Alex Hill and Joe Szabo, in addition to the original LED-to-fiber coupler design. I detail the efforts to redesign the LED-to-fiber coupler to increase the coupled power, in which I spearheaded creating and testing multiple design iterations along with important contributions from Daniel Sanchez-Rosales. Daniel and I worked in close collaboration on the measurements and experimental design for the spectral and tempo-

Chapter	Contributions
Chapter 2 QKD Source Design	• LED-to-fiber coupler design and fabrication • Collaborated on spectral and temporal data collection • Derived quantitative indistinguishability metrics
Chapter 3 Free-Space Optics Bench	• Carried out tabletop QKD demonstrations • Helped choose optics and modify optical bench layout • Modeled polarization properties of the overall system, including non-ideal effects • Modeled system and created procedure for polarization fidelity compensation
Chapter 4 Field Demonstrations	• Compiled and developed procedures for tabletop and field testing, numerous mini-experiments • Collaborated on developing custom solutions to simplify field testing procedures Performed mini-experiments during field testing: • LED-to-fiber coupling • Internal transmitter alignment • Mean photon number balancing • Polarization fidelity compensation
Chapter 5 Data analysis and Processing	• Advised and helped troubleshoot FPGA data structure Developed extensive analysis software, including: • Receiver data stitching • Course synchronization methods • Bayesian qubit-based synchronization: theory, simulation, and practical implementation • Key sifting and noise filtering
Chapter 6 Discussion and Results	• Collaborated in performing QKD for all field applications • Generated performance plots, supporting figures-of-merit

Table 1.2: Summary of my contributions to the Drone QKD project.

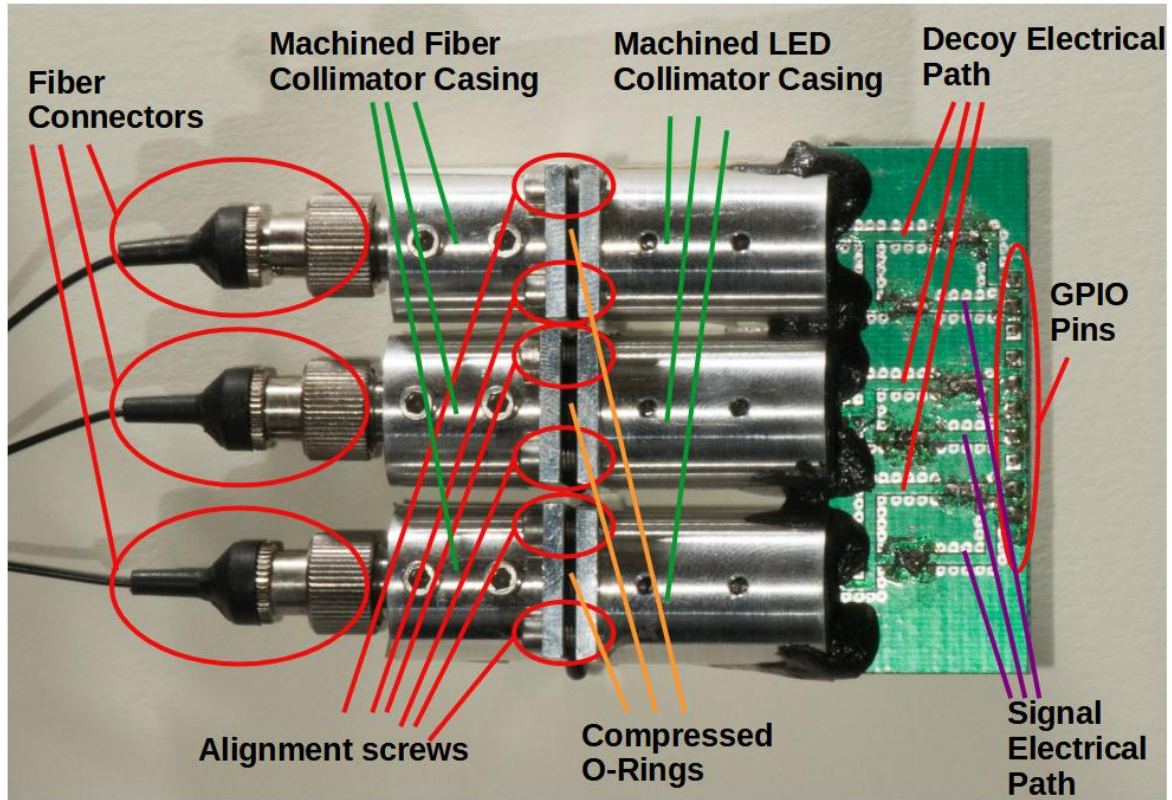


Figure 1.1: Photograph of the LED driving and fiber coupling apparatus. The right side of the photo shows the PCB used to drive the LEDs from the DE10-Standard GPIO pins through the two different resistive paths used to generate signal and decoy intensities. The LEDs are soldered to the PCB, while the custom-machined LED-to-fiber coupling apparatus (shown in the center) is epoxied to PCB as well. On the left, the fibers connect to this apparatus, from which they carry the optical pulses to the in-fiber attenuators and to the state preparation optical bench. Photo credit: Timur Javid.

ral indistinguishability. I took a lead role in collecting the temporal data, determining the necessary electrical pulse adjustments, and deriving the framework for quantifying the indistinguishability of the transmitted states. Daniel created the FPGA architecture that drives the source and allows the electrical pulse adjustments required for the temporal indistinguishability, while I assisted with conceptual discussions and pseudocode.

Chapter 3: Free-Space Optics Bench

The photonic source discussed in Chapter 2 handles the generation of wavepackets, the fiber collection, and subsequent attenuation. The next system in the optical train is the transmitter optical bench (shown in Fig. 1.2), which uses free-space optics to prepare the three polarization states along different optical paths before combining them so they can be sent out to the receiver. The receiver optics bench uses free-space optics to sort and measure all of the incoming states so they can be coupled into fiber and measured by the single-photon detector. The state preparation and measurement fidelity play a key role in determining the error rate of the system, which in turn has strong ramifications for the generation of secure key.

In Chapter 3, I give an overview of the system layout for both transmitter and receiver, along with a detailed discussion of the non-ideal polarization effects that impact the state preparation and measurement fidelity. I go on to explain the polarization compensation system used to counteract the birefringent effects of the spatial filter fiber. Additionally, I detail the procedures for tuning the polarization compensation, the internal transmitter and receiver alignments, and the alignment between the two. I close by mentioning the measures taken on the receiver bench to filter out noise and prevent stray light from overwhelming the quantum signal.

While many of the general design aspects and optical components for the transmitter

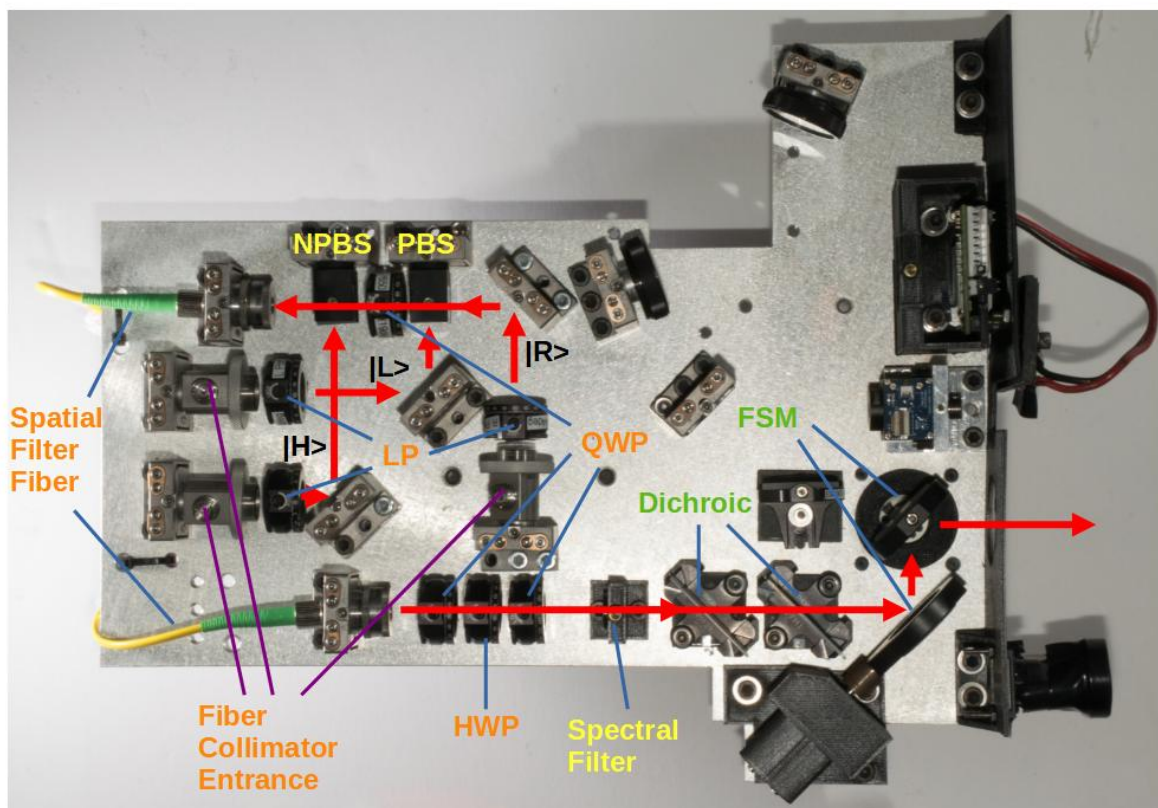


Figure 1.2: Photograph of the transmitter optics bench. State preparation optics along the three optical paths encode the R, L, and H polarization states. Beam splitters combine these three paths before they enter the spatial filter fiber. After exiting the fiber, the combined beam passes through the polarization compensation waveplates, the spectral filter, and then the dichroic mirrors that split off the pointing-and-tracking beacons. The fast steering mirrors on the right side of the bench perform fine control on the attitude of the beam exiting the optics bench. Photo credit: Timur Javid.

and receiver were chosen by former project members Alex Hill and Joe Szabo, I made the first effort to set up the tabletop version of the state preparation and measurement. I led the effort to create models of the polarization fidelity and its interactions with the various optical components. These processes led me to introduce many changes to the design in terms of finding more suitable polarization optical components, but also in terms of identifying the necessary degrees-of-freedom and the physical space required to tune the system in practice. I worked extensively on the tuning and alignment of the system in preparation for experiments and created procedures for others to use in these tasks. Samantha Isaac created the large majority of the transmitter and receiver design models with CAD, while Daniel Sanchez-Rosales and I provided important insights and feedback on changes between different design iterations. The pointing-and-tracking system that mediates the alignment between the transmitter and receiver was designed primarily by Andrew Conrad, along with contributions from Daniel Sanchez-Rosales and Timur Javid.

Chapter 4: Field Demonstrations

Thus far I have introduced the subsystems of a QKD system with high fidelity and indistinguishable states designed to operate in low SWaP conditions; however, a key obstacle in these experiments is dealing with environmental factors associated with outdoor flight testing, such as non-ideal or variable temperature, humidity, wind, ambient light, and vibration. Chapter 4 details the experimental efforts our team has undertaken to maintain this high level of performance in these non-ideal outdoor environments. In order to highlight and explore how the experiment must be adapted from a tabletop laboratory setting to an outdoor one, I compare and contrast the procedures in the two different settings, highlighting the mini-experiments that must be performed on testing days to prepare for full QKD data collection, why they are performed in a specific order, and the solutions engineered to make

these demonstrations possible. To conclude, I detail the procedures for the full data collection in the various mobile QKD configurations. Figure 1.3 shows the final data collection in the air-to-air testing configuration.



Figure 1.3: Photograph showing the two drones in flight during an air-to-air data collection at the testing field beside the Seed House. Photo credit: Timur Javid.

My contributions in Chapter 4 include developing and compiling the overarching procedures for tabletop and field testing. I developed the procedures for the LED-to-fiber alignment, the internal transmitter and receiver alignments, the mean photon number balancing, and the polarization fidelity compensation (for which I wrote a data acquisition GUI in Python that significantly improved the adjustment procedure). I worked closely with Daniel Sanchez-Rosales on the procedure for the full optical temporal compensation. Daniel wrote the FPGA code that allowed the adjustment of the electrical signals and adapted the code to also output the necessary reference signals. I wrote the data acquisition code in Python,

developed the metrics that informed which adjustments to make, and wrote a rudimentary automation of the process that interfaced with Quartus using an auto clicker package. Daniel adapted my data acquisition code to measure and save the electronic waveforms and wrote a better interface between Quartus and the data acquisition in Python.

On a typical night of field testing, I performed the LED-to-fiber alignment, the internal transmitter alignment (multiple times), the optical temporal pulse adjustment (or Daniel would perform the electronic temporal pulse adjustment), the mean photon number balancing, and the polarization fidelity compensation. Typically, Daniel and Samantha Isaac performed the internal receiver alignment, and Daniel and Andrew Conrad collaborated on the pointing-and-tracking tuning. During QKD data collection, Samantha and Andrew piloted the drones and/or drove the cars, Daniel handled the FPGA data acquisition, while I activated the control loops and monitored the flight status and informed the pilots when power was running low.

Chapter 5: Data Analysis and Processing

After the QKD data is collected, it must be synchronized and processed so that the relevant performance metrics can be calculated and a sifted key can be extracted, which can then be distilled into a secure key after the security analysis is performed. Chapter 5 covers the structure of the data and some of its associated issues, the pre-processing steps that synthesize the asynchronous receiver data into a single data stream, the Bayesian qubit-based synchronization algorithm that allows the reconciliation of the transmitter and receiver data, and the subsequent noise filtering and key sifting. Figure 1.4 shows how well my synchronization algorithm performs as a function of transmission loss. The output of this data processing is ready for the security analysis performed by our collaborators at the University of Waterloo, as well as any subsequent privacy amplification.

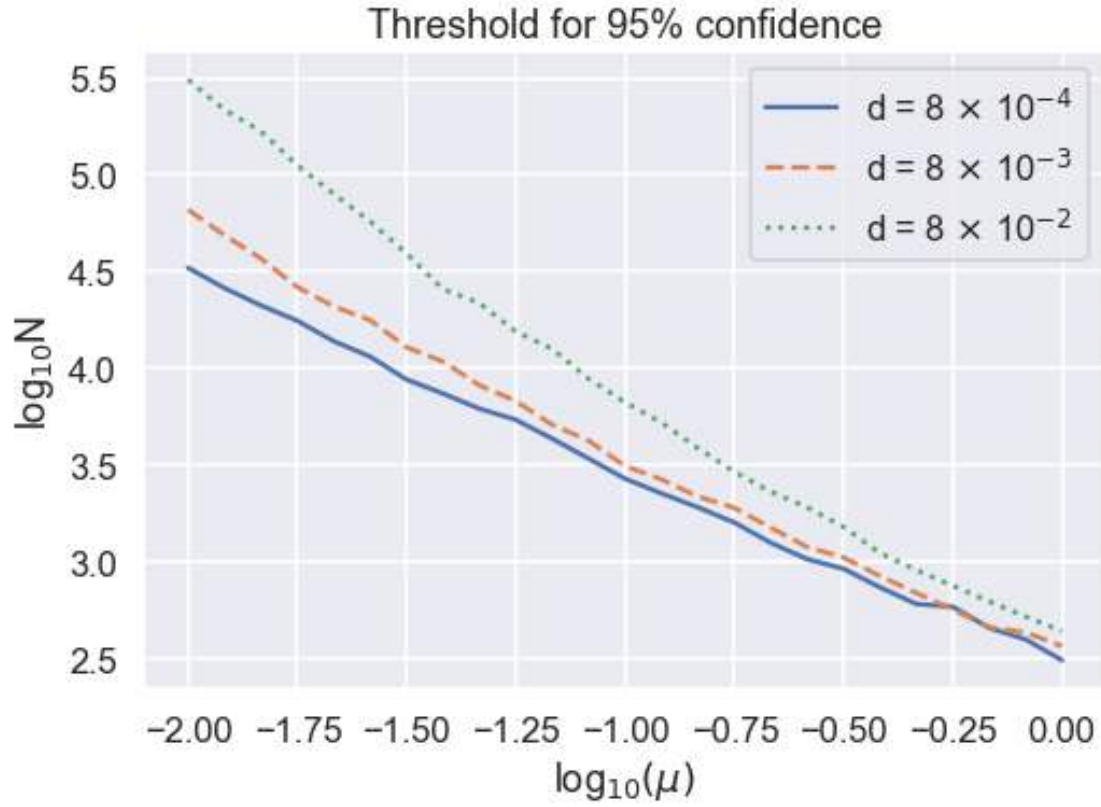


Figure 1.4: Dependence of string length threshold to achieve 95 percent synchronization confidence on received mean photon number on a logarithmic scale, parameterized by different dark count probabilities.

The data structure from the FPGA data collection described in the first part of Chapter 5 was pioneered by Daniel Sanchez-Rosales, with contributions to prototypes from Akash Gutha, and continual input and troubleshooting efforts from me. Additionally, Daniel and I collaborated on the measurement of the clock drift rate between PLLs on different FPGAs. All other sections pertaining to the processing of the data after it is extracted from the FPGAs up through the creation of the final summary table and the sifted key were my exclusive domain and contribution.

Chapter 6: Discussion and Results

Having introduced the system hardware, procedures, and data processing, I now present the results of the QKD data collection along with figures-of-merit. I highlight data sets that demonstrate the progression of difficulty in data collection from the laboratory to field experiments. These configurations of interest are **tabletop**, **ground-to-ground**, **air-to-air**, **air-to-car**, **car-to-car (low speed)**, and **car-to-car (high speed)**. The figures-of-merit and associated plots give insight into the particular challenges encountered in each configuration and the degree to which the system was able to address them. Table 1.3 gives a summary of the key metrics and figures-of-merit for the experiments in all the configurations discussed in Chapter 6.

Daniel Sanchez-Rosales, Andrew Conrad, Samantha Isaac, and I collaborated to collect these data sets. The plots and supporting figures-of-merit (*e.g.*, QBER, synchronization confidence, etc.) are outputs of my data analysis code. The finite secure key rates, which are the primary figures-of-merit for these data sets, were calculated by Lars Kamin at the University of Waterloo.

	Collection Time (s)	Raw Key Rate (kbits/s)	Average QBER (%)	Finite Secure Key Rate (kbits/s)
Tabletop	93.7	501	3.06	41.3
Ground-to-Ground	209.1	165	2.83	6.05
Air-to-Air	166.4	247	2.21	7.98
Air-to-Car	129.8	113	3.31	1.52
Car-to-Car (5 mph)	93.7	235	2.57	19.6
Car-to-Car (70 mph)	144.2	9.26	3.74	0

Table 1.3: Summary table of the relevant figures-of-merit for QKD demonstrations in different mobile platform configurations.

Chapter 2

QKD Source Design

A QKD transmitter consists of three main subsystems: the control system, the photonic source, and the state preparation. The control system drives the photonic source, which sends the photonic wavepackets to the state preparation. From there, the encoded photonic qubits exit the transmitter and are sent to the receiver for measurement (see Fig. 2.1). In this chapter, I focus on the photonic source design and properties such as the mean photon number and the spectral, spatial, and temporal indistinguishability. I also discuss the systems that connect the photonic source to the control system and the state preparation, and the options in photonic sources along with their advantages and drawbacks in the context of this QKD system.

The basic design choices for these systems including the photonic source were made by former project members Alex Hill and Joe Szabo, in addition to the original LED-to-fiber coupler design. I detail the efforts to redesign the LED-to-fiber coupler to increase the coupled power, in which I spearheaded creating and testing multiple design iterations along with important contributions from Daniel Sanchez-Rosales. Daniel and I worked in close collaboration on the measurements and experimental design for the spectral and temporal indistinguishability. I took a lead role in collecting the temporal data, determining the

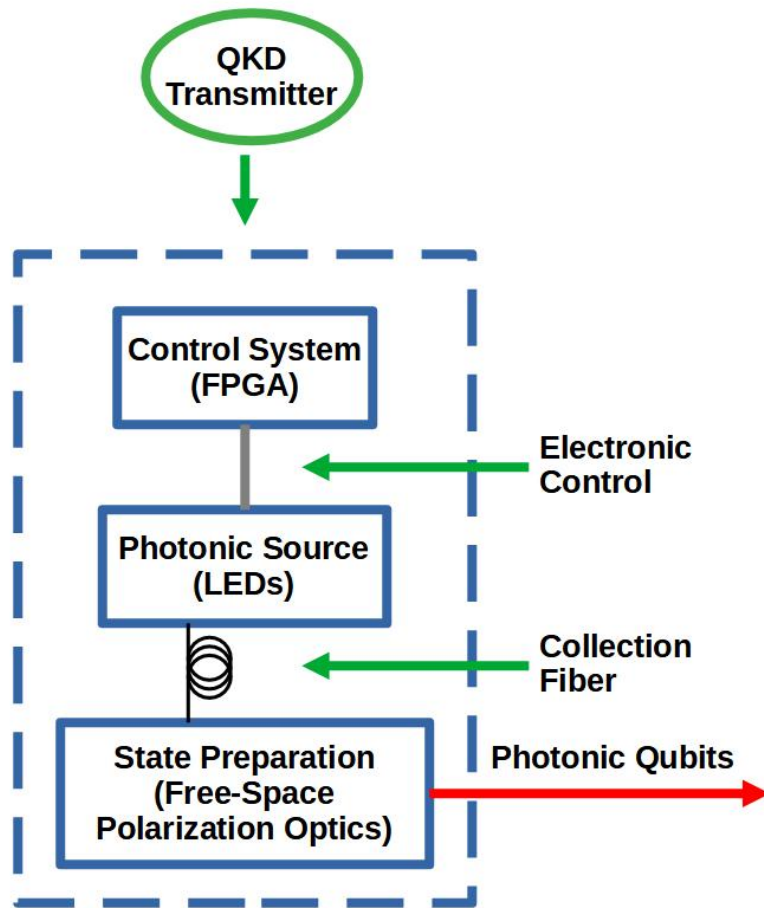


Figure 2.1: Block diagram of the overall QKD transmitter architecture for this system. The FPGA drives the LEDs to generate photonic wavepackets, which are conveyed to the polarization optics bench via fiber to generate encoded qubits for the QKD protocol.

necessary electrical pulse adjustments, and deriving the framework for quantifying the indistinguishability of the transmitted states. Daniel created the FPGA architecture that drives the source and allows the electrical pulse adjustments required for the temporal indistinguishability, while I assisted with conceptual discussions and pseudocode. In this chapter, “we” and “our” refers to my efforts in collaboration with Daniel Sanchez-Rosales.

2.1 Important Qualities in a QKD Source

Designing a high quality source that can produce secure, high-fidelity photonic qubits is a key aspect of a QKD system. Polarization is the information encoding degree-of-freedom for this system; however, it is crucial in order to have a fully secure QKD system that the other degrees-of-freedom be indistinguishable across the different polarization states. In essence, any variation in the spectral, temporal, and spatial degrees-of-freedom must be uncorrelated with the state of polarization. The security of QKD rests on the assumption that the eavesdropper must measure the quantum system in the information encoding degree-of-freedom in order to glean any information related to the transmitted key. As an extreme example, one can consider a QKD system sending two polarization states, horizontal and vertical, but the horizontal polarization state always arrives earlier in time. In this case the eavesdropper could learn the contents of the transmitted key by measuring the arrival time of the photons, and the other two parties would be unaware because temporal measurements need not affect the polarization measurement statistics. Strategies in which the eavesdropper exploits correlations in the peripheral degrees-of-freedom are known as side-channel attacks [22]. Another important aspect of the design choice for the source in this system is reducing the SWaP requirements. Because the system is intended for mobile platforms including airborne drones, the weight and power constraints significantly influence the photonic source

for this system.

2.2 Choosing a Photonic Source

There are multiple options when choosing a photonic source for a QKD. The original BB84 method uses a true single-photon source, but these sources typically have limited photon generation rates and/or require bulky, power intensive supplementary hardware. With the introduction of the decoy state protocol [11], one can deploy attenuated classical light sources instead and generate photonic wavepackets at much greater rates. This increase in repetition rate provides significant advantage in the secure key generation.

There are multiple viable attenuated classical sources that can be used in a decoy state protocol with different advantages and drawbacks. The primary choices considered were LEDs and lasers, and whether to have one source with a polarization modulator or to use multiple sources to produce the different polarization states.

Many QKD systems use weak coherent pulses [10, 16, 17, 18], which are typically produced using attenuated pulsed lasers. In designs that use multiple lasers with different optical paths to encode the different polarizations, it can be difficult to guarantee spectral indistinguishability between the separate source because lasers have very narrow bandwidth that can vary with manufacturing differences.

Designs that instead use a single laser to produce all the polarization states require a high-speed polarization modulator to produce the different polarization states. One drawback for these types of systems is the tendency for the modulator itself to induce changes in the spectral and temporal degrees-of-freedom [23], potentially compromising the information security. In addition to these drawbacks, lasers and polarization modulators are heavier and more power intensive in comparison to LEDs, especially if temperature controllers are

employed.

The main disadvantage of LED sources is the low spectral brightness. While the broader spectrum of the LEDs compared to lasers makes spectral indistinguishability easier to achieve (with the help of spectral filters), the low spectral brightness limits the ability to achieve higher mean photon numbers. However, the combination of resonant cavity light emitting diodes, which have relatively higher spectral brightness, and engineering solutions to efficiently collect light from these sources makes it possible to combat this issue effectively. Ultimately, former project members Alex Hill and Joe Szabo chose resonant cavity LEDs (RC-LED-650-02) as the best available light source for this system. These are a lightweight, low-power solution that also presents a few advantages with regard to side-channel attacks.

2.3 LED-to-Fiber Coupling

2.3.1 Early Designs

The light from the LEDs is carried to the polarization preparation optics bench via single-mode fibers, including in-fiber attenuators that provide control over the intensity of each state so one can balance the mean photon numbers at the desired level. However, there must be an LED-to-fiber coupling system in order to efficiently collect the light from the LEDs into the fiber. One of the biggest challenges associated with the LED light source is coupling a sufficient amount of light into these fibers to supply a reasonable mean photon number for the QKD protocol.

The original static coupler design collected light by butting the fiber directly against the LED. The LED itself was epoxied into the 3D printed LED-to-fiber coupler, shown in Figure 2.2a, while the collection fiber was screwed into the threaded side of the coupler. However,

this design had drawbacks in the fact that the part was 3D printed, including the threading on which the optical fiber was tightened. These printed threads were not durable and would wear out quickly after being tightened and loosened several times, further complicating the capacity for consistently coupling in power from the LEDs. Additionally, I found that these 3D printed versions of the coupling assembly were subject to noticeable mechanical creep as well as temperature dependence and they functioned poorly in outdoor settings. This ultimately led to a design change using a custom-machined aluminum coupling assembly instead.

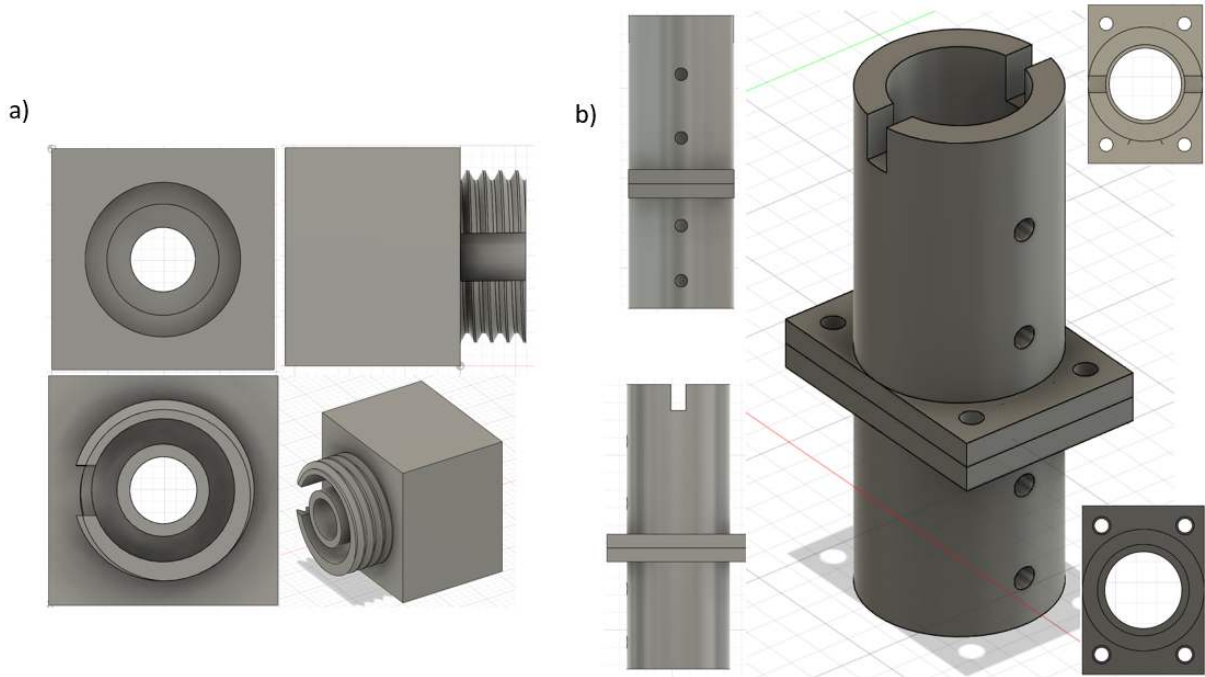


Figure 2.2: CAD renderings of the original and current LED-to-fiber coupler designs. The original static coupler design (a) was 3D printed and the bare LED was epoxied into the plastic coupler. The optical fiber connector threads directly onto the 3D printed part, placing the optical fiber directly in front of the LED. The current design (b) is a machined dynamic two-piece LED-to-fiber coupler. The LED collimator and the fiber collimator slide into the two aluminum cylinders, which are mated and aligned using four screws along with an o-ring pressed between the two components. The holes along the cylinder are tapped and set screws are used to hold the collimators in place.

The diode emission element emits light in many directions, and the single-mode fiber can only accept light over a narrow range of incident angles. The presence of a wire directly above the LED emission element limited how close the fiber could be placed to the element in the original design, thus reducing the fraction of rays from the emitter that were incident on the collection fiber (see Fig. 2.3a). This severely limited the maximum fraction of the LED light that could be collected into the fiber. The emission element can be thought of as a point source, and the introduction of a collimation lens allow the light to be collected, collimated, and refocused onto the collection fiber at a more manageable distance from the emission element. Furthermore, in the original design, when the fiber was off-center with respect to the diode element or not oriented perfectly toward the emission element it collected significantly lower power. The original coupler did not provide the degrees-of-freedom necessary to adjust this alignment. This ultimately led to a design change with collimators for both the LED and the fiber to gather and refocus the light, and a compressed o-ring and adjustment screws that could provide the necessary tip-tilt control.

2.3.2 Improved Design

After multiple design iterations, we settled on a new LED to fiber coupler that houses the LED in an adjustable focus collimator, which is butt-coupled to a fixed focus fiber collimator (Thorlabs F220FC-B) with a machined aluminum housing that compresses an o-ring between them (see Fig. 2.2b). The collimators allow the refocusing of the light at a more practical distance from the diode element, giving the opportunity to couple in more light from the LED (see Fig. 2.3b). The compressed o-ring and screws that compress it provide the tip-tilt control necessary to optimize the coupling in the face of angular dependencies. This design allows significantly higher power collection from the LEDs, and including all other losses in

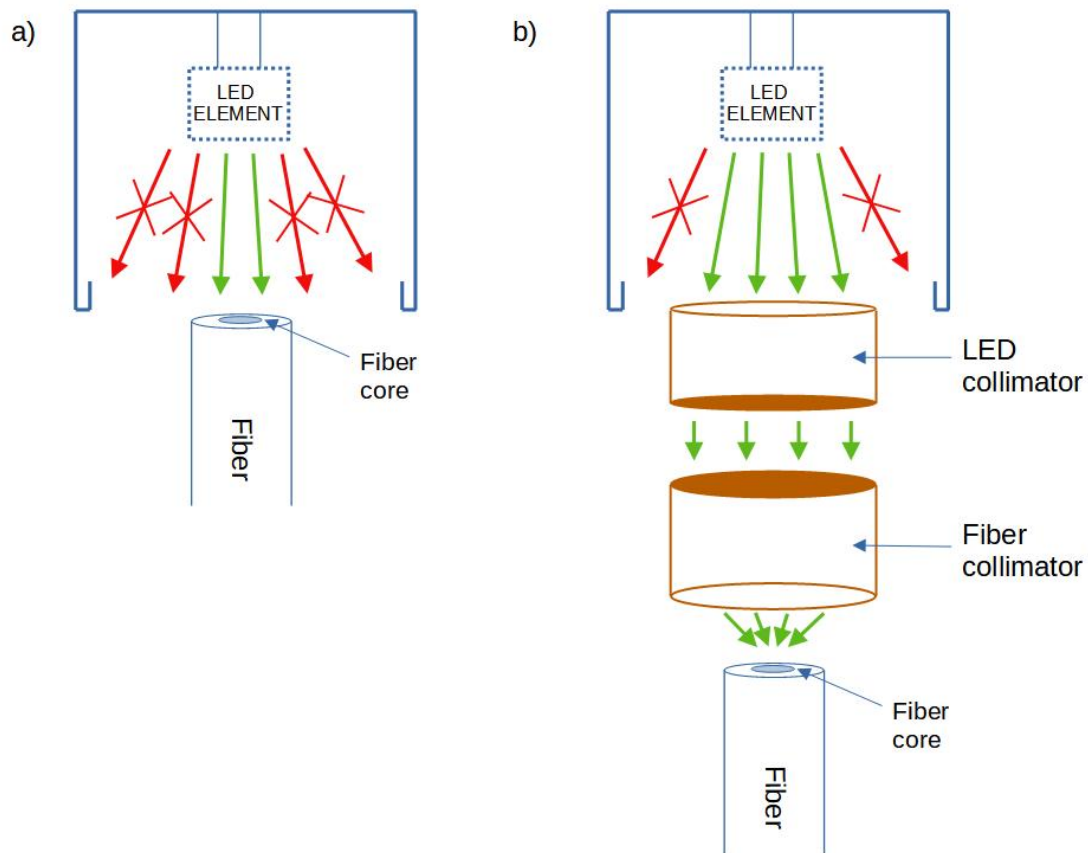


Figure 2.3: Earlier design iterations involved statically securing the fiber to the LED housing without any focusing elements (a). The current design employs collimating lenses that collect the light coming from the LED element so it can be refocused onto the fiber at a more manageable distance (b).

the transmitter system, I have been able to achieve mean photon numbers on the order of unity. This is as high an intensity as is optimal for most QKD protocols [11, 24].

The LED source is driven by a DE10 Standard field programmable gate array (FPGA) board. The LEDs are soldered to a custom PCB with a GPIO connector and two different resistive paths (see Fig. 2.4). Each LED has two GPIO pins that can drive current to it. By alternately setting one pin to the driving signal and the other to high impedance, the current is channeled through a particular resistive path. In this fashion the FPGA supplies different currents to the LED to create the signal and decoy states for the QKD protocol.

The alignment procedure for one of the LED-to-fiber couplers consists of iteratively adjusting each of the four screws in turn until the power is maximized. While it is true that adjusting three of the four screws is sufficient to control the angle between the two collimators, adjusting the fourth helps counterbalance tension in the o-ring when screws are tightened unequal amounts. Additionally, I adjust the focus on the LED collimator to improve the mode matching and increase the maximum coupled power. Unfortunately, the LED collimator focus can only be adjusted while the two halves of the LED-to-fiber coupler are separated. To tune this focus, I optimize the coupler alignment (as described above) then make a small adjustment to the focus, then iteratively repeat these two steps until a focus has been found that maximized the coupled power after the alignment. I typically tune this alignment while running the LEDs in CW mode, which increases the power output enough that I can use a Thorlabs fiber-coupled power meter to optimize the coupling rather than a single-photon detector. In this configuration I typically couple 15-25 nW once the LED-to-fiber package is fully optimized.

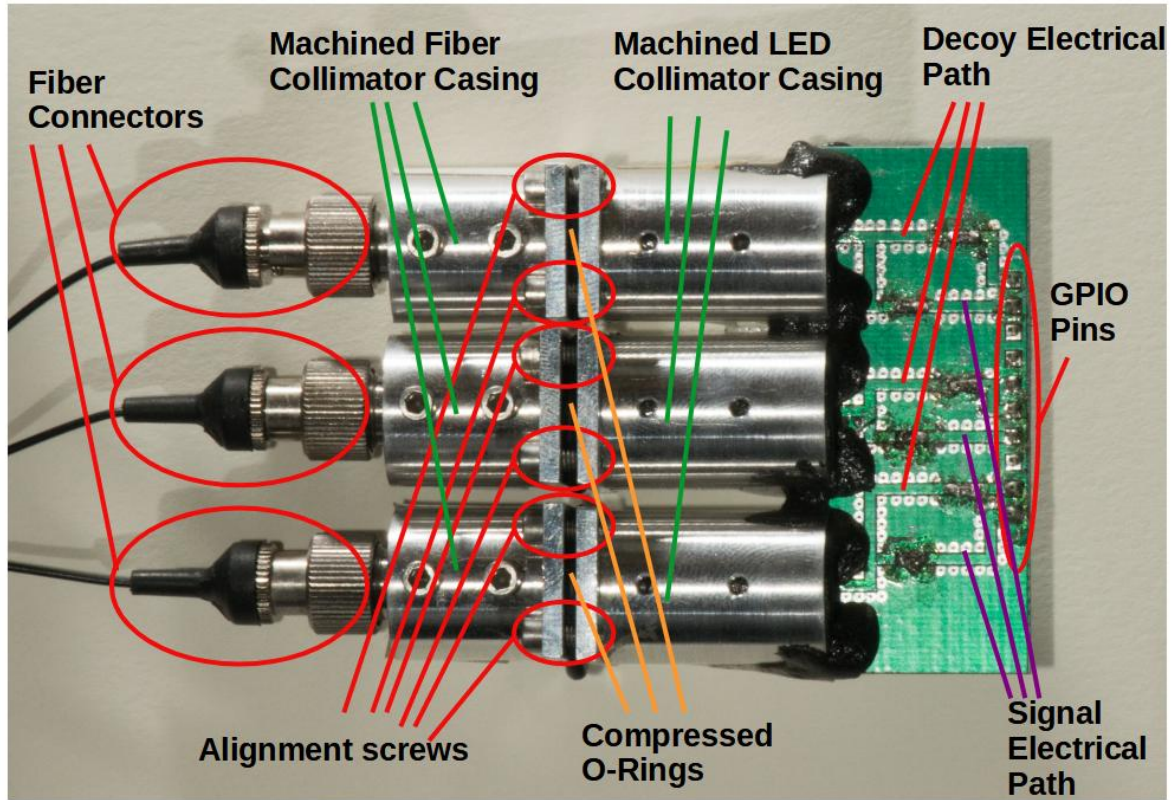


Figure 2.4: Photograph of the LED driving and fiber coupling apparatus. The right side of the photo shows the PCB used to drive the LEDs from the DE10-Standard GPIO pins through the two different resistive paths used to generate signal and decoy intensities. The LEDs are soldered to the PCB, while the custom-machined LED-to-fiber coupling apparatus (shown in the center) is epoxied to PCB as well. On the left, the fibers connect to this apparatus, from which they carry the optical pulses to the in-fiber attenuators and to the state preparation optical bench. Photo credit: Timur Javid.

2.4 Spectral Indistinguishability

Differences in the spectra of the different QKD states presents a potential side-channel attack whereby the eavesdropper can measure the spectral degree-of-freedom to determine the state without disturbing the polarization. Because the system consists of three separate LEDs that produce the different polarization states, there is a potential information security leak arising from the non-overlap of the different LED spectra. For the sake of rigor, we also consider spectral differences between the signal and decoy states for a given LED; *i.e.*, differences in spectrum resulting from driving the LED with different currents. However, we find these differences to be largely negligible compared to the native spectral distinguishability between separate LEDs resulting from manufacturing differences.

We employ two main strategies to combat the spectral indistinguishability issues: LED curation and spectral filtering. LED curation is simply choosing a set of three LEDs that happen to already have high native spectral overlap. This is a relatively simple matter of measuring the spectra for all the remaining LEDs in the ordered batch, choosing the three with the highest overlap, and soldering them to a new PCB. The unfiltered curated LED spectra is shown in Figure 2.5. I quantify the indistinguishability of the sources using the classical mutual information, which is discussed in detail in Section 2.7.

The unfiltered spectra have a full width half maximum of about 12 nm, and their native overlap gives a leaked mutual information of 1.77×10^{-4} (with a 1.23×10^{-6} measurement bias and a 1.55×10^{-5} uncertainty). The spectral filtering employs a 1 nm bandwidth spectral filter (Andover 656FS02-12.5), which is inserted near the end of the optical path after all states have been combined into the same spatial mode. The spectral overlap after the spectral filter is shown in Figure 2.6, giving a leaked mutual information of 2.4×10^{-5}

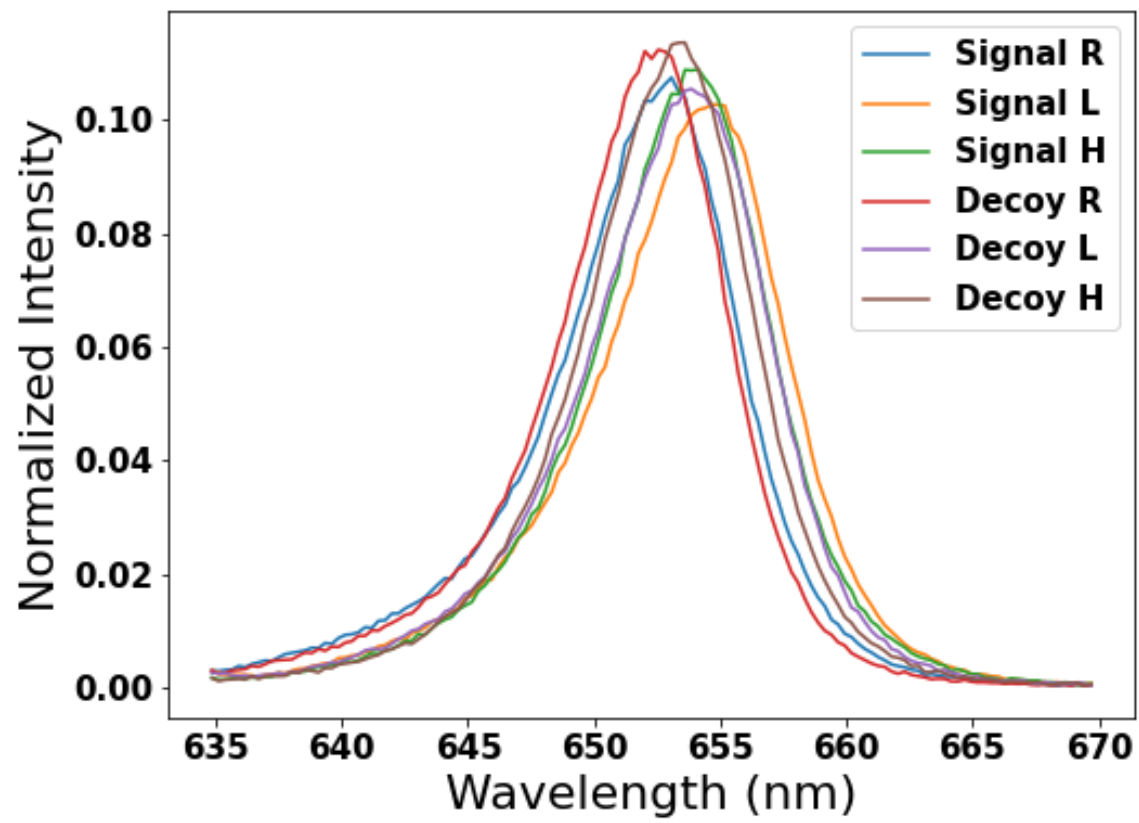


Figure 2.5: The native spectra for the 3 LED sources, without any spectral filtering.

(with a 2.44×10^{-5} measurement bias and uncertainty of 1.20×10^{-5}).

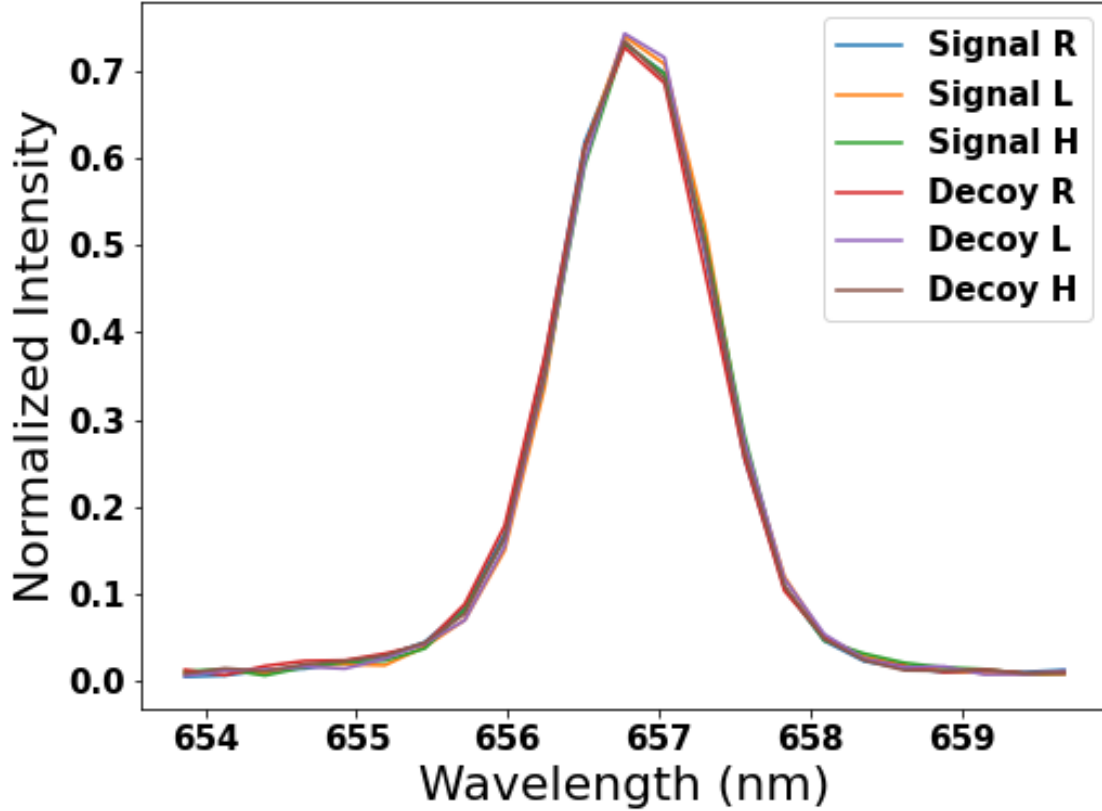


Figure 2.6: The spectra of the 3 LED sources after passing through a 1 nm narrow band spectral filter (Andover 656FS02-12.5).

We made the choice of a 1 nm spectral filter in the context of enforcing indistinguishability, transmitting sufficient intensity, and serving as a noise filter (because there is an identical spectral filter on the QKD receiver). The most significant differences in the native LED spectra tend to be in the tails of the distributions, thus a narrow spectral filter near the center of the native spectra (where they are more similar) noticeably improves the spectral indistinguishability of the states. There is significant advantage in using a narrow spectral

filter on the receiver in order to reject light at other wavelengths (typically from ambient light or from the laser beacons required for the pointing-and-tracking system), which would noticeably affect the signal-to-noise ratio. However, a narrower spectral filter also corresponds to a lower maximum intensity that the transmitter can provide, as a lower fraction of the spectrally-broad LEDs is allowed to transmit. We ultimately choose a 1 nm filter as the option that provides the best overall solution with these considerations. Another parameter that varies between different spectral filters is the maximum transmission efficiency (*i.e.*, the fraction of the light in the designed wavelength range that is transmitted). Lower maximum transmission can reduce the maximum achievable mean photon number and hurt the overall channel transmission, because all the prepared states must pass through the spectral filters on both the transmitter and receiver, respectively. We seek to strictly maximize this parameter with our choice of filter because higher maximum transmission is always helpful in our system. The filter we select (Andover 656FS02-12.5) has a maximum transmission of around 40%.

2.5 Temporal Indistinguishability

Differences in the temporal waveforms of the different states also expose the system to side-channel attacks. The six states sent by the transmitter are each generated with different electrical pulses on the FPGA. The different logical and electrical paths these pulses travel through can create timing differences between the resulting optical pulses. In addition, The LED current response is somewhat non-linear, and because the decoy state electrical pulses provide a lower current to the LEDs, the resulting optical pulses are typically decreased in width and delayed in comparison to signal states. Another factor that can affect temporal pulse overlap is the different lengths of the free-space optical paths for the different

polarizations.

We compensate for all of these timing differences using dynamic shifting of the FPGA phase-lock-loops, which provides control over both the width and relative offset of the electrical pulses sent to the LEDs. In order to measure the temporal overlap of the different states, we intercept the beam path after it exits the state preparation bench and align it into a single-mode fiber, which in turn feeds into a single-photon detector. We use an output of the FPGA to send a copy of the 12.5 MHz clock to the time-tagger to be used as a reference signal. We also connect the electrical output of the single-photon detector to the time-tagger. Using these two inputs we record the time difference for each received signal at the single-photon detector. Each electrical response from the single-photon detector provides one sample of the temporal waveform of the optical pulse, and histogramming these results aggregated over time makes a measurement of the temporal profile of each state with an uncertainty determined by the total number of signals received. Our Swabian Time Tagger Ultra has a reported jitter of 9 ps and our single-photon detector (SAP-500) has a reported jitter on the order of 170 ps, which broadens the resulting temporal waveform measurement. While it is still beneficial to use a measurement resolution below the jitter in order to get an accurate measurement of the broadened waveform, it is also important to strike a reasonable balance between measurement resolution and total data collection time. Finer resolutions require more collection time to obtain the same number of counts per bin, and thus to obtain the same relative error. We find that the most practical balance of these considerations is achieved by collecting data with a time resolution of 20 ps.

The presence of multi-photon events creates a bias in the temporal waveform measurement because the detector dead time precludes the measurement of any subsequent photons within a 20 ns window. In order to mitigate this bias, we tune the mean photon number to be 0.1 for each state measured. While a lower mean photon number would reduce this bias

further, it would also increase the data collection time and reduce the signal to noise ratio. Because all the states are tuned to the same mean photon number, they are all subject to the same bias, albeit a small one, and the derived overlap metrics are still valid. We record data with a collection time of 300 s for the signal states and 1200 s for the decoy states, which allows the collection of sufficient data to give repeatable values in the indistinguishability metric to the precision of the adjustment resolution.

By measuring the temporal profile for each state and comparing it to a target temporal profile, we iteratively adjust the phase-lock-loops until maximal overlap is achieved. The resulting temporal waveforms we achieve after performing this procedure are shown in Figure 2.7, giving a leaked mutual information of 4.31×10^{-5} (with a 3.18×10^{-6} measurement bias and a 7.99×10^{-7} uncertainty).

2.6 Spatial Indistinguishability

In order to ensure indistinguishability in the spatial degree-of-freedom for the different polarization states, I use beam splitters to combine the three paths into a single spatial mode such that they all couple into the same single-mode fiber. However, the birefringence of a single-mode fiber exposes the polarization states to a unitary transformation, which severely impacts the state fidelity. In earlier iterations of the system design, I attempted to connect custom single-mode fibers at short enough lengths (less than 7 cm) that the state fidelity impact would be negligible. Nevertheless, neither my in-house efforts nor short fibers I purchased were able to avoid significant impacts to the polarization state fidelities. Because the transformation from the single-mode fiber is unitary, it can be corrected using a combination of two quarter waveplates and one half waveplate. The procedure for this polarization fidelity compensation is discussed in later sections. With the polarization fidelity

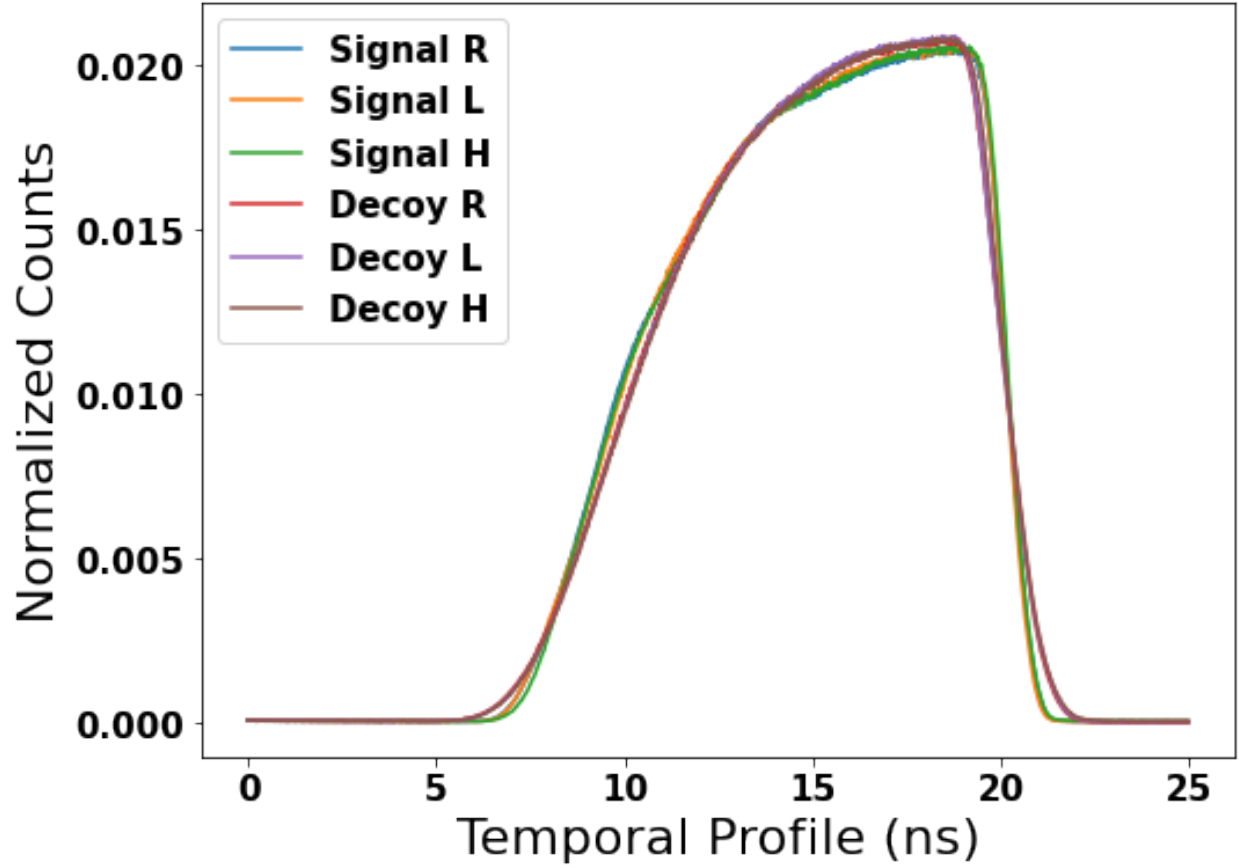


Figure 2.7: The temporal profiles of the 6 different prepared polarization states after making electronic adjustments to maximize overlap. We perform the measurement with a mean photon number of 0.1 and a collection time of 300 s for the signal states and 1200 s for the decoy states.

compensation in place, I am no longer constrained to use the shortest possible fiber, so I instead employ a 30-cm-long Thorlabs 630HP single-mode fiber for spatial filtering [25].

2.7 Quantifying Indistinguishability

In order to quantify the indistinguishability of Alice's various states, I use the mutual information to calculate the fraction of the final sifted key that an eavesdropper would know after making temporal and/or spectral measurements on every state that is sent. In an ideal experimental setup, Alice and Bob would have identical sifted keys, but even in the non-ideal case, their sifted keys are equally relevant, so I choose to compare with Bob's sifted key for convenience. I take E to be the set of Eve's side-channel measurements, B to be the set of Bob's sifted bits, and S to be the statement that the channel access attempt matriculated into a sifted bit. The mutual information between Eve's measurements and Bob's sifted key is given as

$$I(B; E|S) = H(B|S) + H(E|S) - H(B, E|S) \quad (2.1)$$

where H is the Shannon entropy:

$$H(X) := - \sum_{x \in X} p(x) \log p(x)$$

Thus,

$$\begin{aligned} I(B; E|S) = & - \sum_k p(B_k|S) \log p(B_k|S) - \sum_j p(E_j|S) \log p(E_j|S) \\ & + \sum_{k,j} p(B_k, E_j|S) \log p(B_k, E_j|S) \end{aligned} \quad (2.2)$$

In order to compute these entropies, I must find $p(B_k|S)$, $p(E_j|S)$, and $p(B_k, E_j|S)$ in terms

of known probabilities.

$p(B_k|S)$ can be expanded by applying marginalization,

$$p(B_k|S) = - \sum_i p(B_k|A_i, S)p(A_i|S)$$

where A is the set of states Alice can send. Applying Bayes' Theorem I get

$$p(B_k|S) = - \sum_i p(B_k|A_i, S) \frac{p(S|A_i)p(A_i)}{p(S)}$$

And applying marginalization again, I get

$$p(B_k|S) = - \frac{\sum_i p(B_k|A_i, S)p(S|A_i)p(A_i)}{\sum_\ell p(S|A_\ell)p(A_\ell)}$$

With all terms now being known probabilities. Now, $p(E|S)$ can be derived identically:

$$p(E_j|S) = - \frac{\sum_i p(E_j|A_i, S)p(S|A_i)p(A_i)}{\sum_\ell p(S|A_\ell)p(A_\ell)}$$

but can be simplified further by the knowledge that $p(E_j|A_i, S) = p(E_j|A_i)$ because the distribution of Eve's measurements is fully determined by the knowledge of Alice's state, and S is irrelevant.

$$p(E_j|S) = - \frac{\sum_i p(E_j|A_i)p(S|A_i)p(A_i)}{\sum_\ell p(S|A_\ell)p(A_\ell)}$$

I can rewrite

$$p(B_k, E_j|S) = p(B_k|E_j, S)p(E_j|S)$$

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, E_j, S)p(A_i|E_j, S)p(E_j|S)$$

$p(B_k|A_i, E_j, S) = p(B_k|A_i, S)$ because Bob's measurement outcome is fully determined by Alice's input state and the knowledge that it matriculates into a sifted bit.

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, S)p(A_i|E_j, S)p(E_j|S)$$

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, S) \frac{p(E_j|A_i, S)p(A_i|S)}{p(E_j|S)} p(E_j|S)$$

$$p(B_k, E_j|S) = \sum_i p(B_k|A_i, S)p(E_j|A_i, S) \frac{p(S|A_i)p(A_i)}{p(S)}$$

$$p(B_k, E_j|S) = \frac{\sum_i p(B_k|A_i, S)p(E_j|A_i, S)p(S|A_i)p(A_i)}{\sum_\ell p(S|A_\ell)p(A_\ell)}$$

At this point, I have written all relevant quantities in terms of known probabilities that can be plugged into Equation 2.2 to find the mutual information the Eavesdropper gains from the side-channel measurements.

However, I must also account for the systematic bias in my mutual information calculation. Because measurement noise inherently makes a variable more random, said noise creates a systematic bias toward higher entropies. Roulston [26] shows that the observed entropy H_{obs} can be written as

$$\langle H_{obs} \rangle = H_\infty + \sum_i^B \frac{\langle \epsilon_i^2 \rangle p_i}{2} \quad (2.3)$$

where H_∞ is the “true” entropy of the system, B is the number of states, ϵ_i is the fractional

measurement error in the state probability p_i , with the measured probability q_i such that

$$\epsilon_i = \frac{q_i - p_i}{p_i}$$

Thus, Equation 2.3 can be rewritten as

$$\langle H_{obs} \rangle = H_\infty + \sum_i^B \frac{\langle (q_i - p_i)^2 \rangle}{2p_i}$$

and using the definition of the variance and approximating the denominator as $2q_i$, I get

$$\langle H_{obs} \rangle = H_\infty + \sum_i^B \frac{\sigma_i^2}{2q_i} \quad (2.4)$$

where σ_i is the error in the measurement of p_i . I now have the entropy bias in terms of known quantities, and I can add the biases of the constituent entropies to easily find the bias for the mutual information as a whole.

To quantify the uncertainty of the mutual information, I used two approaches, the first of which being the standard propagation of errors technique using first derivatives:

$$\sigma_f = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2}$$

That is, for some function f , the uncertainty σ_f in terms of the uncertainties of and the derivatives with respect to the constituent variables x_i .

However, this formula is an approximation that assumes $\frac{\partial f}{\partial x_i}$ is relatively constant in the neighborhood of σ_{x_i} . While this approximation is accurate for evaluating the temporal degree-of-freedom (for which the measurement has smaller error), it is not accurate for the

spectral degree-of-freedom.

Instead, I employ a Monte-Carlo sampling approach. By randomly sampling the constituent variables from their distributions and evaluating the mutual information function many times, I am able create a histogram representing the probability distribution of the mutual information. The standard deviation of this distribution is the uncertainty of the mutual information measurement.

Chapter 3

Free-Space Optics Bench

The photonic source discussed in Chapter 2 handles the generation of wavepackets, the fiber collection, and subsequent attenuation. The next system in the optical train is the transmitter optical bench, which uses free-space optics to prepare the three polarization states along different optical paths before combining them so they can be sent out to the receiver. The receiver optics bench uses free-space optics to sort and measure all of the incoming states so they can be coupled into fiber and measured by the single-photon detector. The state preparation and measurement fidelity play a key role in determining the error rate of the system, which in turn has strong ramifications for the generation of secure key.

In this chapter, I give an overview of the system layout for both transmitter and receiver, along with a detailed discussion of the non-ideal polarization effects that impact the state preparation and measurement fidelity. I go on to explain the polarization compensation system used to counteract the birefringent effects of the spatial filter fiber. Additionally, I detail the procedures for tuning the polarization compensation, the internal transmitter and receiver alignments, and the alignment between the two. I close by mentioning the measures taken on the receiver bench to filter out noise and prevent stray light from overwhelming the quantum signal.

While many of the general design aspects and optical components for the transmitter and receiver were chosen by former project members Alex Hill and Joe Szabo, I made the first effort to set up the tabletop version of the state preparation and measurement. I led the effort to create models of the polarization fidelity and its interactions with the various optical components. These processes led me to introduce many changes to the design in terms of finding more suitable polarization optical components, but also in terms of identifying the necessary degrees-of-freedom and the physical space required to tune the system in practice. I worked extensively on the tuning and alignment of the system in preparation for experiments and created procedures for others to use in these tasks. Samantha Isaac created the large majority of the transmitter and receiver design models with CAD, while Daniel Sanchez-Rosales and I provided important insights and feedback on changes between different design iterations. The pointing-and-tracking system that mediates the alignment between the transmitter and receiver was designed primarily by Andrew Conrad, along with contributions from Daniel Sanchez-Rosales and Timur Javid. In this chapter, “we” refers to the full-time drone team (consisting of myself, Daniel, Andrew, and Samantha) in relation to our identification and evaluation of optical components used on the transmitter and receiver optical benches.

3.1 Overview

After the optical pulses are generated by the LEDs, coupled into single-mode fiber, and attenuated to the desired mean photon number, they are routed to the state preparation optics bench. Here, the fibers corresponding to the three different polarization states are connected to their respective collimators. These three separate optical paths allow the encoding of the appropriate polarization state for the different pulses using passive free-space optical

components. The transmitter bench contains the optics necessary to encode polarizations as well as the beam splitters used to recombine the three beams into the same spatial mode so they can all be coupled through the spatial filter fiber.

As shown in Figure 3.1, the left- and right-circular polarizations are prepared by first passing them through linear polarizers (LP) that set them to be vertical and horizontal, respectively. From there, the two beams are combined at the polarizing beam splitter (PBS), which efficiently combines the two beams because horizontal light is transmitted and vertical light is reflected. After the PBS, these two beams pass through a quarter wave plate (QWP) with the fast axis rotated 45 degrees with respect to the horizontal axis. This transforms the vertical polarization into left-circular and the horizontal into right-circular. Meanwhile, the horizontal polarization is created with a linear polarizer. The horizontal path is combined with the left- and right-circular path at the 50/50 non-polarizing beam splitter (NPBS).

With the three optical paths now combined, the full path is directed into a variable focus collimator, which couples the beam into the single-mode spatial filter fiber (see Fig. 3.1). The variable focus allows improved mode matching and thus improves the coupling efficiency. There is also a variable focus collimator when exiting the spatial filter fiber, which provides control over the outgoing beam size. The spatial filter fiber birefringence induces a unitary polarization transformation that impacts the state fidelity, so after exiting the fiber, the beam passes through a quarter waveplate, a half waveplate, and a second quarter waveplate, as shown in Figure 3.1. These three compensating waveplates can produce an arbitrary unitary transformation, which is used to counteract the birefringence of the spatial filter fiber, as well as the polarization transformations caused by any other optics in the system.

After the compensating waveplates, the beam passes through the narrowband spectral filter (Andover 656FS02-12.5). From there, the beam transmits through two different dichroic

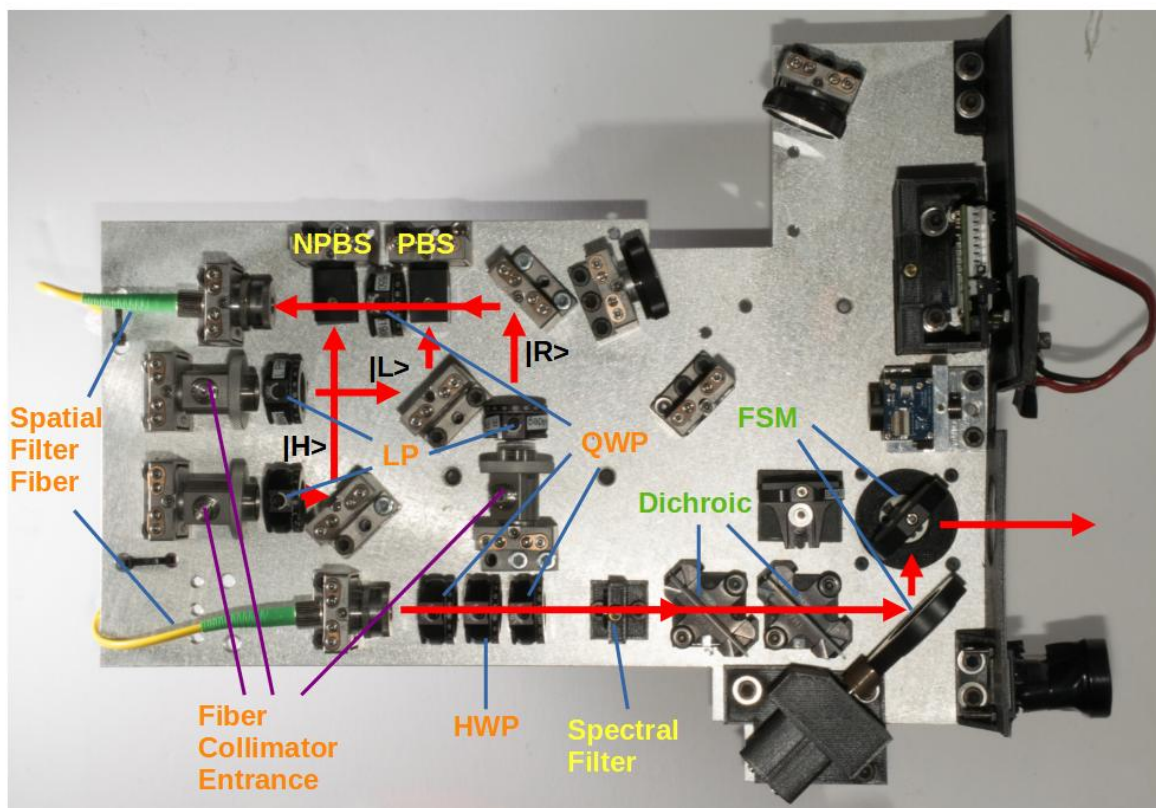


Figure 3.1: Photograph of the transmitter optics bench. State preparation optics along the three optical paths encode the R, L, and H polarization states. Beam splitters combine these three paths before they enter the spatial filter fiber. After exiting the fiber, the combined beam passes through the polarization compensation waveplates, the spectral filter, and then the dichroic mirrors that split off the pointing-and-tracking beacons. The fast steering mirrors on the right side of the bench perform fine control on the attitude of the beam exiting the optics bench. Photo credit: Timur Javid.

mirrors, IDEX FL-007100 and Thorlabs DMLP567 that are used to redirect the alignment beacons, and finally reflects off the two fast-steering mirrors (FSM) that provide the fine control for the pointing-and-tracking system before travelling through free-space to the receiver drone.

The receiver optics bench (shown in Fig. 3.2) is largely symmetrical to its transmitter counterpart, with a few important differences. Upon entering the receiver bench, the beams reflect off two fast steering mirrors before passing through the two dichroic mirrors. The state measurement optics are enclosed by a 3D printed box used to reduce background light, with an aperture blocked by a narrowband spectral filter (Andover 656FS02-12.5) and an additional longpass filter (Thorlabs FELH0650) that ensure that both background light and light from the pointing-and-tracking beacons do not reach the state measurement apparatus. The quantum signal enters the state measurement optics through these filters.

The measurement optics begin with a 50/50 NPBS, which passively chooses an optical path that defines the measurement basis. The light that reflects is sent directly to a PBS, which allows horizontal states to transmit, vertical states to reflect, while circular states are split evenly, thus measuring in the H/V basis. The other optical path from the NPBS passes through a quarter waveplate with the fast axis 135 degrees from the horizontal before reaching a PBS. In conjunction, these two components cause the right-circular states to transmit, the left circular to reflect, while the linear states are split evenly, thus measuring in the R/L basis.

After the polarization states are sorted, the four resulting optical paths are sent into 1 inch diameter, 0.25 numerical aperture collimators (F810FC-635), which collect the light into multi-mode fibers that carry the light to the four-channel single-photon avalanche diode (Perkin-Elmer SPCM-AQ4C). We choose 200 μm core multi-mode fibers and 1 inch optics for the receiver so that the alignment between the two drones is less sensitive and higher

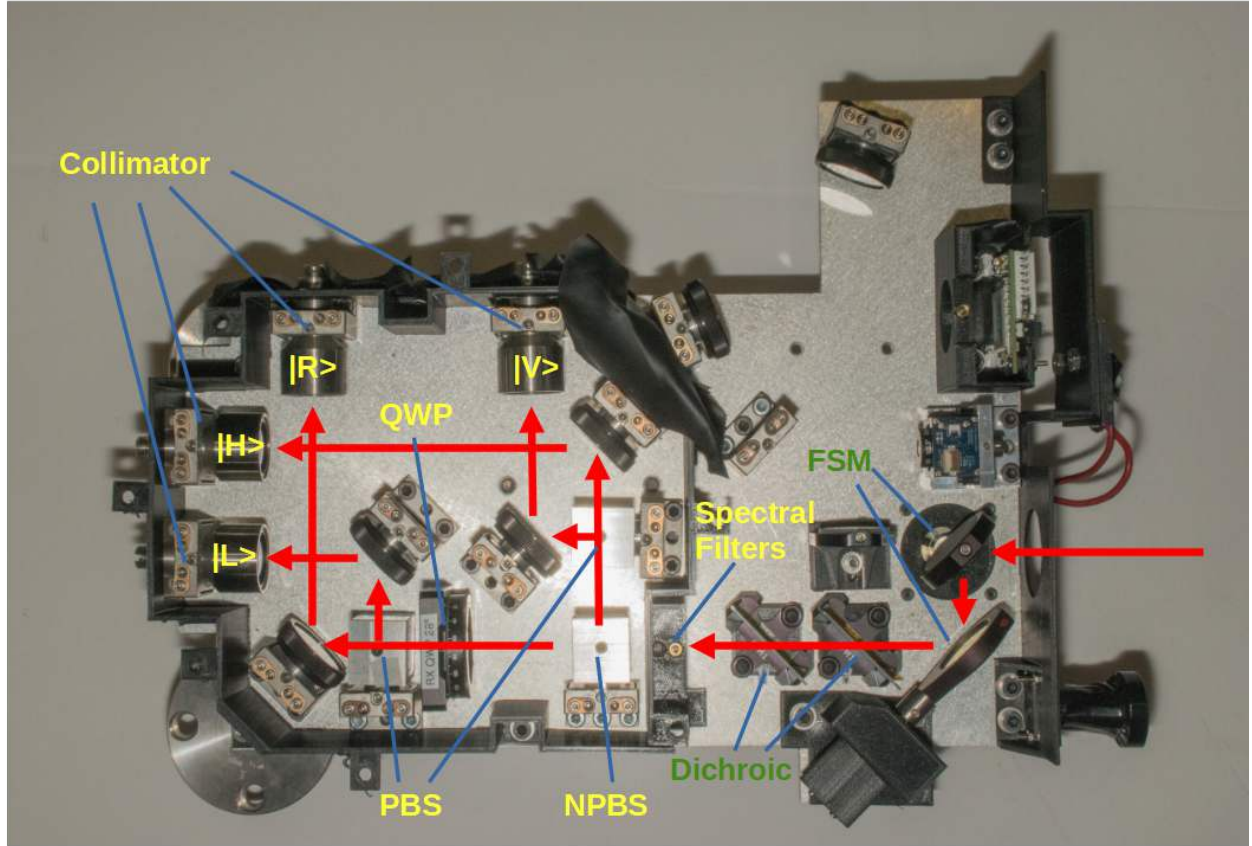


Figure 3.2: Photograph of the receiver optics bench. The beam enters through the aperture on the right side, where it reflects off the fast steering mirrors that perform fine control on the attitude of the entering beam. Subsequently, the beam passes through the dichroic mirrors that split off the pointing-and-tracking beacons. The beam transmits through spectral filters to enter the 3D printed box that contains state measurement optics. The NPBS chooses the measurement basis, and the PBS and quarter waveplate sort the polarizations to make the desired measurements in the R/L and H/V bases. Collimators collect the pulses so they can be conducted to the four-channel single-photon detector via multi-mode fibers. Photo credit: Timur Javid.

signal throughput can be achieved. The four-channel single-photon detector registers the photons and sends the electrical pulses to the receiver FPGA for time-tagging.

3.2 Non-Ideal Polarization Effects

Creating high-fidelity polarization states with free-space optical components depends on a number of factors including the quality of the components themselves and their orientation and alignment. Lightweight commercial off-the-shelf components are preferable and help mitigate SWaP concerns; however, this means that the system must contend with non-ideal properties in some of the components. For example, we choose to use 1/2 inch optics on the transmitter bench that are relatively lightweight and easily purchased, including the Thorlabs RSP05 optics rotation mounts that are both lightweight and form factor. However, the small size that makes these mounts ideal for this application also makes it naturally harder to set their orientation as accurately as on a larger rotation mount. Additionally, when setting the orientation of these particular mounts, they can sometimes rotate slightly while the set screw is being tightened, increasing the inaccuracy of the angle of the optic. These orientation errors reduce the overall state fidelity.

Another aspect of these non-ideal components revolves around the availability of components with certain design wavelengths. The LED spectra are mostly strongly peaked at around 653 nm; however, the best option for the spectral filter (balancing factors such as spectral width, wavelength, transmission, size, and cost) is at 656 nm, resulting in slightly lower spectral brightness in the ultimately produced pulse. Finding polarization optics at 656 nm is also a challenging task. The most common options for waveplates near this wavelength are designed for 670 nm or 633 nm light. For the state preparation and measurement quarter waveplates, whose properties the state fidelity is especially sensitive to, we identify

options at a 650 nm design wavelength. This is the best option available without resorting to expensive custom-designed waveplates. Using a waveplate at a different wavelength than the design wavelength changes the effective retardance of the waveplate. In this case this means that instead of a retardance of 0.25 (that of a true quarter waveplate), the retardance using the 650 nm waveplate becomes 0.248, which only has a minor impact on the state fidelity, and in fact is within specified retardation tolerance for the part at the design wavelength. A further complication in deploying quarter waveplates is their sensitivity to the angle of incidence (AOI) of the beam. Small changes in the AOI can cause dramatic shifts in the effective retardance of the waveplate. When installing the waveplate rotation mounts in the setup, I attempt to minimize the AOI, but this process is prone to small errors that may change the effective retardance noticeably.

The properties of the non-polarizing beam splitters used to combine optical paths on the transmitter and to choose the basis on the receiver can noticeably impact the state fidelity and measurement fidelity. An idealized version of this component would split the intensities of all incoming beams 50/50, regardless of the polarization. Even if a beam splitter were to divide non-polarized light at a perfect 50/50 ratio, a small polarization dependence to that splitting would greatly affect the fidelities. We identify an NPBS (Optosigma NPCH-10-6700) with specifications of approximately 56.7/43.3 transmission vs. reflection for the s polarization and 53.4/46.6 for the p polarization. When the s and p transmissions and reflections are different, it causes the circular states to become slightly elliptical along one axis, and does not preserve their orthogonality. This reduces state preparation and measurement fidelity for these states. Additionally, it can introduce differences in the basis measurement probability between the different input states, which can reduce the effectiveness of the protocol [27].

Another complication with both the NPBS and the PBS components is that their

reflection and transmission properties for different polarization are affected by the angle of incidence of the incoming beam. Samantha places all the optical components in the bench design so that the beams are normal to the beam splitters when all the alignment mounts are at their neutral position. However, the system already requires a sensitive alignment into the single-mode spatial filter fiber, and adjusting that alignment can cause small changes in the beam splitter sorting, in turn lowering the fidelity. Additionally, on the receiver bench, the angle of the incoming beam can be variable depending on the performance of the pointing-and-tracking system. This effect is limited by the spatial filtering defined by the multi-mode receiver fibers, but within those parameters there are still small perturbations to the beam splitter sorting. These perturbations also affect the AOI for the receiver quarter waveplate, and thus cause instability in the effective retardance and in turn the measurement fidelity.

As outlined above, multiple components in this setup have known non-ideal behaviors not easily controllable and in the context of these complications, achieving high state preparation and measurement fidelity hinges on achieving a precise orientation of the polarizers and waveplates. Small errors in these alignments can reduce the fidelity, but the magnitude of the reduction depends both on the size of the error and on which component is misaligned. For example, I use the linear polarizers (Thorlabs LPNIRE050-B), which are broadband and have extinction ratios of $\approx 2500:1$ at the wavelength of the quantum signal, to set the initial horizontal and vertical polarizations. Figure 3.3 shows how these linear polarization fidelities vary as a function of the polarizer orientation error. In order to achieve extinction ratios of $>50:1$, the polarizer must be correctly oriented to within 5.75 degrees of the correct angle.

The procedure for setting the linear polarizers is relatively simple because the PBS reliably transmits high fidelity horizontal light ($>1000:1$ extinction ratio, Thorlabs PBS122). It is easy to set up a laser passing first through the PBS, then through the linear polarizer

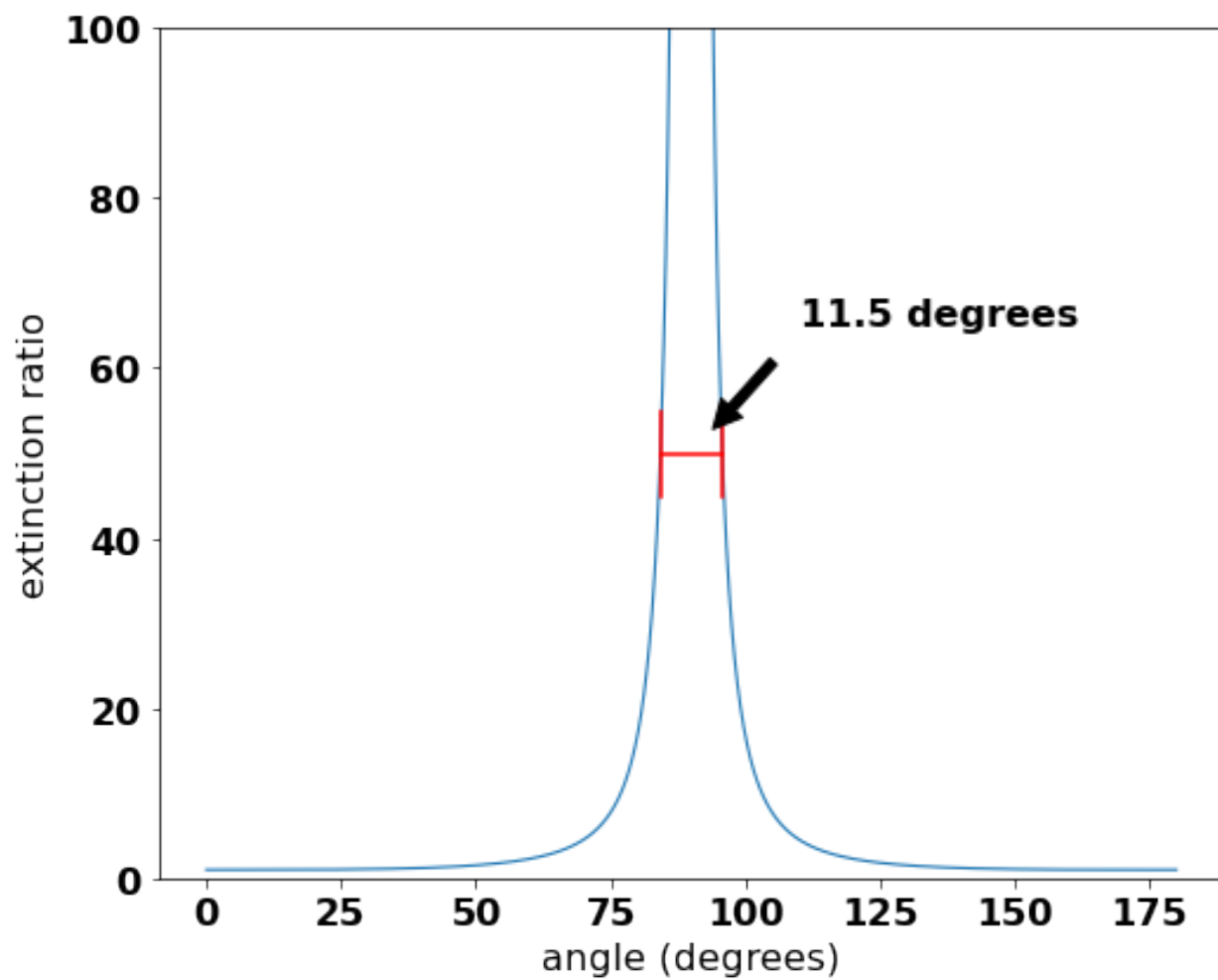


Figure 3.3: Extinction ratio for the vertical polarization produced by a linear polarizer. The maximum extinction ratio is produced when the polarizer is oriented at 90 degrees from the horizontal, and extinction ratios of $>50:1$ can be found over a range of 11.5 degrees.

being tuned, and then intercept the beam with power meter. By rotating the linear polarizer until the power incident on the power meter is minimized, the polarizer is set to vertical. Because minimizing the power is more precise than maximizing the power, the horizontal polarizer can be set by simply passing the laser through the vertical polarizer, the unset polarizer, intercepting with a power meter, and minimizing the power.

The orientation of the quarter waveplates has the potential to be more sensitive than that of the linear polarizers. In Figure 3.4 I model the state fidelity as a function of the transmitter quarter waveplate orientation (assuming the receiver quarter waveplate is at the ideal angle), but include some of the previously mentioned non-ideal effects, such as the wavelength dependence of the quarter waveplate retardance over the source's spectrum, the imperfect sorting of the NPBS at the quantum signal wavelength, and a non-zero AOI for the beam with respect to the quarter waveplate. It is clear to see that the sensitivity of the QWP orientation depends highly on these non-ideal factors. Even a 2 degree difference in the QWP AOI severely reduces the maximum achievable extinction ratio, and reduces the range over which a $>50:1$ extinction ratio can be achieved from 17.2 degrees to 13.2 degrees, and a 5 degree AOI renders this unreachable. This simulation neglects effects such as poor sorting at the beam splitters due to nonzero AOI, possible orientation error and nonzero AOI on the receiver QWP, in addition to any issues caused by the spatial filter fiber, compensating waveplates, and dichroic mirrors. With all these factors considered, the QWP orientation can be highly sensitive, but proper alignment of the surrounding components makes the problem more tractable.

The procedure for setting the QWP orientation is more complicated than that of the linear polarizer. Again, I use the transmitting port of the PBS to produce a reliable horizontal polarization. I send this horizontal polarization through an NPBS, which will allow me to pick off a beam for minimization at the end. From there, the transmitted beam passes

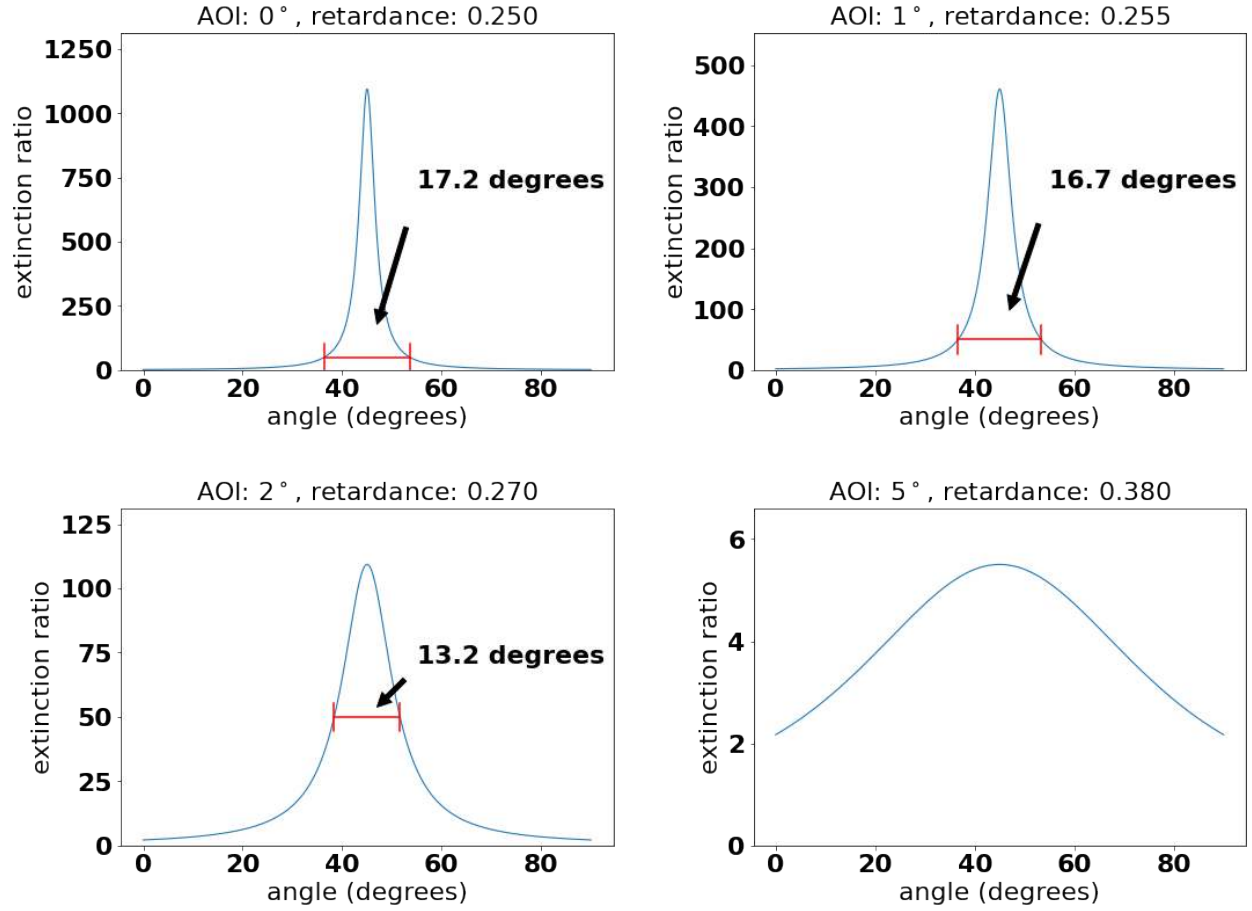


Figure 3.4: Extinction ratio for the left-circular polarization as a function of the quarter waveplate orientation, shown for different angles of incidence of the beam. The maximum extinction ratio is always produced when the waveplate is oriented at 45 degrees from the horizontal, but the maximum extinction ratio and the range over which a $>50:1$ extinction ratio can be found vary with the angle of incidence.

through the QWP under calibration, which, when properly oriented with the fast axis at 45 degrees with respect to the horizontal, produces right-circular polarization. This beam reflects off a mirror at normal incidence, causing the polarization to change to left-circular, before passing back through the QWP, which from the opposite direction has the fast axis at 135 degrees with respect to the horizontal, transforming the beam to vertical polarization. The beam arrives back at the NPBS, and I pick the reflected beam, which I then pass through another PBS. The light that transmits through this PBS is minimized when the QWP under calibration is optimized at 45 degrees.

3.3 Polarization Fidelity Compensation

As previously mentioned, the transmitter employs a 30-cm-long single-mode spatial filter fiber to address indistinguishability concerns. This fiber causes a unitary transformation to be applied to the polarization states, which left unaddressed would severely impact the state fidelity. The dichroic mirrors used in the pointing-and-tracking system also appear to induce a unitary transformation. Fortunately, a unitary transformation can be undone by the proper application of compensating waveplates. In order to construct an arbitrary unitary transformation, three waveplates are necessary to supply the necessary degrees-of-freedom. The typical waveplate combination used is a quarter waveplate, followed by a half waveplate (HWP), followed by a second quarter waveplate. The capability of this configuration to achieve any polarization transformation is easy to see by construction: the first QWP removes any ellipticity from the incoming polarization, the HWP rotates the resulting linear polarization to the desired axis, and the second QWP adds back the desired amount of ellipticity for the final state.

However, the QWP-HWP-QWP combination, while straightforward, is not the only wave-

plate configuration that can achieve any arbitrary unitary transformation. Due to easy access, I choose compensating waveplates with retardances at the quantum signal wavelength of 0.256, 0.531, and 0.256. In order to demonstrate that these waveplates can reverse the unitary transformations applied to the polarization states, I simulate 200,000 random orientations for the three waveplates for the three different polarization states, R, L, and H. For comparison I also compute the same simulation for the standard QWP-HWP-QWP case. I plot the resulting states on the Bloch sphere, which I divide into two hemispheres based on the H projection measurement (see Figures 3.5 and 3.6). There are no empty regions, which demonstrates that the three polarization states can map onto any arbitrary polarization state and vice versa. Furthermore, the density of points in a given region of the Bloch sphere gives insight into the relative sensitivity required in aligning the waveplates to produce states found in that region. In essence, a region with a low density of points in this plot would have fewer waveplate angle combinations that achieve states in that region, meaning that the waveplate alignment to correct a state in that region would need to be more precise. The plots for the non-standard waveplates do not feature any noticeably low-density regions compared to those for the standard waveplates, indicating that the polarization compensation has not been made more difficult by these waveplate choices. While this falls short of demonstrating that the non-standard waveplate combination can map between any two arbitrary polarization states, which would show the ability of the waveplates to perform any arbitrary unitary transformation, it does show that they can recover the original polarization states with high fidelity, which is sufficient for the purposes of this experiment.

Due to the non-standard waveplate values, I adjust the procedure for tuning the waveplates and recovering high-fidelity polarization states. My procedure consists of iterative adjustment of the waveplates while alternately maximizing extinction ratios for the R/L and H/V bases. I start by adjusting the first two waveplates (0.256-0.531) while sending

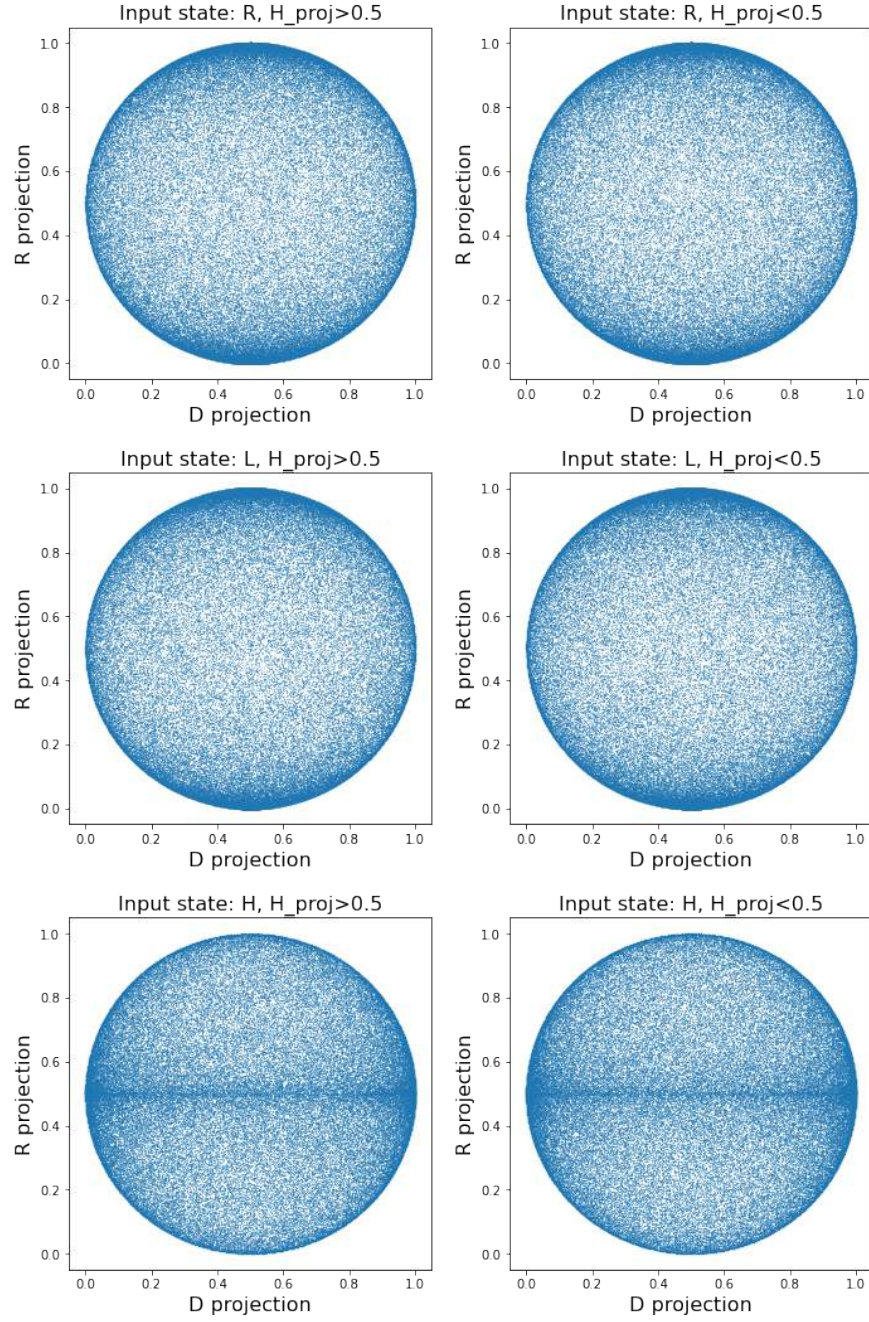


Figure 3.5: Scatter plot of the head-on view for each hemisphere of the Bloch sphere (divided based on the H projection measurement) for 200,000 random waveplate orientations applied to the different input polarizations (R, L, and H). The waveplates are the standard QWP-HWP-QWP configuration.

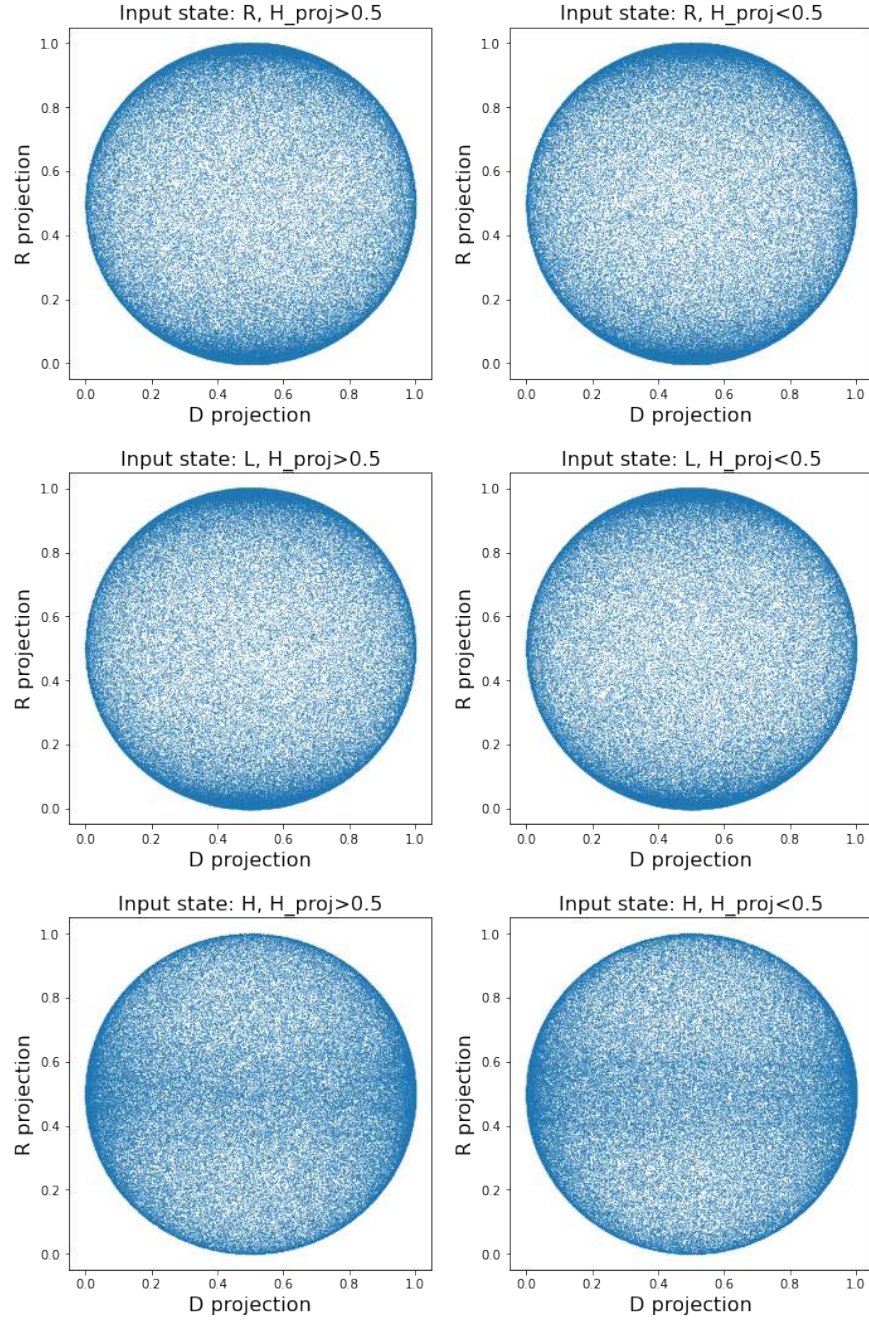


Figure 3.6: Scatter plot of the head-on view for each hemisphere of the Bloch sphere (divided based on the H projection measurement) for 200,000 random waveplate orientations applied to the different input polarizations (R, L, and H). The waveplates are the non-standard 0.256-0.531-0.256 retardances configuration.

in the R or L states and maximize the extinction ratio in the R/L basis with these two degrees-of-freedom. Next, I send in the H state and adjust the third waveplate (0.256) until the extinction ratio in the H/V basis is maximized. I repeat these steps until high fidelity is achieved in both bases simultaneously.

3.4 Alignment Considerations

One of the most technically difficult aspects of the optics benches is the spatial beam alignment. This alignment can be divided into three separate systems: the internal alignment of the transmitter, the internal alignment of the receiver, and the pointing-and-tracking alignment between the transmitter and the receiver.

The most sensitive of these spatial alignments is the internal alignment on the transmitter, due to the presence of the single-mode spatial filter fiber. The spatial filter fiber guarantees indistinguishability in the spatial degree-of-freedom (as shown in Section 2.6), but the alignment must be extremely precise to produce good coupling. Poor alignment leads to low mean photon numbers and thus low overall throughput for the system. The mean photon numbers for the three states need to be balanced for this protocol, which means that the mean photon number is limited by the lowest achievable coupling. So this system requires high coupling efficiencies in all three channels simultaneously. Another complicating factor is the performance of the polarization optics at non-zero angles of incidence, as discussed in Section 3.2. Given the known sensitivity of the polarization properties of the QWP and NPBS to angle of incidence, I am limited in what degrees-of-freedom I can employ to perform the alignment. The tip tilt mount that houses the fiber collimator (see Fig. 3.7) would seem to be an ideal degree-of-freedom for the alignment because it can control the beam path angle without affecting the offset, making the walking the beam procedure very

straightforward; however, it cannot be used as such because the filter fiber collimator angle defines the optical path through the polarization optics, and I must instead use this degree-of-freedom to enforce as small an angle of incidence as possible. Additionally, even if the filter fiber collimator could have been used as a degree-of-freedom, it only would have been able to be used to align one of the three polarization paths, because it would not be able to be adjusted subsequently. By a similar token, the NPBS and PBS would be useful degrees-of-freedom for the alignment because of their proximity to the spatial filter collimator, but they must be reserved for ensuring normal incidence to optimize polarization fidelity.

This leaves me with only the entrance collimator for each polarization and the accompanying mirror to use as degrees-of-freedom for the alignment (indicated by the green circles in Fig. 3.7). Because they are quite close to each other relative to the subsequent path length to the filter fiber collimator, walking the beam to perform the alignment with these degrees-of-freedom is more challenging and can take significantly more adjustment before the correct spatial mode is achieved. In some cases when the other optics are not close enough to their neutral angular position, these degrees-of-freedom can provide inadequate range to successfully align all of the beams into the filter fiber. On an optical table setup, when degrees-of-freedom provide insufficient range, I can often choose a different hole in the breadboard and move a mirror in order to change the alignment range without readjusting the other optics. However, on the optics bench, with designated holes for installing each component, all of which are already tightly arranged, such adjustments prove to be impractical. For these reasons it is important in this setup to ensure that all other optics are roughly aligned and set close to neutral angular positions. This ensures that the alignment degrees-of-freedom have sufficient range to perform the task, and helps avoid situations where difficult alignments must be restarted from scratch after encountering the edge of the alignment range.

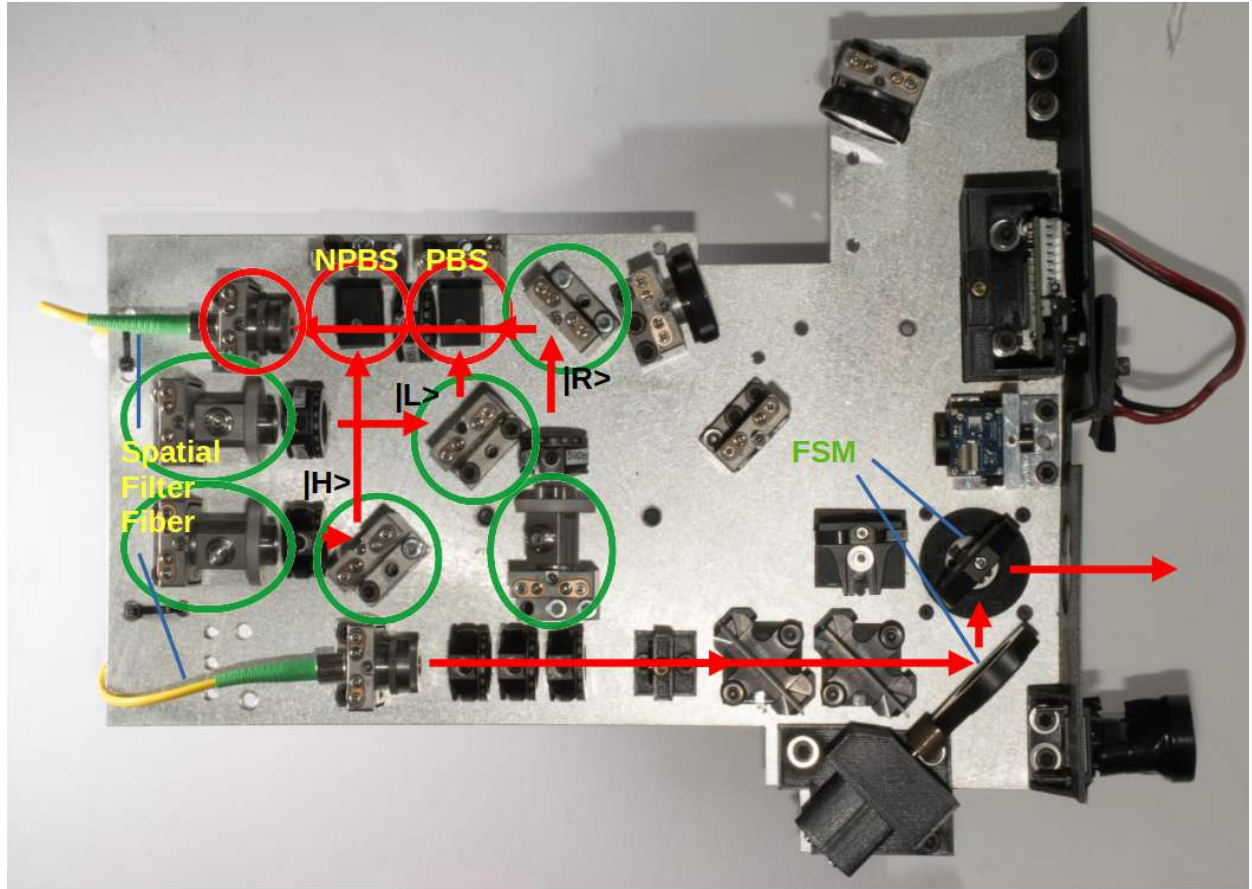


Figure 3.7: Photograph indicating the potential alignment degrees-of-freedom on the transmitter bench. The green circles indicate viable alignment degrees-of-freedom, while the red circles indicate alignment degrees-of-freedom that should be avoided due to polarization fidelity considerations. Photo credit: Timur Javid.

The pointing-and-tracking system is a dynamic architecture that mediates the alignment between the transmitter and receiver benches. It uses a combination of an inner and an outer loop. The outer loop controls the course alignment by reorienting the gimbals that hold the optical payloads. The tracking is informed by an IR camera that images the IR beacon found on the opposite drone and rotates the gimbal to keep the IR beacon centered in the image. Simultaneously, the inner loop control fine alignment using sets of two piezoelectric fast steering mirrors that control the attitude of the beam. These are informed by position sensitive diodes (PSDs) that track the power and location of the two counter-propagating alignment beacons. These systems in tandem successfully mediate locking between the two drones in flight and in other configurations, with performance routinely exceeding 60% average coupling into the multi-mode receiver fibers.

The internal alignment of the receiver is less sensitive than that of the transmitter because the receiver fibers are multi-mode, but ensuring spatial mode overlap is still important. The fact that the incoming quantum signal is single-mode and the receiver fibers are multi-mode means that there is a wider range of possible angles that can achieve high coupling. However, if the mode profiles of the four receiver fibers are not well overlapped, there may or may not be alignments between the transmitter and receiver that produce high coupling in all channels simultaneously, but there will certainly be alignments that produce high coupling in some channels and low coupling in others. Not only does this interfere with the throughput, as there are often times when the drone locking fluctuates and the receiver loses counts in particular channel, but it exposes the protocol to Eavesdropper attacks where she can take advantage of the relative detection probabilities depending on the alignment that she can manipulate [28].

The most straightforward method for ensuring the modal overlap of the multi-mode collection fibers is through back-propagation of the receiver bench. I can send laser light

backwards through the multi-mode receiver fibers and observe the resulting beam profile that exits the bench at various distances. As I align the receiver, I can ensure that these back-propagated beams are well-aligned, which is the preferred figure-of-merit for the internal receiver alignments. Similar to the transmitter, the degrees-of-freedom that would seem easiest to use, such as the NPBS and the two PBS (see Fig. 3.8), should not be used because of the relation between AOI and polarization measurement fidelity. Instead, I use the tip-tilt mounts on the receiver collimators and the mirrors that accompany them. It is important to make sure the counter-propagated beams exit the receiver bench close to the designed beam axis. This ensures that the waveplate and beam splitters encounter the beam paths at close to zero AOI for polarization purposes, and also eases the calibration of the pointing-and-tracking system, which would otherwise have to correct for such an off-axis beam path.

3.5 Noise Filtering

Given the quantum nature of the system, signal-to-noise ratio (SNR) is a significant concern. The mean photon numbers are on the order of unity, which means stray light is often of comparable or greater intensity and can contribute significantly to noise in the data. Stray light incident on the detectors contributes random events to the raw key, which can become quantum bit errors if the transmitted signal is lost and a stray photon is registered at the wrong detector. In an extremely high noise environment, the QBERs can approach 50%, which is far beyond the error threshold to exchange secure key in this protocol. This experiment avoids the vast majority of environmental noise by operating the system at night, but it is still subject to some ambient light. Additionally, there is unwanted light from the pointing-and-tracking alignment beacons due to scattering and the small fraction of the

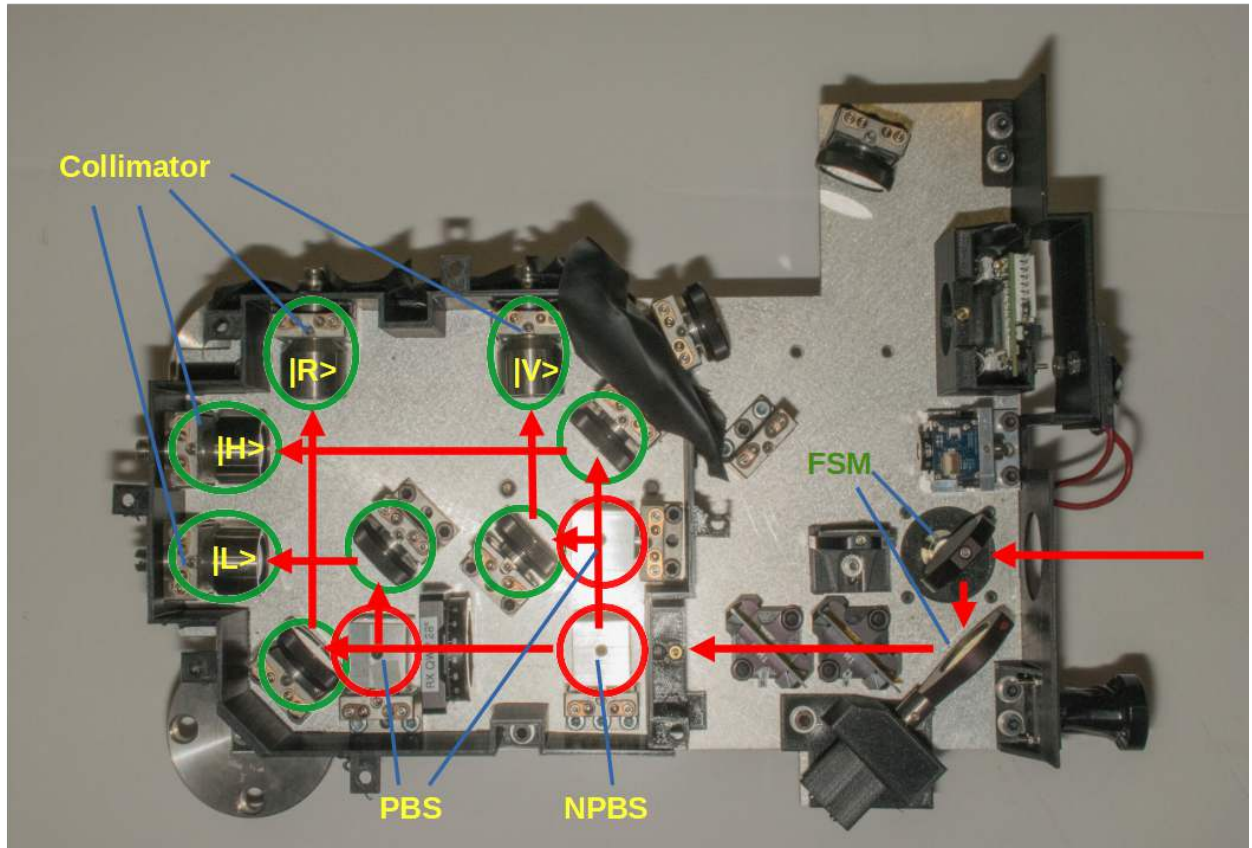


Figure 3.8: Photograph indicating the potential alignment degrees-of-freedom on the receiver bench. The green circles indicate viable alignment degrees-of-freedom, while the red circles indicate alignment degrees-of-freedom that should be avoided due to measurement fidelity considerations. Photo credit: Timur Javid.

beacon power that transmits through both dichroic mirrors.

A significant fraction of environmental noise is rejected by the acceptance angle defined by the numerical aperture of the multi-mode fibers. Said fibers are black jacketed, so the only significant light that reaches the detectors is light that couples in through those collimators. While this effectively reduces environmental noise (which is largely isotropic), it is much less effective at reducing noise from the beacons, which are aligned in the same optical path as the quantum signal beam at 520 nm and 705 nm. Additionally, we place a black, 3D printed box around the polarization measurement optics, further reducing the contributions from ambient light. The entrance to the black box is an aperture occupied by two spectral filters: the narrowband filter matching that on the transmitter bench (Andover 656FS02-12.5) and a longpass filter (Thorlabs FELH0650). The narrowband spectral filter brings down both the ambient and beacon noise, but because of the high beacon power, the receiver needs additional filters to reduce the residual beacon noise to a manageable level. The narrowband spectral filter allows more transmission of the 705 nm beacon light, so we configure the system so that the 520 nm beacon is mounted on the transmitter, thus avoiding a situation where the 705 nm beacon is incident directly on the measurement setup. The receiver also employs a high-transmission long pass filter at 650 nm, which allows the further reduction of the beacon noise without noticeably impacting the transmission of the quantum channel. Other methods for reducing noise using post-processing techniques are discussed in Chapter 5.

Chapter 4

Field Demonstrations

Thus far I have introduced the subsystems of a QKD system with high fidelity and indistinguishable states designed to operate in low SWaP conditions; however, a key obstacle in these experiments has been dealing with environmental factors associated with outdoor flight testing, such as non-ideal or variable temperature, humidity, wind, ambient light, and vibration. This chapter details the experimental efforts our team has undertaken to maintain this high level of performance in these non-ideal outdoor environments. In order to highlight and explore how the experiment must be adapted from a tabletop laboratory setting to an outdoor one, I compare and contrast the procedures in the two different settings, highlighting the mini-experiments that must be performed on testing days to prepare for full QKD data collection, why they are performed in a specific order, and the solutions engineered to make these demonstrations possible. To conclude, I detail the procedures for the full data collection in the various mobile QKD configurations.

My contributions in this chapter include developing and compiling the overarching procedures for tabletop and field testing. I developed the procedures for the LED-to-fiber alignment, the internal transmitter and receiver alignments, the mean photon number balancing, and the polarization fidelity compensation (for which I wrote a data acquisition GUI in

Python that significantly improved the adjustment procedure). I worked closely with Daniel Sanchez-Rosales on the procedure for the full optical temporal compensation. Daniel wrote the FPGA code that allowed the adjustment of the electrical signals and adapted the code to also output the necessary reference signals. I wrote the data acquisition code in Python, developed the metrics that informed which adjustments to make, and wrote a rudimentary automation of the process that interfaced with Quartus using an auto clicker package. Daniel adapted my data acquisition code to measure and save the electronic waveforms and wrote a better interface between Quartus and the data acquisition in Python.

On a typical night of field testing, I performed the LED-to-fiber alignment, the internal transmitter alignment (multiple times), the optical temporal pulse adjustment (or Daniel would perform the electronic temporal pulse adjustment), the mean photon number balancing, and the polarization fidelity compensation. Typically, Daniel and Samantha Isaac performed the internal receiver alignment, and Daniel and Andrew Conrad collaborated on the pointing-and-tracking tuning. During QKD data collection, Samantha and Andrew piloted the drones and/or drove the cars, Daniel handled the FPGA data acquisition, while I activated the control loops and monitored the flight status and informed the pilots when power was running low.

4.1 Procedures Overview

Tabletop QKD procedures:

- Mount and align transmitter and receiver optical benches on the tabletop with two mirrors in the optical path between to align the benches
 - Internally align transmitter with a laser so the three optical paths couple into the

- single-mode spatial filter fiber
- Internally align receiver with back-propagated lasers through the multi-mode receiver fibers so the four beams are well overlapped
- Use the intermediate mirrors to maneuver the transmitter output into the receiver with good coupling with an alignment laser into all four receiver fibers
- Optimize LED-to-fiber coupling with LEDs in CW mode
- Program transmitter FPGA and perform temporal adjustments to guarantee indistinguishability of quantum states
- Use the in-fiber attenuators to balance the mean photon numbers of the three quantum states
- Perform polarization fidelity compensation on quantum states
- Program the receiver FPGA and proceed to collect QKD data

Meanwhile the procedures for collecting outdoor data have additional steps related to tuning the pointing-and-tracking system, but it is also marked by extra repetitions of some tuning steps from the tabletop procedure as well as a specific ordering of the steps to ensure the most sensitive tunings are performed last. Additionally, some methods and procedures that are common to both tabletop and outdoor QKD procedures were developed out of necessity because old strategies were untenable in the outdoor environment.

Outdoor QKD procedures:

- Optimize LED-to-fiber coupling with LEDs in CW mode
- Internally align transmitter bench with a laser so the three optical paths couple into the single-mode spatial filter fiber
- Program transmitter FPGA and perform temporal adjustments to guarantee indistin-

guishability of quantum states

- Use the in-fiber attenuators to balance the mean photon numbers of the three states
- Internally align receiver with back-propagation through the multi-mode receiver fibers so the four beams are well overlapped
- Set up the drones between two tables at the desired locking distance and tune pointing-and-tracking parameters until high coupling is achieved between the two drones
- Re-tune internal transmitter alignment with a laser
- Perform polarization fidelity compensation on quantum states
- Program receiver FPGA and move both drones to flight position
- Re-tune transmitter alignment with a laser
- Fly drones and collect QKD data

Both QKD experimental procedures can be largely divided up into a series of smaller experiments: **LED-to-fiber alignment, internal transmitter alignment, temporal pulse adjustment, mean photon number balancing, internal receiver alignment, transmitter-to-receiver alignment, and polarization fidelity compensation.** The order of these mini-experiments in the tabletop setup is mostly interchangeable and the order in this procedure is determined by convenience. In contrast, for the outdoor setup, the order of these mini-experiments is more driven by necessity, as some alignments and tunings are not as robust to temperature and should not be performed too long before the data collection. In addition, some experiments require an Eavesdropper, which is set up on the portable optical breadboard, while some are best performed with the drones in the ground-to-ground configuration.

4.2 Mini-Experiments

4.2.1 LED-to-fiber Alignment

Working from the beginning of the field testing procedure, the first step is to optimize the LED-to-fiber coupling. Because the updated machined aluminum coupling assembly design is relatively robust and temperature insensitive (see Section 2.3), its tuning is fairly stable and can be trusted to remain constant during standard operation and handling. Furthermore, the LED-to-fiber coupling does not need to be performed with an Eavesdropper or ground-to-ground configuration, does not have any prerequisite alignments, and is in fact easiest to perform with the LED PCB removed from the gimbal and optimized at a table. Lastly, the LED-to-fiber package must be aligned in order to perform the later procedures that involve the quantum system: the temporal pulse adjustment, the mean photon number balancing, and the polarization fidelity compensation. For these reasons, the LED-to-fiber alignment is performed first when taking outdoor data.

4.2.2 Internal Transmitter Alignment

One of the most obvious differences between these two procedures is the repetition of the transmitter alignment in the transmitter in the outdoor QKD procedures. The alignment of the three states into the single-mode spatial filter fiber is highly sensitive in the best of circumstances, but once achieved in a laboratory setting, it remains stable. However, in the face of constantly changing temperatures in an outdoor environment, as well as the vibrations and shocks associated with moving the apparatus from location to location and flying and landing the drone, frequent realignment is necessary. The alignment of the transmitter impacts both the overall signal throughput and the relative balance of the mean photon

numbers of the quantum states, an imbalance of which impacts the secure key rate calculation [21]. Because the internal transmitter alignment is arguably the most sensitive aspect of the system, it follows that it requires realignment it just before taking data. Several of the intermediate tuning steps also require good coupling on the transmitter, which is why I re-tune the transmitter alignment several times throughout the procedure.

The procedure for performing the internal transmitter alignment is fundamentally the same in the field as it is in the lab (see Section 3.4). I send an alignment laser (housed on the gimbal tray) through the transmitter bench and use a free-space power meter placed after the spatial filter fiber to optimize the coupling.

4.2.3 Temporal Pulse Adjustment

The temporal pulse adjustment experiment required good internal transmitter coupling and LED-to-fiber coupling so that the 0.1 mean photon number states can be generated by the transmitter, which means this experiment must be performed after the internal transmitter alignment and the LED-to-fiber alignment. Furthermore, the width of the temporal pulses is a determining factor in the mean photon number. Because the width of the temporal pulses is significantly decreased during this procedure (thus lowering the mean photon number of said pulses), it is important to perform it before the mean photon number balancing experiment. Additionally, the temporal pulse adjustments have been shown to remain stable over time and temperature fluctuations, which argues strongly that it should be tuned earlier than some of the more volatile systems in the procedure.

As described in Section 2.5, I use an Eavesdropper intercepting the beam exiting the transmitter bench to measure the temporal profile of each state and use electronic adjustments on the transmitter FPGA to perform the temporal pulse adjustment. When con-

ducting optical measurements, each signal state temporal profile collection is 300 s and each decoy state temporal profile collection is 1200 s. Because temporal profiles must be collected between each adjustment, these times are multiplied and the total time for making full adjustments to maximize overlap for all states typically exceeds five hours. Once the adjustment is complete, the achieved temporal overlap remains viable as long as the transmitter FPGA remains powered on.

The field version of the temporal pulse adjustment experiment is performed with the transmitter gimbal on the ground above the portable optical breadboard with optics posts to brace the bench so the gimbal cannot rotate. I install a mirror in the path of the beam exiting the transmitter and a collimator connected to a single-mode fiber. I send the alignment laser through the transmitter bench and maximize the coupling into this fiber using the mirror and collimator mounts as degrees-of-freedom. I subsequently connect the single-photon detector to serve as the Eavesdropper intercepting the states for measurement of the temporal profile. This experiment must be performed in a dark environment because stray light coupled into the Eavesdropper overwhelms the quantum signal being adjusted. The necessity to maintain power to the FPGA precludes performing this experiment in a laboratory in advance of field testing, so a dark environment at the testing site is necessary.

While the simplest strategy would be to wait until after sunset, waiting five hours after sunset before beginning the rest of the procedure would be nearly prohibitive to collecting any QKD data before sunrise. Instead, our team received access to a basement room in the Seed House building on the agricultural campus at the University of Illinois (until better solutions were developed). While this situation was not an ideal scientific environment (with high humidity and significant dirt and dust), it was dark enough to conduct the temporal pulse adjustment during the day, reserving the use of the time after dark for the rest of the procedures and for data collection. I was able to automate the iterative adjustments to some

extent by interfacing the data acquisition Python script with the FPGA pulse controls in Quartus, at first using an auto clicker, and later using a package that interfaces with Quartus directly through Python.

The key drawback of this approach, however, was that the time consuming temporal pulse adjustment needed to be performed on wall power (because the gimbal batteries do not last long enough, nor would it be prudent to discharge them in advance of outdoor data collection), but the FPGA needed to be transferred onto and off of battery power before and after performing the other procedures and taking outdoor data. While this power transfer was simple in concept, in practice our team had to transfer the power frequently enough while sleep deprived during nighttime experiments that someone would periodically accidentally disconnect the power to the FPGA. These unfortunate mistakes would typically preclude any data collection for the rest of the night, because the temporal pulse adjustment was so time consuming. So while this method for performing the temporal pulse adjustment was technically feasible, the drawbacks were such that our team was highly motivated to develop an alternative experiment.

Instead of collecting the temporal profile of the optical pulse and measuring and adjusting the pulses from scratch each time, it is feasible to record the electrical signal waveform for each state after high temporal indistinguishability has been achieved and use adjustments to match those electrical waveforms subsequently. To measure the electrical waveforms, the same electrical signal that would be sent to the LEDs is sent directly to the time tagger, along with the reference signal. By histogramming the time differences between the reference pulse and the rising edge of the electrical pulse, as well as the time differences between the reference pulse and the falling edge of the electrical pulse, the width and relative shift of the electrical waveform can be fully characterized. This is advantageous because it avoids the need for a carefully aligned optical setup in a dark environment. With fully electronic

data collection, signals can be acquired at the full repetition rate (12.5 MHz) and only need to acquire enough signals to characterize the rising and falling edges (distributions that are less than 0.5 ns wide). Meanwhile, with optical data collection the data is attenuated by the 0.1 mean photon number, the Eavesdropper fiber collection efficiency, and the single-photon detector quantum efficiency, in addition to needing to acquire enough data for a much wider histogram, given that the optical pulse is 10 ns wide. The result is that the electronic temporal pulse adjustment is drastically faster than the optical procedure, and it does not require alignment into an Eavesdropper fiber out in the field. This new procedure was primarily developed to alleviate field testing challenges, but due to its speed, reliability, and simplicity, our team has since incorporated it into the tabletop procedures as well.

Additional Temporal Testing

I tested the electronic temporal pulse adjustment numerous times in a lab setting and showed that it could achieve comparable values with the indistinguishability metrics to the performing the full optical temporal pulse adjustment. Because the native temporal pulses (before the adjustments are applied) can be different based on how the signals are routed on the FPGA for different compiled files, it is important to always redo the full optical temporal pulse adjustment when a new compiled file is generated for the transmitter. After this, Daniel collects the electronic waveforms, which can subsequently be used whenever the FPGA is reprogrammed with that same file to achieve the same level of temporal indistinguishability.

I also tested how robust the temporal system would be over time in a static environment. After performing temporal adjustments to achieve high indistinguishability metrics I left the system undisturbed in the laboratory over a long weekend and found the indistinguishability metrics unchanged within measurement error upon retesting.

I also conducted tests to show that the temporal adjustments are robust against tem-

perature changes. I set up the transmitter FPGA and the LED-to-fiber coupler with the attenuator fiber connected directly to the single-photon detector and collected the optical waveform. I collected these temporal optical waveforms in this configuration both at room temperature and in a small office refrigerator to simulate the cold evening temperatures the transmitter encounters during field testing. (This would not be sufficient to show that the states exiting the transmitter bench are indistinguishable because the free-space optical paths may be different lengths. Measuring the optical pulse after the beam exits the bench is necessary to rigorously show that the states are indistinguishable, because this is where an Eavesdropper would have the ability to measure them.) I found that the optical waveforms for the different states do not noticeably change in width or relative shift under these temperature changes, which shows that the temporal indistinguishability characterizations remain robust against temperature fluctuations during field testing.

4.2.4 Mean Photon Number Balancing

The prior three procedures (LED-to-fiber alignment, internal transmitter alignment, and temporal pulse adjustment) are all prerequisites for the mean photon number balancing experiment. The LED-to-fiber alignment and internal transmitter alignment must be maximized so that mean photon numbers on the order of unity are within reach, but importantly, they should not be adjusted further after the mean photon numbers are tuned because it may cause an imbalance in the mean photon numbers. (The obvious caveat here is that the internal transmitter alignment drifts on its own and must be re-tuned. I record the internal transmitter coupling efficiencies during mean photon number balancing and aim to recreate those same coupling efficiencies when I realign, rather than simply maximizing those coupling efficiencies.) The temporal pulse adjustment changes the pulse width and

thus the mean photon number, so it too must be performed before the mean photon number balancing experiment.

The mean photon number balancing stability rests on three relatively stable systems (the LED-to-fiber alignment, the temporal pulse adjustment, and the in-fiber attenuators used to effect that balance), and one highly volatile system (the internal transmitter alignment). However, because the internal transmitter alignment does not require supplementary experiments such as an Eavesdropper or alignment between the transmitter and receiver, and the coupling efficiencies can be accurately measured, it can easily be re-tuned just before the final data collection, restoring a reasonable balance between the mean photon numbers. Because much of the drift in the internal transmitter alignment is caused by temperature change, it is important to try to minimize the temperature change the bench must undergo by performing the experiments at the threshold of the Seed House that serves as the base of operations for field testing. While performing the experiment fully outdoors would be preferable, I did not have access to any sufficiently flat and stable ground outdoors to set up the portable optical breadboard on. The concrete floor in the Seed House served that purpose better and the large double doors at the entrance made it possible to bring the experiment much closer to the outside temperature.

As in the temporal pulse adjustment, the field version of the mean photon number balancing experiment is performed with the transmitter gimbal on the ground above the portable optical breadboard with optics posts to brace the bench so the gimbal cannot rotate. I install a mirror in the path of the beam exiting the transmitter and a collimator connected to a single-mode fiber. I send the alignment laser through the transmitter bench and maximize and record the value of the coupling into this fiber using the mirror and collimator mounts as degrees-of-freedom. I subsequently connect the single-photon detector to serve as the Eavesdropper. I measure the count rate at the single-photon detector for the different states and

calculate the mean photon number for each assuming Poisson statistics and dividing by the detector quantum efficiency and the measured fiber collection efficiency. With the in-fiber attenuators fully un-attenuated, I identify the state with the lowest mean photon number and use the in-fiber attenuators to lower the other states to the same value. I record that value as well as the values of the internal transmitter coupling efficiencies. This experiment requires a dark environment, but rather than set up in a basement during daylight hours, I perform this experiment after dark so as not to allow too much time to elapse between this tuning and the final data collection.

4.2.5 Internal Receiver Alignment

The internal receiver alignment does not require any prerequisite experiments, but it is itself a prerequisite for the transmitter-to-receiver alignment and the polarization fidelity compensation because the receiver must be internally aligned to get high throughput with the composite system. Additionally, the internal receiver alignment does not require any extra setup to align (*e.g.*, an Eavesdropper). Because its setup is simple and the earlier procedures are not prerequisites for this experiment, our team often performs it in parallel with those earlier procedures during field testing.

The field testing procedure for the internal receiver alignment is essentially the same as the tabletop procedure discussed in Section 3.4. In the field Daniel and Samantha typically perform this procedure with the receiver drone suspended between two tables with one person adjusting the degrees-of-freedom on the receiver while a second person holds up a sheet of paper at the different distances to gauge the overlap of the back-propagated beams. This is most easily performed after sunset because the combination of the multi-mode fibers and the 0.25 NA collimators cause the back-propagated beams to expand quickly and they can

be difficult to see well in sunlight with the gimbal lasers. This does not typically delay data collection, however, because the mean photon number balancing experiment is also performed after sunset and they can be performed in parallel.

4.2.6 Transmitter-to-Receiver Alignment

The transmitter-to-receiver alignment requires the internal transmitter alignment and the internal receiver alignment, and it is a prerequisite for the polarization fidelity compensation experiment. This procedure involves adjusting the setpoints for the inner and outer loops of the pointing-and-tracking system. The setpoints can sometimes drift with temperature, but not as rapidly as the internal transmitter alignment or the polarization fidelity compensation, which also makes it important to perform this experiment before those. The pointing-and-tracking outer loop uses an IR beacon and camera along with the gimbals for the course alignment of the drones (see Section 3.4). However, sunlight contains significant IR light and even twilight conditions can confuse the target acquisition software and prevent the outer loop from locking on. For these reasons, the transmitter-to-receiver alignment experiment also cannot be performed until after dark. This experiment is performed with the two drones suspended between tables and placed at the separation intended for the actual data collection. The transmitter drone is shown in this configuration in Figure 4.1.

Daniel and Andrew use an iterative alignment procedure to adjust the inner and outer loop setpoints until high coupling is achieved in all four receiver fibers, either using an alignment laser or with the pulsed LED source. The alignment laser presents the advantage of higher signal-to-noise ratio, which can be critical for tuning the system when the coupling is very low. However, after even low-quality coupling is achieved, a quantum alignment can be advantageous because feedback can be gathered from all four receiver channels simulta-

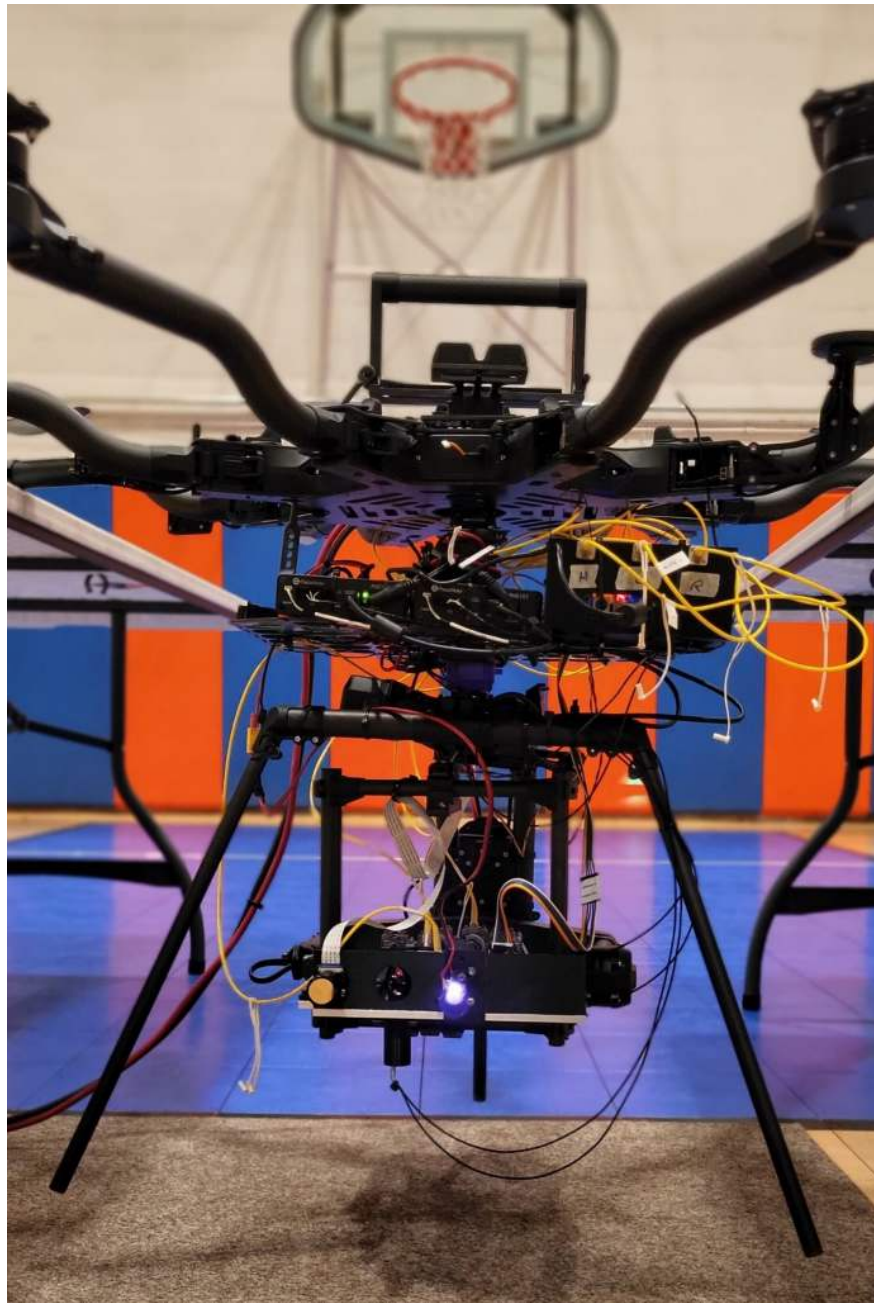


Figure 4.1: Photograph of the transmitter drone suspended between two tables during an indoor ground-to-ground test. The drone arms rest on the tables, allowing the gimbal to rotate freely so that the pointing-and-tracking system can lock to the other drone and the experiments that require this configuration can be performed. Photo credit: Andrew Conrad.

neously using the single-photon detector, as well as the added benefit that the signal used for the tuning is the one to be used in the data collection.

4.2.7 Polarization Fidelity Compensation

As discussed in Section 3.3, the polarization fidelity compensation serves to correct for birefringent effects in the optics train. While this is also necessary in tabletop demonstrations, it presents additional challenges with temperature dependent birefringence effects (most strongly observed in the spatial filter fiber) because it means that the polarization fidelity compensation must be frequently re-tuned as the temperature shifts over time. The polarization fidelity is one of the most temperature sensitive aspects of the system, and the lowering of the fidelity directly impacts the QBER, which is a critical metric in determining how much secure key the protocol can generate. For these reasons, the polarization fidelity compensation is the second-to-last experiment that should be performed before final data collection. Typically, at this point in the procedure, the transmitter bench alignment has slipped enough that I must realign it, ensuring enough throughput with the quantum signal to perform the polarization fidelity compensation easily.

In Section 3.3 I explain how the polarization compensation waveplates are tuned based on the measured extinction ratios in the H/V and L/R bases. When originally conducting this experiment in a laboratory setting, I would measure these extinction ratios by sending an alignment laser through the transmitter bench and record the power received at the different collimators on the receiver using a power meter. With only one power meter to make measurements, I would have to measure the power at each port in turn to calculate accurate extinction ratios. After performing the calculation, I would make an adjustment with the compensating waveplates then make another set of power measurements. This

procedure was very time consuming, often taking several hours. In light of the temperature dependent birefringence, this strategy would be untenable in an outdoor setting because the unitary transformation being corrected would change before good QBERs were even achieved.

To satisfy this need for faster feedback on the extinction ratio measurement, I turn to the four-channel single-photon detector. Using the quantum signal instead of an alignment laser, I connect the multi-mode receiver fibers to the single-photon detector and connect the electrical output to the Swabian Time Tagger Ultra, which can read out the detector count rates directly into Python. From here, I use a GUI I wrote in Python that calculates the relevant extinction ratios in real time (in addition to overall count rates and other useful statistics). This allows me to perform the polarization fidelity compensation on a fast enough timescale to realistically achieve high QBERs during field testing, and it also simplifies the procedure for tabletop demonstrations.

The two main differences between the tabletop procedure for the polarization fidelity compensation and the field testing procedure are the physical distance between the transmitter and the receiver and the quality of the signal throughput. I use the splitting of the polarization measurement optics on the receiver bench to calculate the extinction ratios needed to optimize the polarization fidelity compensation, and I use the polarization compensation waveplates on the transmitter bench to perform the actual tuning. However, during the ground-to-ground portion of field testing, the two drones are placed at the separation intended for the actual data collection (which is 10 m apart for most of the data sets) and the extinction ratio must be communicated to the person performing the adjustments. Our team originally dealt with this by setting up one laptop at the receiver and sharing a screen over WiFi to a laptop on the transmitter side, but the network latency slowed down the measurement feedback enough to make the optimization somewhat more difficult. We later

found better performance by using USB extenders to connect the laptop at the transmitter directly to the time tagger at the receiver. In the latest experiments, Daniel has improved the design further by using the receiver FPGA to stream the count values directly to the laptop at the transmitter over the closed router. This is advantageous because it allows this procedure to easily be performed with the drones at greater separations in future experiments. It also removes the need to temporarily connect the electrical outputs of the single-photon detector to the time tagger for this experiment and disconnect them afterwards. This simplification saves time, which helps avoid drift in the temperature dependent subsystems and thus collect higher quality data, and it simplifies the procedure, removing an opportunity for human error (*e.g.*, forgetting to switch the electrical outputs back to the receiver FPGA).

4.2.8 Drone Flights and Air-to-Air Data Collection

After the polarization fidelity compensation experiment is completed, the drones are moved from the ground-to-ground testing station out to the flight field and the transmitter optical bench is realigned for a final time. After a brief series of checks to guarantee that each subsystem is connected and configured correctly, Andrew and Samantha perform the takeoff and begin hovering in “position hold mode” (see Fig. 4.2). At this point I activate the pointing-and-tracking acquisition remotely from the control station, and once the logs report that the locking is successful, Daniel remotely activates the quantum data collection on both the transmitter and receiver FPGAs. After the drones reach 15% battery remaining, I signal to the drone pilots to land. From there Daniel and Andrew extract the collected data from the FPGAs over the router and we begin preparations for the next flight.



Figure 4.2: Photograph showing the two drones in flight during an air-to-air data collection at the testing field beside the Seed House. Photo credit: Timur Javid.

4.3 Back-to-Back Flights

When performing air-to-air or air-to-car tests, our team often attempts to perform back-to-back flights with the system. We have enough batteries for the drone and gimbal systems to perform two sets of data collections. This is appealing because it allows the collection of twice as much data while only spending a little more time than is required to conduct the experiments and procedures detailed above. Because most of the systems subject to drift in performance owe that drift to changes in temperature, and over the short time between flights the temperature does not change significantly, the performance of these subsystems should be comparable from one flight to the next. The internal transmitter alignment and to a lesser extent the internal receiver alignment are the only systems that are noticeably susceptible to reduced performance due to the drone vibration and the shocks associated with takeoff and landing. Because the multi-mode fibers are less sensitive, and because their realignment would also require a re-tuning of the pointing-and-tracking system, it is not prudent to realign the receiver bench between flights. The transmitter bench is much more sensitive due to the single-mode filter fiber alignment. By realigning the transmitter to the same coupling values recorded during the mean photon number experiment, I successfully re-tune the mean photon number balance and overall throughput of the transmitter. Because I can do this without impacting any other subsystem (which might be more time consuming to re-tune), it is only beneficial to realign the transmitter bench between back-to-back flights. After two flights however, our team repeats the whole procedure in its entirety.

4.4 Vehicle Configurations

While most of these field tests have featured air-to-air QKD demonstrations, our team has also implemented air-to-car and car-to-car configurations. The procedures for these tests are largely the same, with a few important differences. For procedures that would otherwise take place at the ground-to-ground testing area with the drones suspended between tables, the drone or drones that are to be placed in the vehicle for data collection are placed there at this stage. This helps avoid any unnecessary bumps or shocks that might occur when transferring the drone from the ground-to-ground testing station to the vehicle. The gimbal that is placed in the vehicle is secured in the back seat with the window down (so it cannot interfere with the signals). In order to allow the gimbal to rotate freely (which it cannot with the legs resting on the ground), the optical bench is mounted in an inverted configuration. Andrew, Daniel, and Samantha perform this inversion before any of the other procedures. However, this complicates the experiments that require an Eavesdropper, because it becomes very difficult to brace the gimbal and prevent rotation in its inverted configuration. To get around this, Samantha bolted together two small optical breadboards and attached the mirror and collimator needed for the Eavesdropper. This portable Eavesdropper can be bolted directly to the output port of the transmitter, allowing the mean photon number balancing experiment to be performed without bracing the gimbal or setting it up over the larger portable optical breadboard.

In both the air-to-car and car-to-car testing configurations, the entire experiment is in motion, so the control station must be brought along in the car, including the laptops remotely controlling and monitoring the FPGA data collection, the pointing-and-tracking, and the drone status and power, as well as the router to keep the network connected during

the test. It necessary for all personnel to ride along in the vehicle(s) to run these systems as well as driving the car(s).

For the air-to-car data collection, our team uses a 0.5 mile road near the testing site. Beginning at the parking lot of the Seed House, we set up the car at the entrance to the road and set the drone on the pavement. Andrew performs takeoff and I activate the pointing-and-tracking between the drone and the car. Samantha begins driving down the road at 10 mph while Andrew flies the drone alongside to maintain locking (see Fig. 4.3). At the end of the road we land the drone, transfer the collected data, replace the batteries, re-tune the transmitter alignment, and prepare to collect another data set along the same road back to the Seed House.



Figure 4.3: Photograph showing the transmitter drone in flight while the receiver drone sits in the car during an air-to-car data collection flight on the road near the Seed House. Photo credit: Timur Javid.

For car-to-car data on a closed test track, the procedure is similar except both gimbals are inverted and placed in cars facing each other. The cars are parked next to each other

and the pointing-and-tracking acquisition is activated, and after locking is achieved the data collection on the FPGAs begins. The drivers operate the two cars alongside each other at 5 mph until the data collection is complete. Our team has also conducted car-to-car testing on the highway at 70 mph. It is not possible to safely use the lasers from the inner loop on a public road at high speeds, so our team deploys the system with only the outer loop activated. This results in less consistent locking and therefore lower throughput, but the system is still successful in transmitting some raw key. Because two separate vehicles are required during these tests, our team uses cell phones to maintain contact between the two cars to signal when locking is ready and data collection should begin. Communicating via cell phones while driving 70 mph with the windows down is challenging, but ultimately successful. The stretch of highway used for these demonstrations is far enough from the Seed House that it is not practical to return for re-tuning steps between data collections, so our team brings along the equipment necessary to perform the critical experiments in the rest stop parking lots along the highway that served as the turn-around points between data collections.

Chapter 5

Data Analysis and Processing

After the QKD data is collected, it must be synchronized and processed so that the relevant performance metrics can be calculated and a sifted key can be extracted, which can then be distilled into a secure key after the security analysis is performed. This chapter covers the structure of the data and some of its associated issues, the pre-processing steps that synthesize the asynchronous receiver data into a single data stream, the Bayesian qubit-based synchronization algorithm that allows the reconciliation of the transmitter and receiver data, and the subsequent noise filtering and key sifting. The output of this data processing is ready for the security analysis performed by our collaborators at the University of Waterloo, as well as any subsequent privacy amplification.

The data structure from the FPGA data collection described in the first part of this chapter was pioneered by Daniel Sanchez-Rosales, with contributions to prototypes from Akash Gutha, and continual input and troubleshooting efforts from me. Additionally, Daniel and I collaborated on the measurement of the clock drift rate between PLLs on different FPGAs. All other sections pertaining to the processing of the data after it is extracted from the FPGAs up through the creation of the final summary table and the sifted key were my exclusive domain and contribution.

5.1 Data Structure Overview

For the QKD protocol, it is necessary to prepare and send random polarization states on the transmitter, Alice. However, it is also necessary to save all the data sent for future data processing and ultimately the key exchange. The FPGA used to drive the source also provides a robust platform for saving the random state choice sequences. The system sends and save states at a repetition rate of 12.5 MHz. The data saving structure involves temporarily saving data into 64 KB banks on the FPGA, which are then passed to the hard processor and saved onto the SD card.

5.1.1 Receiver Data Saving

The receiver FPGA code works by continuously running a counter at 100 MHz, and when the single-photon detectors register a detection, the current value of that counter is written to a bank corresponding to that detector. Once each bank fills up, the detector identification is written as the last value and the bank is saved to memory. This means that the time tags are saved in order for each detector, but asynchronously with respect to each other. These must later be stitched back together so all data can be processed together on the same timescale. Additionally, the maximum value of the counter determines how many bits must be saved for each time tag and thus what the saving throughput must be. Because the counter resets to zero after reaching its maximum, I use the modular time differences between subsequent tags to infer the global time. One potential pitfall to this strategy is the possibility of the counter cycling all the way around without recording a received time tag. These so-called rollover errors create timing ambiguities and cause offsets in the data, which completely derail the analysis of data that comes after they occur. In the earliest

versions of the FPGA receiver code the data was very vulnerable to these rollover errors, and I could only successfully process data that had high enough count rates and rollover events were very unlikely. At that time, the counter was only 16 bits long, which meant the counter would reset every 655 μ s. This meant that when the signal levels were low there was a significant probability of a rollover error.

In order to combat this, Daniel changed the subsequent version of the FPGA code so that there would be an automatic time tag event recorded at the beginning of each counter cycle, regardless of whether the single-photon detector received a detection at that time. This guaranteed there is always at least one time tag for each counter cycle, removing the possibility of a rollover event. It also added a small amount of extraneous noise to the data, but because the extraneous time tag was always the same counter value, I was able to remove this noise in the post processing by simply throwing away all tags received with that counter value. Thus, the cost of this approach was that it reduced the effective transmission by a factor of $(2^{16} - 1)/2^{16} = .9999847$, which is an extremely negligible loss. However, this intermediate version of the receiver architecture was prone to bit saving errors, which would typically happen several times per data collection flight. If a saved time tag has one bit flipped from the correct value, it can cause large and extraneous time differences for the time expansion of the data before the full processing. For example, consider a system with a 4 bit counter that records 3 time tags with counter values:

$$0001 = 1$$

$$0011 = 3$$

$$0100 = 8$$

Here, I would infer that these 3 subsequent time tags had modular time differences of 2 and 5. However, consider if one bit were flipped on the second time tag:

$$0001 = 1$$

$$0111 = 11$$

$$0100 = 8$$

Instead, I would assume that the counter had reset between the second and third tag, and I would infer time differences of 10 and 13. This induces an offset for all data processed after this point, making proper analysis impossible until the offset is identified and corrected.

The current version of the FPGA code uses a 30 bit counter, which gives a counter cycle of 10.7s. With a rollover time of this length, going an entire cycle without registering a detection is highly improbable even without an automatic time tag event. Additionally, Daniel has improved the saving architecture to the point where I no longer encounter bit flip errors.

5.1.2 Transmitter Data Saving

The original version of the transmitter FPGA code (which our team used for data collected in 2021 and earlier) did not save any of the states it sent. Rather, to generate states, it used a linear feedback shift register (LFSR), which is a pseudo-random number generator that could be easily recreated with the analysis code in Python. This version also included a duty cycle where it would send the pseudo-random data for 4.87 seconds, and send nothing for 0.5 seconds, which served several purposes at the time. The 5.37 second period of the data transmission was helpful in identifying the true signal when optimizing coupling between

the transmitter and receiver drones. Additionally, these so-called blips provide divisions in the data in the post-processing for reasonably sized segments that could be processed together without overwhelming the processing computer. While this served as a very simple data generation method, it would be fundamentally insecure as an adversary could trivially recreate the transmitted data sequence without making any measurements of the quantum system whatsoever.

The next iteration of the transmitter FPGA code introduced a true random number generator, which employs hybrid Boolean networks [29]. While this improved the security of the information transfer, it also necessitated saving all of the states sent by the FPGA, as the states could no longer be independently recreated in a Python script. These states were saved as subsequent 4 bit strings corresponding to which LED was being pulsed and whether or not a decoy or vacuum state was used. This intermediate version of the transmitter (used primarily for data collected in 2022) also included the same duty cycle, with random data for 4.87 seconds, and nothing for 0.5 seconds. During the 0.5 seconds when nothing was sent, the transmitter FPGA would repeatedly save the last tag sent, which would result in regions of the data with a constant state rather than different states varying constantly. This structure in the data served as a delimiter for dividing the transmitted data into the respective blips.

However, this intermediate version of the transmitter FPGA code was prone to significant data loss, which usually manifested as dozens or hundreds of subsequent data banks not being written to memory. Daniel and I estimated that about 10% of the transmitted data was not saved due to these issues. While the data loss itself was a significant issue, the most damaging aspect of these saving errors was the timing offsets induced by the missing data. This resulted in data sets that would process correctly up to a certain point, and the subsequent results would be nonsensical until I identified how much data had been lost

and at what time in the data set the loss had taken place. I could usually infer how much data had been lost because I could see how much data was missing from a particular blip; however, sometimes the data loss subtended multiple blips, deleting the 0.5 second delimiter region as well, which caused some neighboring blips to appear as one extra long blip in the transmitted data. Because the data was completely random, I had no indication of where precisely the data loss had occurred when analyzing the transmitted data on its own. Only in the context of the receiver data, which is probabilistically correlated with that of the transmitter, could I correctly identify when these data losses occurred.

The large numbers of dropped banks only manifested in the transmitter saving architecture (which must handle a significantly higher data throughput). However, the new version of the transmitter FPGA code is more robust and does not have any data loss issues. Additionally, the separate blips are now automatically saved as separate files, simplifying the post-processing. The new architecture uses a 6.7 second transmission with a 0.5 second pause, and it now takes advantage of the 0.5 second data pause to keep saving backlogged data banks, allowing for higher throughput saving. Figure 5.1 shows the mean counts from the receiver data from a tabletop QKD data collection with the new transmitter and receiver architectures in place, and the 7.2 second duty cycle can be clearly seen in all four channels.

5.2 Received Data Stitching

I use the modular differences between successive time tags to expand the recorded data for each channel into a vector of ones and zeros corresponding the presence or absence of a detection at each time step. However, I must merge these four time expanded vectors together into a single $4 \times N$ array for further processing. As mentioned in the previous section, the receiver FPGA stores the data from different detectors separately. While all

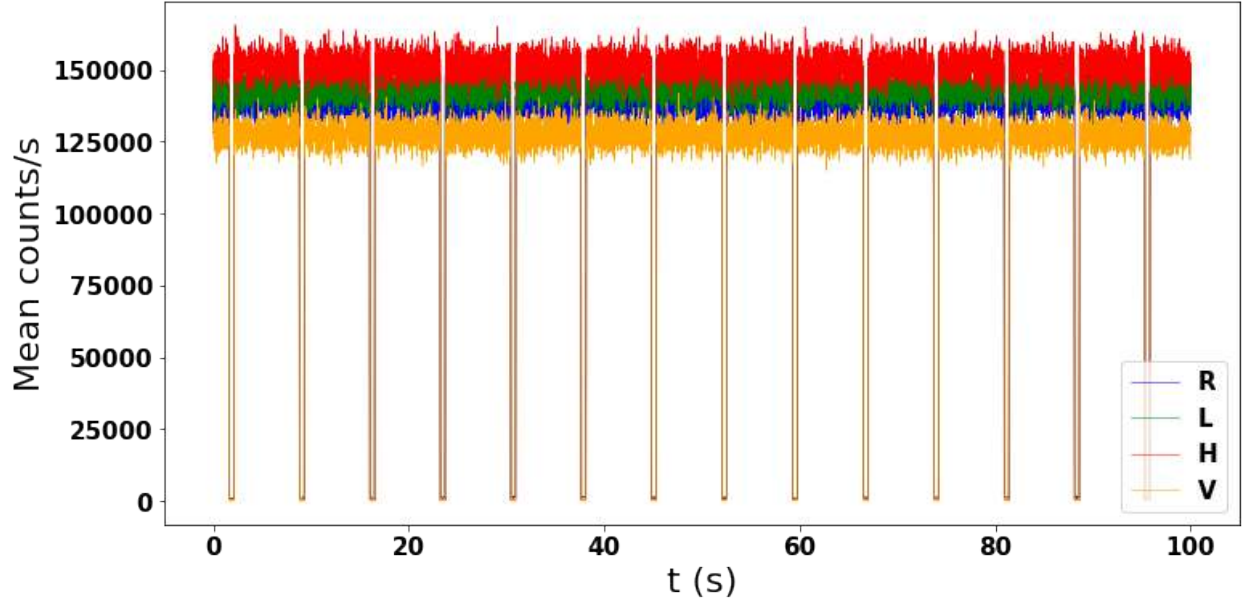


Figure 5.1: The mean counts per second for a tabletop QKD data set taken on August 8th, 2023. All four channels exhibit the expected 7.2 second duty cycle, including the 0.5 second period where the transmission stops and only background counts are observed.

these events are time-tagged with the same counter, the fact that the detectors accrue data at differing rates means that the different detectors record different numbers of banks. This also means that the n th bank for each of the four detectors does not necessarily correspond to anywhere near the same global time. Because the counter can roll over many times in a data set (or even in a single bank, depending on the counter size and the count rate on the detectors), it can be very challenging to use the counter value to infer the global time.

The most obvious strategy would be to use the counter values to track the global time starting at the beginning of the data collection. However, the FPGA is actually constantly recording data in the banks even before the data collection begins saving those banks to the disk. This means that even for the first time tag of the first recorded bank, the 4 channels are not necessarily near the same global time. Additionally, the beginning of the collected data often has very low detector count rates, which can exacerbate these timing offsets because

it takes longer to collect the data to fill one bank.

Instead, I rely on identifying a common feature in the data streams of all four detectors to find a common time point. The 0.5 seconds out of each 7.2 second data blip is an ideal feature for this task. The transition between no signal to high signal region is very easily recognizable and the four channels are exhibited to this transition at precisely the same global time. I wrote a blip auto-finder to roughly identify the blip pauses based on a rolling average of the gaps between successive detection events, (see Fig. 5.2). Blip pauses can sometimes be difficult to identify because drop outs in the locking system can appear very similar to pauses in the transmitter itself, (see Fig. 5.3). However, I can take advantage of the periodic nature of the blip pauses. The auto-finder identifies various candidate times with high average detection gaps (sufficiently separated from each other so as not to double count), and then takes those times modulo 7.2 and histograms them. Thus I can filter the results of the auto-finder based on whether the pauses come at integer multiples of the cycle period compared to each other.

Once the blip pauses have been roughly identified, I find a step function model based on the click probabilities to identify the time-tag at which the transmission resumes after the blip pause. I can estimate the received mean photon number for the four detectors as well as the average background count rate, which allows me to build a model of the expected distribution of the gaps between successive time tags for the two different regions. From there, I can find a maximum likelihood estimate for the transition point for each of the four channels, as shown for one of the channels in Figure 5.4. The model can typically find these transitions to within less than half of the counter rollover time, which allows me to assume the counter values for the four detectors are on the same timescale. I can use the counter value to determine which of the four channels has the earliest identified transition. While the true transition feature should be simultaneous across the four detectors, due to

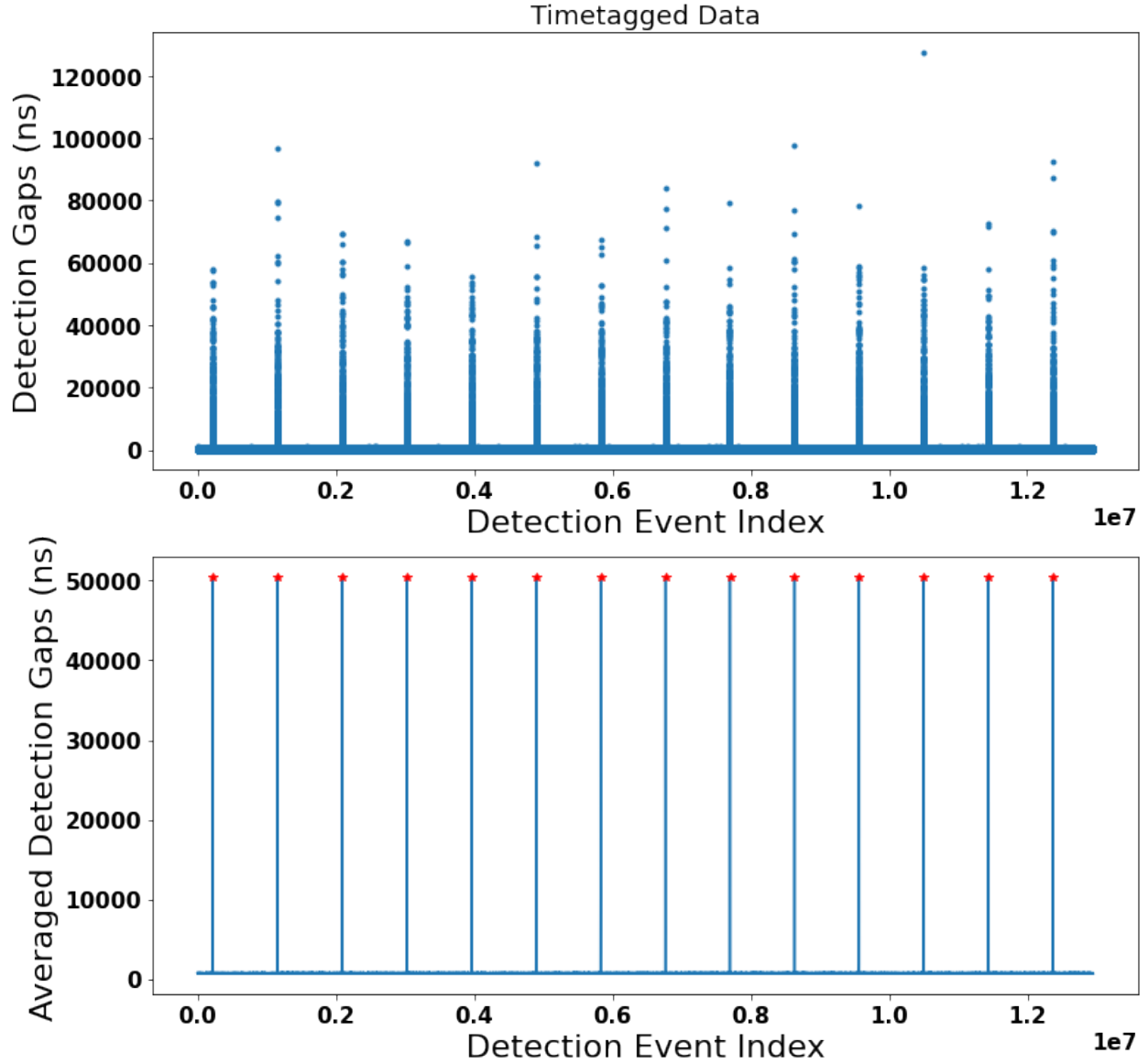


Figure 5.2: The gaps and averaged gaps between successive time-tags for the right-circular detector data from a tabletop QKD data set taken on August 8th, 2023. The detection gaps increase dramatically during the 0.5 second period when the transmission stops and only background counts are observed, which is an easily found marker in the data of all four detectors. The red stars in the second plot show the indices of the blip pauses identified by the blip auto-finder routine.

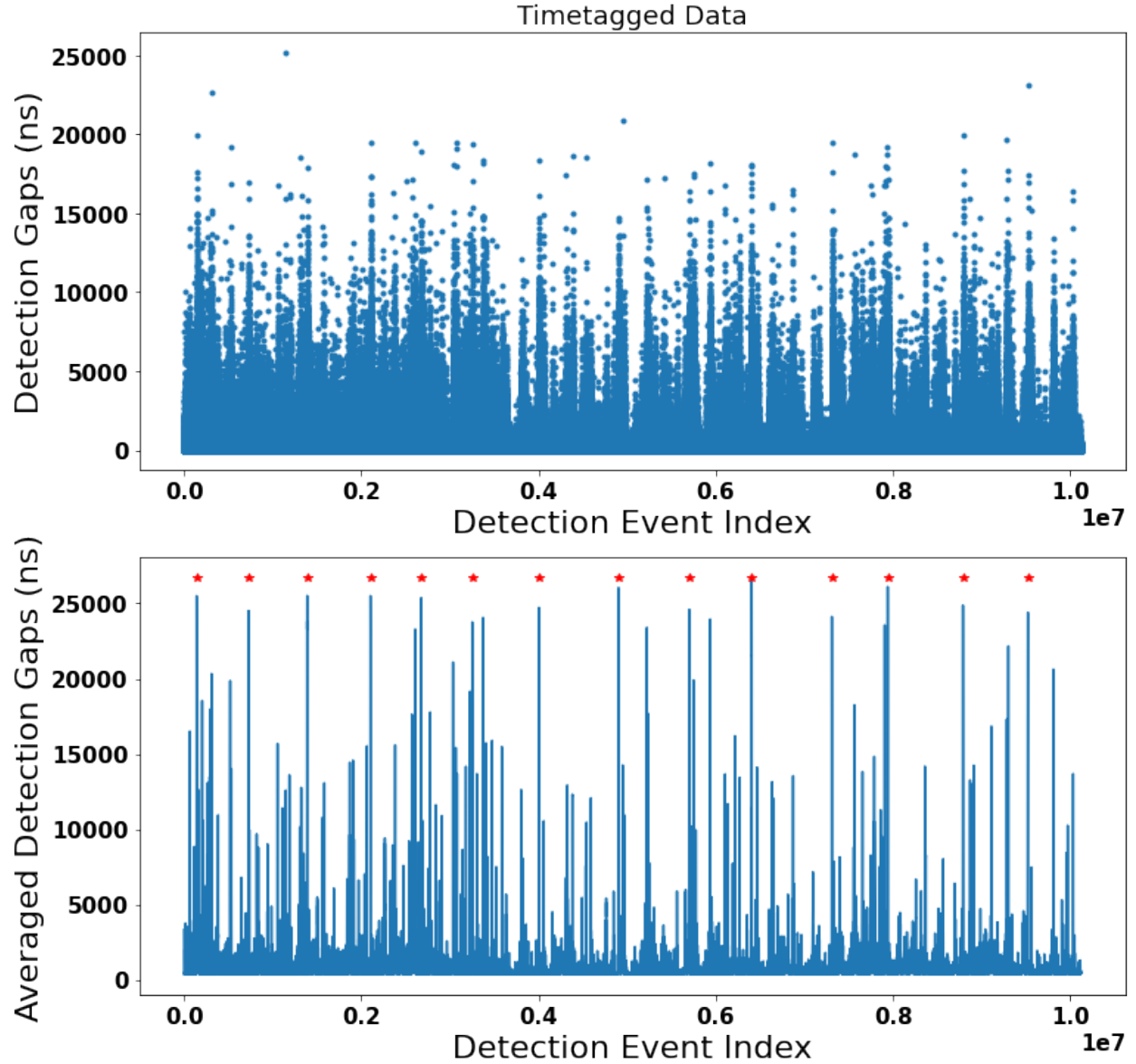


Figure 5.3: The gaps and averaged gaps between successive time-tags for the right-circular detector data from an air-to-air QKD data set taken on August 28th, 2023. While the detection gaps increase dramatically during the 0.5 second blip pauses, the drop outs due to alignment difficulties also increase the detection gaps, making the blip identification more challenging. However, the auto-finder is still robust enough to identify the blip pauses correctly for all four detectors. The red stars in the second plot show the indices of the blip pauses identified by the blip auto-finder routine.

the imprecision of my transition identification method, it is found at slightly differing times across the four channels. I then pad the other channels with the appropriate number of zeros to ensure the simultaneity of the resulting columns. Thus, using these 4 transition points as a common time point, I can stitch together the data streams of the four detectors. This yields the $4 \times N$ matrix needed for further processing.

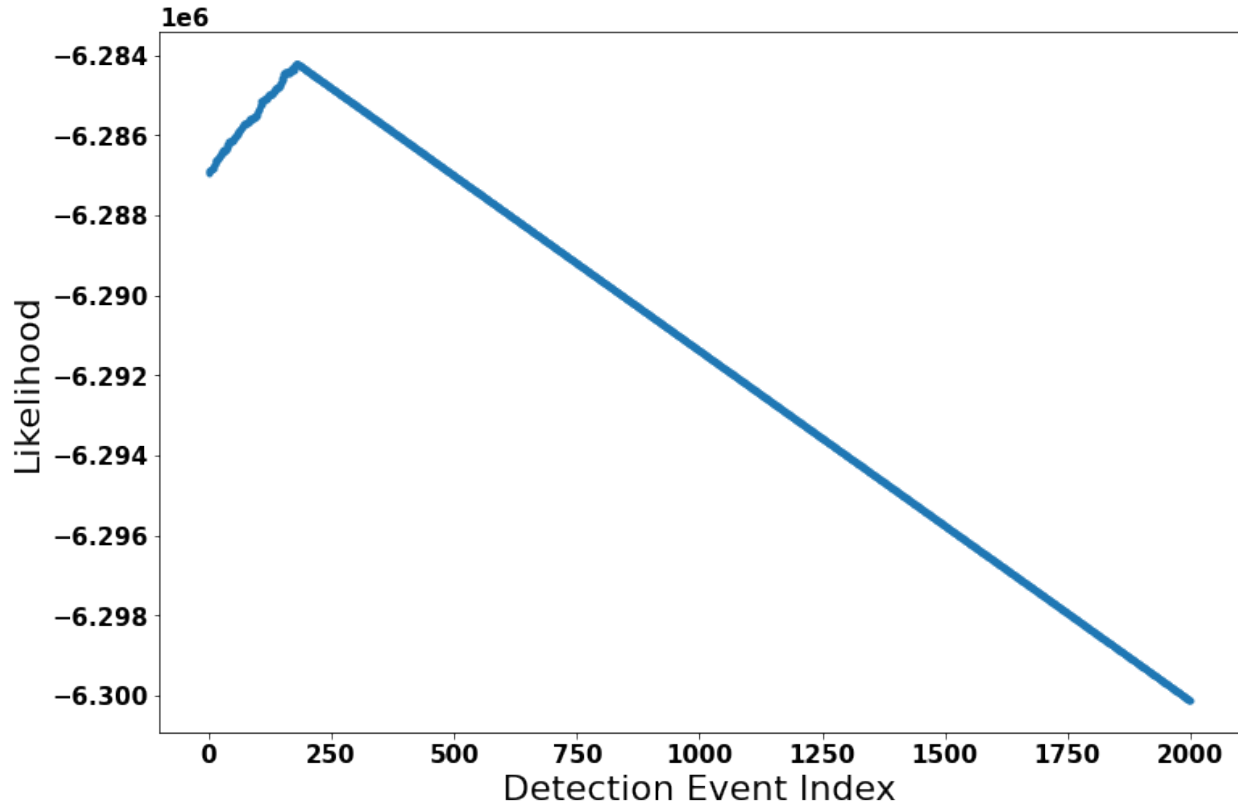


Figure 5.4: Plot of the calculated likelihood distribution for the transition point from the blip pause into the transmission region. The transition point is identified by simply taking the maximum of this distribution. Data shown is from the first blip of the right-circular detector data set from a tabletop QKD data set taken on August 8th, 2023.

The current version of the receiver saving architecture is robust enough that once a common stitching point has been successfully identified for the four different detector data sets, it can be reliably extrapolated to any other blips in the data set that might be more difficult

to stitch independently. This extrapolation is achieved by using the gaps between successive time-tags to count up to a multiple of the 7.2 seconds that separates each blip. This allows the movement from the index of the known transition point up to that of the desired blip. However, as discussed earlier, the previous version of the receiver saving architecture was prone to minor data corruption, which led to timing offsets between the four data channels in multiples of the clock rollover time (which was $655\text{ }\mu\text{s}$ for that version). Extrapolation was still necessary for some blips in these data sets, but the stitching needed to be adjusted afterwards. Because the shifting always came in integer multiples of the clock rollover time, fixing the stitching largely consisted of determining which channel or channels had become offset, and applying different compensating offsets until the channel(s) processed correctly. My diagnostic for whether or not a channel is correctly processed relies on comparing to a nominal value the fraction of all detection events in a channel that are received when the vacuum state is sent. Because the times when vacuum states are sent must already be revealed on a public channel for the decoy security analysis, I do not sacrifice any key in service of this diagnostic. If the aforementioned fraction is too high compared to the nominal value, it indicates that the data in that channel is randomly distributed, rather than correlated with the states being sent. From this I deduce that the channel in question has an offset that needs to be corrected.

5.3 Qubit-Based Synchronization Using a Bayesian Approach

5.3.1 Motivation

After having successfully stitched together the data from the four detectors, I turn my attention to synchronizing the transmitter, Alice, with the receiver, Bob. A practical issue in quantum communication protocols is synchronizing Alice and Bob's two data streams. If Bob does not know precisely when Alice begins data transmission, he must begin recording measurements early or else risk missing some of Alice's transmission. In either case, because some signals do not arrive at Bob due to channel loss, and extraneous events are caused by stray light and detector dark counts, the first event Bob records is unlikely to be the first event Alice sends, resulting in some timing offset that must be determined. Correcting this offset is an essential precursor to sifting: If Alice and Bob do not agree on the timing of the events, they compare basis choices from different events, resulting in a high QBER and likely no shared information. In addition, determining which time bins correspond to Alice's wavepacket arrival and which do not allows timing-based noise filtering.

Further complicating the communication protocol is that the relative clock offset may not be a constant due to drift in the relative phase and frequency between the transmitter and receiver clocks. Alice has a communication protocol temporal bin width τ_A that may be different from Bob's bin width τ_B . The timing offset between their clocks Δ at the n th communication time bin since the most recent clock synchronization is given by

$$\Delta = t_0 + (\tau_A - \tau_B)n + \varepsilon \tag{5.1}$$

for an initial timing offset t_0 and higher-order timing error ε . In this way, small differences in clock frequencies can gradually change the clock offset so that a previously calculated synchronization is no longer valid. Other timing errors, such as clock jitter and frequency drift, also contribute to the need for a more robust synchronization solution. I denote the time over which synchronization is maintained as T_b , *i.e.*, the time over which the error in $\Delta \ll \tau_A$.

Clock synchronization is sometimes achieved by directly sending Alice's clock signal to Bob over a separate channel via an optical link or using a radio-frequency signal [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]. However, this introduces additional hardware requirements and increases the cost and complexity of the setup. One way to avoid these additional resource requirements is to use the quantum channel itself to transmit the information necessary to perform the synchronization [40, 41, 42, 43]. One such qubit-based synchronization protocol was introduced and demonstrated by Calderaro *et al.* [40]. Their protocol uses a dedicated clock-synchronization phase followed by a key distribution phase. In the synchronization phase, a pre-agreed synchronization string is transmitted to Bob and the clocks are aligned during post-processing.

The pre-agreed synchronization string is used to find the initial offset between Alice and Bob's clocks. Because it must be public knowledge, it cannot be used to generate a secure key. If the clock frequencies are not consistent, simple clock offset recovery only temporarily aligns until the clock drift becomes of-the-order-of the communication protocol temporal bin width τ_A . Correcting for this clock frequency drift using only clock offset recovery requires repeated synchronization/key distribution phases with a regularity that depends on the stability of the clocks used in the experiment. This reduces the overall secure key rate because no QKD states can be sent while the synchronization states are being sent, which may result in zero key rate due to finite-key effects [44, 24]. However, to account for this drift without needing

to send regular synchronization strings, Calderaro performs clock frequency recovery using the periodic arrival times of Alice’s qubits. Unlike the clock offset recovery, this does not require a pre-agreed synchronization string and thus does not decrease the amount of key that can be sent.

While my method only uses clock offset recovery at this time, it avoids these potential impacts on secure key rate by synchronizing the clocks using only information that is already publicly sent over the insecure classical channel by Alice and Bob for sifting and security analysis: The basis choices and the mean photon number of the transmitted signal. Because the system only transmit one state in the monitoring basis, the basis choices provide information about which of Bob’s measurement outcomes are more likely. The decoy state choices, which determine Alice’s mean photon number for each wavepacket, also contain information about Bob’s measurement outcomes. For example, Bob is unlikely to record any detections if Alice sends the vacuum decoy state.

By comparing this information to his measurement outcomes, Bob can probabilistically determine the timing offset. To account for potential clock drift, Bob can perform this synchronization in subsets of length T_b . Thus, Bob can find the up-to-date timing offset and ensure that the basis choices he publishes are properly lined up with the ones sent to him by Alice, but this requires an efficient analysis method to reduce the data requirements. Of course, my approach as well as Calderaro’s require low enough channel loss so that there are enough events received by Bob over a drift interval as discussed below.

Another example of a qubit-based synchronization protocol for continuously-pumped entanglement-based QKD systems was introduced by Ho *et al.* [43]. Here, they correlate Alice and Bob’s detection events without considering basis information. Their synchronization method relies on Alice’s knowledge that some communication time bins are empty (assuming essentially unit detection efficiency for Alice’s setup) and hence Bob’s corresponding time

bin should also be empty. There is a single dominant peak in the correlation function that identifies Δ assuming a large enough number of Bob’s detection events. Because the detection timing information must already be shared publicly, this strategy does not sacrifice any secure key. This method fails when the probability of Alice generating a photon per communication time bin approaches unity because every time bin is likely to be filled and hence the correlation function has multiple high-value peaks that create timing ambiguity.

In the next section, I outline my synchronization algorithm and its advantages, and derive a formula for the synchronization probability using Bayesian analysis [45]. In Section 5.3.4 I introduce a model system, and in Section 5.3.5 I simulate data in this model system to demonstrate the effectiveness of my method. In Section 5.3.6 I present my conclusions and the potential applicability of this work to other QKD systems.

5.3.2 Qubit-Based Synchronization Algorithm

Similar to previous approaches, my algorithm uses a cross-correlation of Alice’s periodically transmitted data and Bob’s received data to find the number of each type of event pairing, where the cross-correlation is computed efficiently using a Fast Fourier Transform (FFT). One complication of a prepare-and-measure scheme is that Alice attempts to send a quantum state every communication time bin, corresponding to the high-photon-probability limit of the Ho *et al.* [43] method discussed above. This problem is addressed here using the decoy-state protocol [10], which must be used anyway to prevent a photon-number-splitting attack.

Decoy states are sent by Alice randomly and correspond to wavepackets with a mean photon number smaller than the signal state, which often includes sending the vacuum state. The vacuum state is particularly effective in the synchronization process because Alice has high certainty that she sent no photons, limited by her ability to completely block

the source. Bob should then also see no photons, limited by the presence of detection clicks from non-ideal effects such as detector dark counts, detector afterpulsing, stray light, and the bleed through of light from Alice’s source.

Beyond the decoy states, there are additional sources of correlation that can be exploited to help improve the synchronization process. For example, Alice’s use of the efficient three-state protocol, where she only sends one state in the monitoring basis, gives useful information if Alice and Bob also share basis-state information, which is already required for sifting. I use a Bayesian statistical method, described below, that uses all prior knowledge of the system characteristics, such as the state fidelities, the mean photon numbers, the channel loss, the fractional sorting of Bob’s device for the two bases, and the detector efficiency, to generate a lookup table of Bob’s detection probabilities for Alice’s different inputs. With these, I can easily compute the synchronization probabilities of different possible offsets using Bayesian statistics. Alice and Bob’s data is most correlated when they are synchronized.

A significant advantage of my approach is that it does not sacrifice any secure key: I only use the information that is already sent publicly over the insecure classical channel. This is an improvement over synchronization protocols that share some fraction of the raw data for synchronization purposes, as well as protocols that have a dedicated clock-synchronization phase [40] during which no QKD states can be sent.

Bayesian analysis is a logical choice for synthesizing all available information and using it to make accurate predictions about Δ . It also has the advantage that it predicts the probability that $\hat{\Delta}$ is the best estimate of synchronization offset. This allows me to quantitatively express the level of confidence in the synchronization estimate. Furthermore, the additional information I incorporate in the protocol allows me to make a decision with fewer received qubits, which makes the system more robust to clock drift.

My algorithm uses FFTs to compute cross-correlations between Alice’s inputs and Bob’s outputs, allowing me to count the number of each type of input-output pairing for the different time offsets. The computational complexity of my algorithm is dominated by these FFTs, which go as $\mathcal{O}(N \log_2 N)$ where N is the number of sampling bins. Each cross-correlation requires three FFT computations, so in order to count the number of each different event pairing, the number of FFTs that must be performed in the naïve strategy would be $3 \times n_{in} \times n_{out}$ for a number of distinct inputs n_{in} and distinct outputs n_{out} . In this protocol, $n_{in} = 5$ (H/V signal, H/V decoy, L/R signal, L/R decoy, and vacuum) and $n_{out} = 4$ (H,V,L, and R), thus maintaining the computational complexity of $\mathcal{O}(N \log_2 N)$. This would give a 5×4 table of event pairings for each candidate Δ , which I would combine with a 5×4 lookup table for the probabilities of those event pairings to compute the synchronization probability. However, Benjamin Cochran pointed out that because the FFT is a linear operation, the lookup table can actually be applied to Alice’s data array before running the cross-correlation [46]. Alice’s array is of the form $n_{in} \times N$, but I can matrix multiply it by the $n_{in} \times n_{out}$ lookup table to form a new array derived from Alice’s data called $\tilde{A}$ with dimensions $n_{out} \times N$. Each row of $\tilde{A}$ contains all the relevant correlations with the corresponding rows of Bob’s array, which means that instead of running $n_{in}n_{out}$ cross-correlations for all combinations of the rows of Alice’s and Bob’s arrays, I can simply run n_{out} cross-correlations between the corresponding rows of Bob’s array and $\tilde{A}$. This matrix multiplication has negligible computational cost compared with the FFTs, meaning that Benjamin’s strategy, which gives identical results, saves a factor of n_{in} for the relevant computing time. Additionally, I was able to achieve noticeable processing time improvements with minor adjustments to the processing batch length N . By choosing a highly factorizable N , I improved the efficiency of the FFT computation by taking advantage of existing “under-the-hood” NumPy FFT algorithms.

5.3.3 Synchronization Probability

Here, I use the strings of Alice and Bob's data. A string of Bob's data consists of the results of each of his detectors at each sampling bin. Typically, Bob's strings are very sparse because there are many sampling bins in which he registers no detections. A string of Alice's data consists of her published information at each sampling bin. If the communication time bin width is greater than the sampling time bin width, Alice will have multiple string entries for each state she sends, each corresponding to what she is sending at that part of her duty cycle. Determining the synchronization probability consists of comparing different strings of Bob's data (starting at different temporal offsets) to strings of Alice's data and calculating which of Bob's strings D is most likely to be the one generated by Alice's corresponding string. I determine, for a particular string of Bob's, the probability that it could have been generated by Alice's published string.

Mathematically, I phrase this as the likelihood $p(D|S)$ of generating Bob's string D given the assumption that its generating string is the one Alice has published, denoted by S . The uninformed assumption, which I denote as $\bar{S}$, is that Bob's string D has been generated by a random string other than Alice's published string (from some other portion of Alice's sent data), with the stipulation that the other string is also periodic. This mathematical framework considers a subset of Alice's data of N sampling bin widths compared against a subset of Bob's data of $N + M$ sampling bin widths, meaning there are M possible offsets to consider.

To begin my protocol formalism, I note that D is a string of length $M + N$ of Bob's measurements at each sampling bin (including sampling bins where no detections were received). Each measurement B_i in Bob's string consists of the click or no-click results at all

of Bob's detectors. Bob's string D can be written as

$$D = \{B_1, \dots, B_{M+N}\}, \quad (5.2)$$

which I can rewrite as

$$D = \{B_1, D'\}, \quad (5.3)$$

where

$$D' = \{B_2, \dots, B_{M+N}\}. \quad (5.4)$$

It is preferable to write the likelihood $p(D|S)$ in terms of known quantities such as the $p(B_1|S)$, the conditional probability of a time bin measurement B_1 given S . Using this notation, $p(D|S)$ is given by

$$p(D|S) = p(B_1, D'|S) = p(B_1|D', S)p(D'|S), \quad (5.5)$$

where the final equality is a result of the product rule. Because I have assumed that B_1 is generated from Alice's string, knowing D' gives no additional information about B_1 . At best, it informs me whether S is true, which is already assumed; the bits are otherwise independent because Alice's sequence is random. Using these observations, I obtain

$$p(B_1|D', S) = p(B_1|S), \quad (5.6)$$

and, by extension,

$$p(D|S) = \prod_{i=1}^{N+M} p(B_i|S), \quad (5.7)$$

allowing me to write the likelihood as the product of the measurement probabilities at

each sampling bin. I note that even in the example where Alice only sends one state in the monitoring basis, Bob must still measure both states in each basis to detect potential eavesdropper attacks [13, 12]. For computational ease, I also determine each sampling bin measurement probability as the product of the probabilities of the outcomes at the four different detectors b_ℓ , which are given by

$$p(B_i|S) = \prod_{\ell=1}^4 p(b_\ell|S) \quad (5.8)$$

Again, because the detectors' events are assumed to be generated by independent random processes, these probabilities can be considered independent when the generating string is known.

When the generating string is not known (under the uninformed assumption $\bar{S}$), the detection probabilities can be approximated as independent when the received mean photon number is low. Because the synchronization task is most difficult in low-signal regimes, I use this approximation going forward. Thus,

$$p(D|\bar{S}) = \prod_{i=1}^{N+M} p(B_i|\bar{S}) \quad (5.9)$$

and

$$p(B_i|\bar{S}) = \prod_{\ell=1}^4 p(b_\ell|\bar{S}). \quad (5.10)$$

For a given input from Alice, each of Bob's four detectors has an opportunity to detect a photon due to the detection clicks arising from non-ideal behaviors. Naturally, I use my knowledge of the system (the state fidelities, the quality of the polarization sorting, the dark count rates, the detector efficiencies, and the signal and decoy received mean photon number) to estimate the detection probabilities as accurately and efficiently as possible. Using a

lookup table of the detection probabilities for the different inputs from Alice, these likelihoods can be calculated using standard statistical methods.

However, the likelihood of generating D from Alice's published string is not the same as the probability that Alice's published string is the one that generated D , which is given by $p(S|D)$ and is the most relevant quantity to determine synchronization. Bayes' theorem allows me to rewrite this quantity, called the posterior, as

$$p(S|D) = \frac{p(D|S)p(S)}{p(D)}. \quad (5.11)$$

In addition, I must also include the information that I expect exactly one correct synchronization offset (not just one on average).

To formulate the problem as an exclusive synchronization, I must find the probability that some discrete timing offset, given by the time-bin index j , is the correct synchronization offset, and that all the other offsets are incorrect. In other words, the probability that, for a given string of length N published by Alice, all the measurements before the j th bin are generated randomly, the measurements from j to $j+N$ are generated from Alice's published string, and the measurements after $j+N$ are generated randomly. Under these assumptions, I can write $p(B_1, \dots, B_{M+N}|S_j)$ as a product of the likelihoods of these three sections as

$$p(B_1, \dots, B_{M+N}|S_j) = p(B_1, \dots, B_{j-1}|\bar{S}_j)p(B_j, \dots, B_{j+N}|S_j)p(B_{j+N+1}, \dots, B_{M+N}|\bar{S}_j). \quad (5.12)$$

Here I introduce $\bar{S}_j$, the assumption that the data is produced by a random string other than the synchronization string in question, but one with the same phase (i.e., the signal arrives at the same time bin in each period as it does for S_j).

I can find the conditional probability for matching Alice's string to Bob's string at a

potential synchronization index j in this framework using Equation (5.11), which gives

$$p(S_j|B_1, \dots, B_{M+N}) = \frac{p(B_1, \dots, B_{M+N}|S_j)p(S_j)}{p(B_1, \dots, B_{M+N})}. \quad (5.13)$$

Equation 5.13 is my main result and is the quantity of interest to identify clock synchronization between Alice and Bob. I determine the optimum synchronization index based on the value of j that maximizes this quantity, and the quantity itself denotes the confidence in that choice.

The denominator in Equation 5.13 can be written in terms of known quantities using marginalization. Marginalization consists of rewriting a probability as a sum of the comprehensive conditional probabilities; in this case, the different possible synchronization indices written as

$$p(S_j|B_1, \dots, B_{M+N}) = \frac{p(B_1, \dots, B_{M+N}|S_j)p(S_j)}{\sum_{i=1}^M p(B_1, \dots, B_{M+N}|S_i)p(S_i)}, \quad (5.14)$$

where the i denotes the other potential synchronization indices.

To evaluate Equation 5.13, the likelihoods $p(B_1, \dots, B_{M+N}|S_j)$ and $p(B_1, \dots, B_{M+N}|S_i)$ can be determined using Equations 5.7, 5.9 and 5.12. The quantity $p(S_j)$, called the prior, is the assumed probability that D corresponds to Alice's published string before other evidence is considered. That is, $p(S_j)$ is the probability that j is the correct synchronization index. I use a uniform prior, which assumes each candidate has a naïve $1/M$ probability of being the correct one given that I have M candidate indices, which means that

$$p(S_i) = p(S_j) = \frac{1}{M} \quad (5.15)$$

so that the prior terms cancel, giving us

$$p(S_j|B_1, \dots, B_{M+N}) = \frac{p(B_1, \dots, B_{M+N}|S_j)}{\sum_{i=1}^M p(B_1, \dots, B_{M+N}|S_i)}. \quad (5.16)$$

Next, I apply Equation 5.12 to obtain

$$p(S_j|B_1, \dots, B_{M+N}) = \frac{p(B_1, \dots, B_{j-1}|\bar{S}_j)p(B_j, \dots, B_{j+N}|S_j)p(B_{j+N+1}, \dots, B_{M+N}|\bar{S}_j)}{\sum_{i=1}^M p(B_1, \dots, B_{i-1}|\bar{S}_i)p(B_i, \dots, B_{i+N}|S_i)p(B_{i+N+1}, \dots, B_{M+N}|\bar{S}_i)} \quad (5.17)$$

and use Equations 5.7 and 5.9 (of which the latter uses a low received mean photon number approximation) to write everything in terms of known quantities as

$$p(S_j|B_1, \dots, B_{M+N}) \approx \frac{\prod_{k=1}^{j-1} p(B_k|\bar{S}_j) \prod_{k=j}^{j+N} p(B_k|S_j) \prod_{k=j+N+1}^{M+N} p(B_k|\bar{S}_j)}{\sum_{i=1}^M \left(\prod_{k=1}^{i-1} p(B_k|\bar{S}_i) \prod_{k=i}^{i+N} p(B_k|S_i) \prod_{k=i+N+1}^{M+N} p(B_k|\bar{S}_i) \right)} \quad (5.18)$$

Equation 5.18 is my master equation for the synchronization probability of an index j . The numerator consists of the probability of an N -length string of Bob's data starting at j being produced by Alice's published string, along with the probability that the remaining data was produced by an unknown string of Alice's data. The denominator sums this same quantity over all possible synchronization indices, ensuring normalization. I take the value of j that maximizes this quantity to be the optimum synchronization index, and the value of $p(S_j|B_1, \dots, B_{M+N})$ gives the probability that the synchronization index is correct. I can compute this conditional probability using FFTs to count the number of each unique bin

measurement along with a lookup table of the probabilities of the events.

5.3.4 Model System

To illustrate this protocol, I simulate a model QKD system using a polarization-based prepare-and-measure protocol with decoy states and only sending one state in the monitoring basis, mirroring the full experiment as closely as possible. I set Alice's repetition rate to be $f_A = 1/\tau_A$ and a wavepacket duration of $\Delta t = \tau_A/m$ with $m = 8$ for a duty cycle of 12.5 percent. I set Bob's sampling rate to nf_A with $n = 8$ so that his sample period is matched to the wavepacket duration. These conditions are illustrated in Figure 5.5. Alice generates a random sequence such that four quantum states L/R/H and a vacuum decoy state (a decoy state with mean photon number equal to zero) are sent in equal parts on average.

For my numerical experiments, I simulate a QKD session by generating data that emulates the state preparation and measurement, including aspects such as the *received* mean photon number μ , probability of a detector dark count d over one communication bin width τ_A , and variation in Δ due to clock drift, assumed to be constant over T_b . This allows me to test how these factors impact the synchronization performance. I assume a transmitted mean photon number $\mu_A = 1$ where the received mean photon number $\mu = \eta\mu_A$ for a channel transmission η . While this μ_A is on the upper end of values used in typical experiments, it allows me to explore the performance and limitations of my algorithm at or beyond the greatest received mean photon number one would realistically use: $\mu_A = 1$ with zero loss.

Assuming a Poisson distribution for Alice's source, the probability of Bob registering a click $p(\text{click}, \ell)$ over a period τ_A at a particular detector ℓ is given by

$$p(\text{click}, \ell) = 1 - (1 - d)e^{-\mu_\ell} \quad (5.19)$$

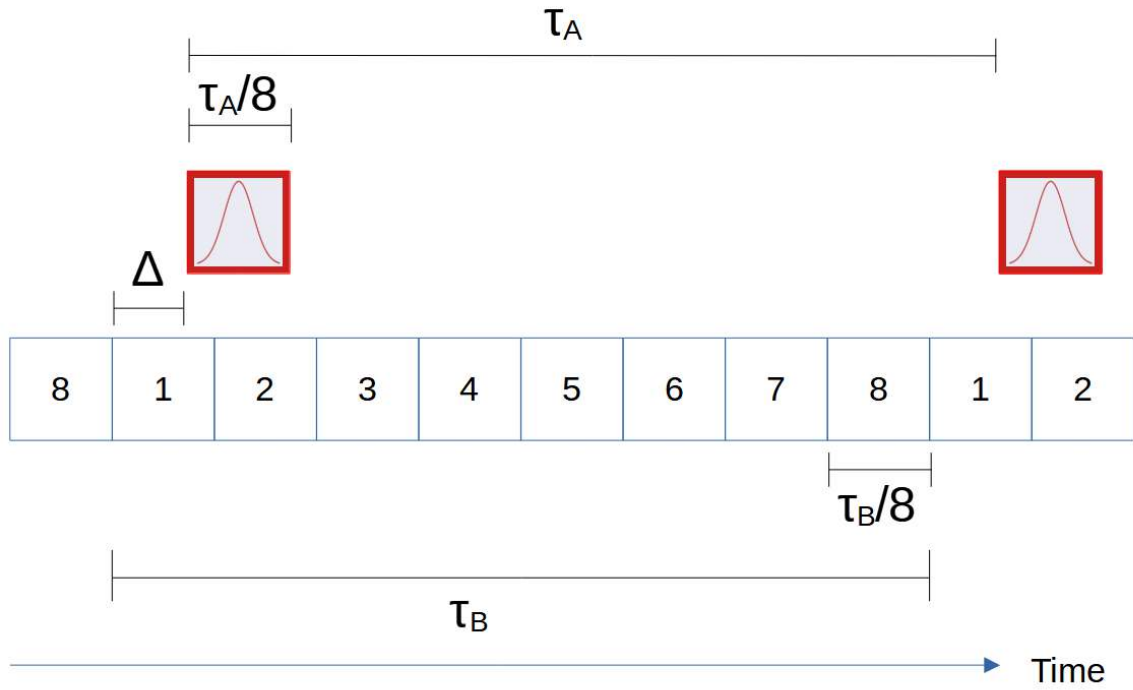


Figure 5.5: Illustration of the relative times used in the QKD protocol. Here, the signal (red) straddles bins 1–2 due to an offset of Δ , and I do not consider bins 3–8. I take $\tau_A = \tau_B$, which is approximately correct for a short enough data subset.

where μ_ℓ is the mean photon number received by detector ℓ . The portion of the total mean photon number μ that goes to the different detectors depends on which polarization state is sent. I use ideal BB84 sorting in the model system so that all states have an equal chance of being measured in either basis. States measured in the same basis as they are prepared are detected accurately, while states measured in the opposite basis have an equal chance of being measured in either opposite-basis state. For example, if Alice prepared an H-state that Bob receives $\mu = 0.8$, Bob's measures $\mu_H = 0.4$, $\mu_v = 0$, and $\mu_L = \mu_R = 0.2$.

I assume that the observation window is long enough so that the p 's and μ 's can be estimated accurately from the finite number of observations. This means the average click probability can be extracted from Bob's raw data and I rewrite Equation 5.19 as a function of $p(click, \ell)$ so that

$$\mu_\ell = \ln\left(\frac{1-d}{1-p(click, \ell)}\right) \quad (5.20)$$

The mean photon numbers of the constituent pulses incident at the four detectors sum to the average mean photon number of the main pulse just before it enters Bob's detection apparatus, so I can estimate the received mean photon number of a signal state as

$$\mu = \frac{4}{3} \sum_{\ell=1}^4 \ln\left(\frac{1-d}{1-p(click, \ell)}\right), \quad (5.21)$$

where the factor of $4/3$ accounts for the fact that the model system sends vacuum states 25% of the time.

I divide the data set into subsets of duration T_b and perform synchronization and sifting on each subset. Bob can record up to eight events (each of which may or may not include a detection event or dark count) assuming that the detector dead time is less than Bob's sampling time. However, because the clocks can only be synchronized to a resolution of

Bob's sampling bin width, I expect Alice's wavepacket to straddle 2 bins as illustrated in Figure 5.5, with the end bins only having a partial wavepacket. I construct a composite event from these 2 bins using a bitwise OR of the presence or absence of a click for the four detectors. The remaining six bins only contain dark counts, which can be discarded to reduce noise after I determine Δ . This amounts to detector time-gating in the post-analysis.

I assume that Bob begins recording before Alice begins transmitting, and continues to record after she stops sending, so the received data is bookended by low signal regions. I use the stitching methods discussed in Section 5.2 to identify where the transmission begins and ends, which gives a coarse approximation of the synchronization index. For a range of different string lengths N that determine the number of sampling bin widths in each synchronization subset, I examine a window of $M = 4000$ nearby potential synchronization indices. This value is chosen based on the typical precision of the coarse approximation of the synchronization given by the best-fit step function.

5.3.5 Synchronization Simulations

To verify that my algorithm returns an accurate probability of synchronization, I run 1000 simulated trials with a known synchronization index and compare the average calculated probability of synchronization $p(S_j|B_1, \dots, B_{M+N})$ to the average rate of finding the correct index, which I denote by $f(S_j|B_1, \dots, B_{M+N})$, in Figure 5.6. If my model is accurate, then $p(S_j|B_1, \dots, B_{M+N}) \sim f(S_j|B_1, \dots, B_{M+N})$, in which case I can take $p(S_j|B_1, \dots, B_{M+N})$ to be a reliable metric for quantifying my confidence in obtaining the correct Δ .

It is clear that $p(S_j|B_1, \dots, B_{M+N}) \sim f(S_j|B_1, \dots, B_{M+N})$ to within the error bars for moderate channel loss (Figure 5.6a). However, $p(S_j|B_1, \dots, B_{M+N})$ is consistently larger than $f(S_j|B_1, \dots, B_{M+N})$ for the case of zero channel loss (Figure 5.6b), a condition that is

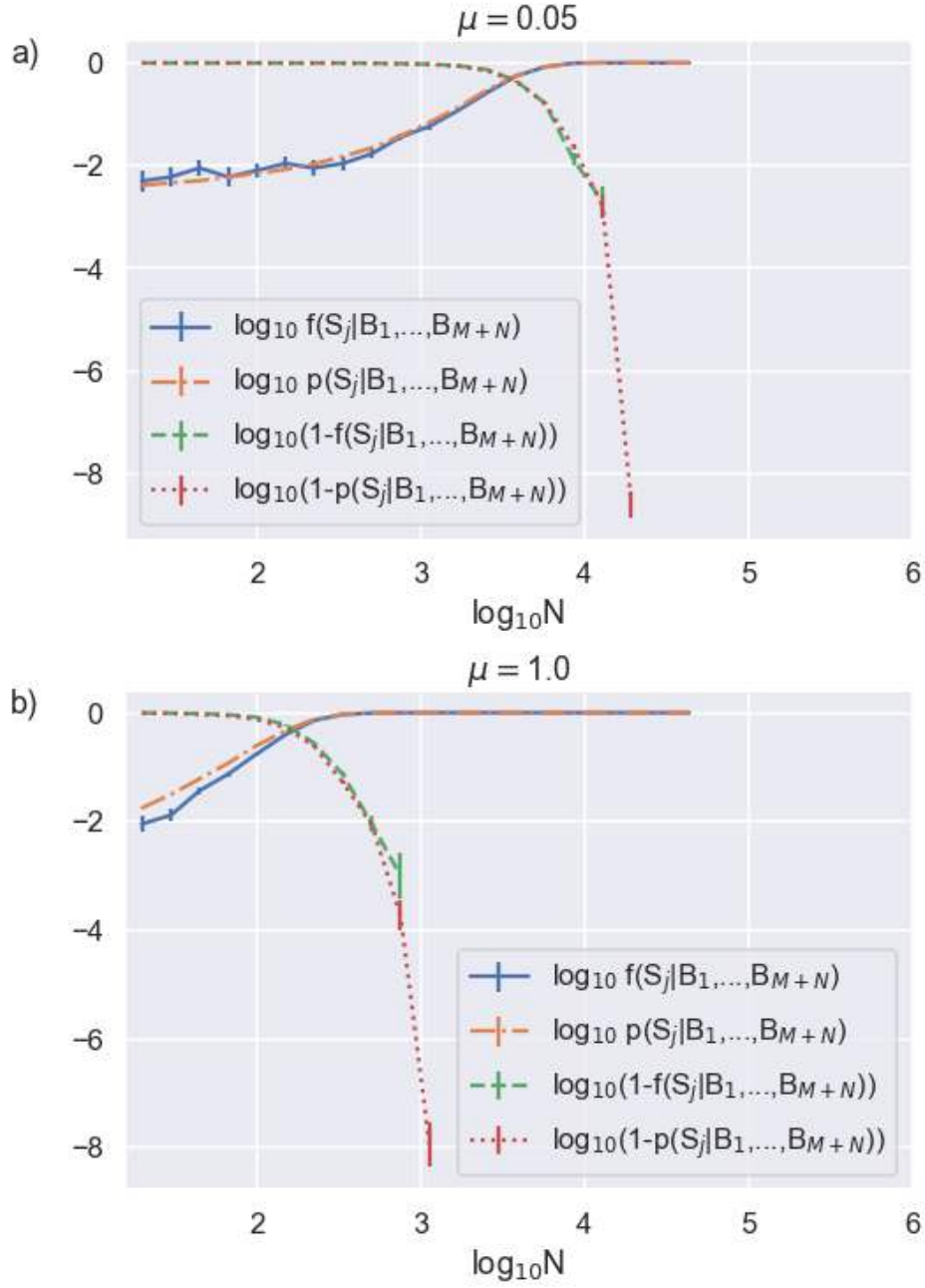


Figure 5.6: Logarithmic plot of relevant synchronization metrics as a function of compared string length for two different channel transmissions, (a) $\eta = 0.05$, corresponding to $\mu = 0.05$ and (b) $\eta = 1$, corresponding to $\mu = 1$. I also show the probability of not obtaining synchronization, which better highlights the transition to high-certainty synchronization.

unlikely to be encountered in an experiment but highlights the limitation of my algorithm. This result is not surprising given that my derivation given in Section 5.3.2 assumes low μ to arrive at Equation 5.9. Assuming a transmitted mean photon number of 1, Figure 5.6b corresponds to a zero channel loss system. This represents an upper limit on μ encountered in a typical decoy state protocol where $\mu_A \lesssim 1$ and thus, also serves as a lower bound on the accuracy of the calculated synchronization probability.

A lower received mean photon number means a lower density of detected events. Because detected events provide more information than no-detection events, a lower μ requires me to consider a larger set of sampling bin widths N to achieve the same synchronization confidence. Despite the fact that $p(S_j|B_1, \dots, B_{M+N})$ does not match $f(S_j|B_1, \dots, B_{M+N})$ as well at higher values of μ , I can still achieve equivalent average values of $f(S_j|B_1, \dots, B_{M+N})$ at lower values of N . This fact is also illustrated in Figure 5.7, where there is a direct correlation between μ and the N at which the synchronization probabilities converge to one. The higher values of μ converge at lower values of N .

Another way to view this relation between μ , N , and $p(S_j|B_1, \dots, B_{M+N})$ is to consider the string length N required to achieve a particular synchronization confidence as a function of μ as shown in Figure 5.8. For high μ and low N , I observe an approximately linear relation between $\log_{10}\mu$ and $\log_{10}N$ with a slope of ~ -1 , which means that $N \sim 1/\mu$. This can be rearranged to observe that $N\mu$, which is the average number of detections, is approximately constant in this domain, indicating that a certain number of detections are required to achieve 95% synchronization. For lower μ , where there are fewer events and dark counts play a larger role, the probability curves exhibit steeper slopes, demonstrating that synchronization becomes increasingly difficult. This data can be used to estimate whether it is possible to synchronize over an experimentally measured temporal block length T_b and, if it is possible, how low a value of μ can be tolerated while still synchronizing reliably. As a

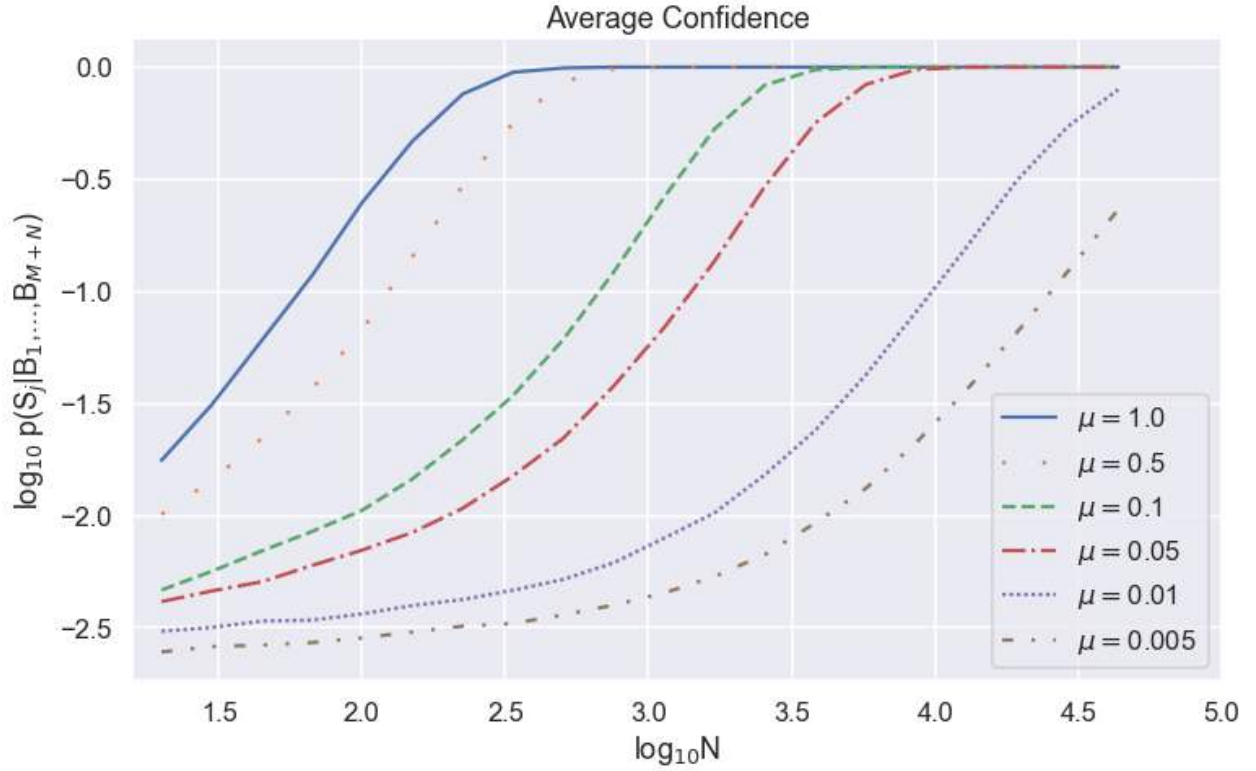


Figure 5.7: Average calculated synchronization probability as a function of string length on a logarithmic scale for different received mean photon numbers. The probability of registering a dark count during one communication bin width is $d = 8 \times 10^{-4}$.

concrete example, Bob needs 33,110 sampling bin widths, or about 4140 communication bin widths, to achieve a 95% confidence for clock synchronization for $\mu = 0.01$ and $d = 8 \times 10^{-4}$. This means the algorithm can tolerate clock drifts approaching 1 part in 4140, or $242 \mu\text{s}$ of drift per second, because my method assumes that the clock drift is much less than one communication bin width.

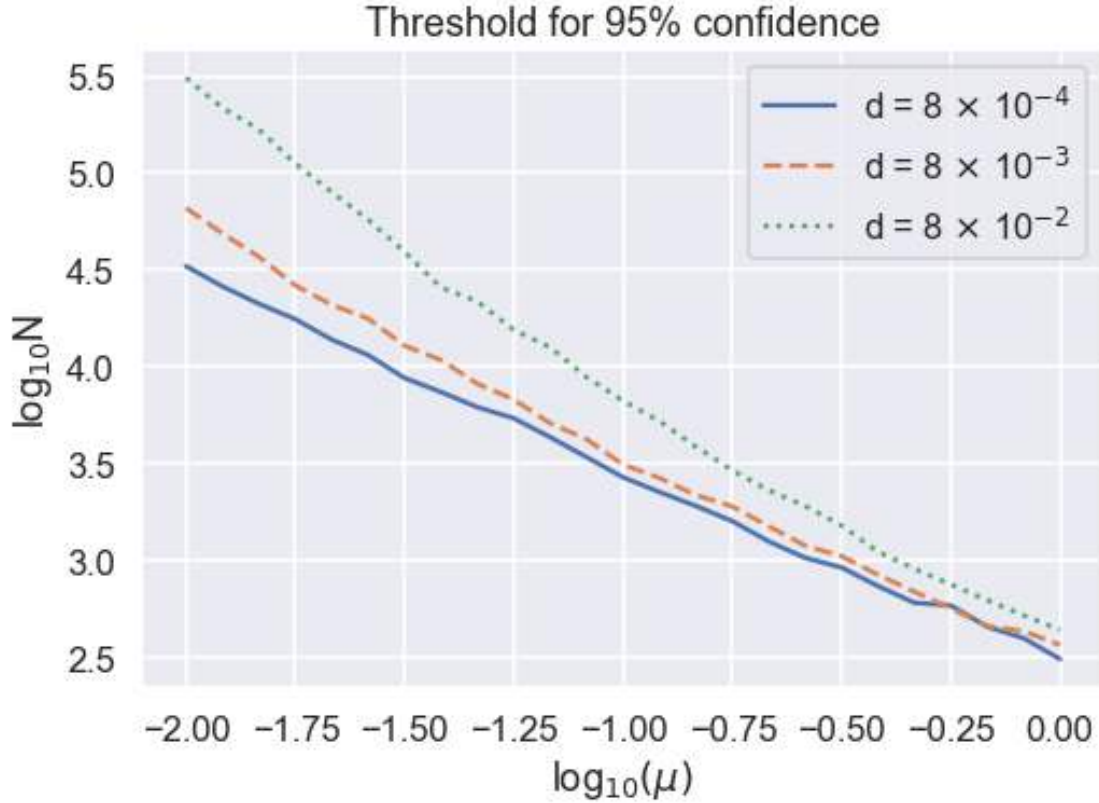


Figure 5.8: Dependence of string length threshold to achieve 95 percent synchronization confidence on received mean photon number on a logarithmic scale, parameterized by different dark count probabilities.

5.3.6 Summary

In summary, I develop a novel probabilistic approach to qubit-based clock synchronization using Bayesian analysis. By exploiting correlations between information Alice shares publicly, such as basis and decoy state choices, and Bob’s detection events, I can find the correct synchronization clock offset without sacrificing any secret key. Additionally, my algorithm is more robust to noise, loss, and clock drift in comparison to other protocols by incorporating all publicly available information using the Bayesian framework. Finally, I demonstrate that my algorithm is successful and robust using a simulated BB84 communication scheme, which confirms that my synchronization metric corresponds to the probability of synchronization, especially in the low- μ limit. my algorithm is applicable to other QKD systems that use other degrees-of-freedom of the photon for which it is possible to divulge some timing information.

5.4 Synchronization Implementations

5.4.1 Realistic Clock Drift for FPGA Systems

I now turn my attention to the application of my synchronization algorithm to the full system. The dominant source of timing drift is the difference in Alice and Bob’s temporal bin widths, as introduced in Equation 5.1. This is primarily caused by small frequency differences between the phase lock loops (PLLs) driven by different FPGAs at the same nominal frequency. Daniel and I measure the rate of clock drift between two PLLs driven by crystal oscillator clocks on DE10 Standard FPGAs in a lab setting (see Fig. 5.9), and find the average clock drift rate to be $13.5\mu\text{s}$ per second of lab time. Thus, my algorithm can tolerate realistic clock drift rates for the experiment. Figure 5.10 shows the change in the

recovered offset over time for a segment of air-to-air data. The plot shows discrete steps because the resolution of the FPGA time-tagger in the full experiment is only 10 ns. The linear drift is the dominant term, but over longer timescales more divergence from the least squares linear fit appears. However, the total drift rate is only $1.56\text{ }\mu\text{s}$ per second of lab time. Based on data collected in the lab and in the field, the drift rate can depend on the particular FPGA used in the experiment and the external temperature, in addition to other factors.

While my algorithm can effectively perform synchronization up to a certain rate of clock drift, higher rates of clock drift violate the assumption that the offset Δ can be taken as a constant over the synchronization batch interval, which leads to increased error rates or the synchronization failing altogether. However, the dominant term in the clock drift is known to be the linear term resulting from small differences in the FPGA clock frequencies. These differences can even be identified before the bulk of the data processing takes place. Because Alice’s communication bin width is 80 ns, compared with Bob’s 10 ns, I would expect in the absence of clock drift for Bob’s raw time tags to all be equal to each other modulo 8. Instead, the average mod 8 time-tag gradually cycles in the presence of clock drift (see Fig. 5.11). This provides an opportunity to calculate the linear rate of clock drift and apply it to the data preemptively, adjusting the time-tag values to account for these effects.

The simplest method I use to estimate the rate of clock drift is to graphically identify a region where the mod 8 time-tag is drifting from 0 to 8 and calculate the slope in that region. The rate at which the mod 8 time-tag changes tells me how quickly the clocks are falling 10 ns further out of synchronization. While this approach is very simple to implement, the drawbacks are that I only consider a short segment of the data, making the estimate vulnerable to noise, and that the slope of the mod 8 time-tag reverses when it crosses from 8 back to 0, which can sometimes bias the result. My subsequent and more refined approach

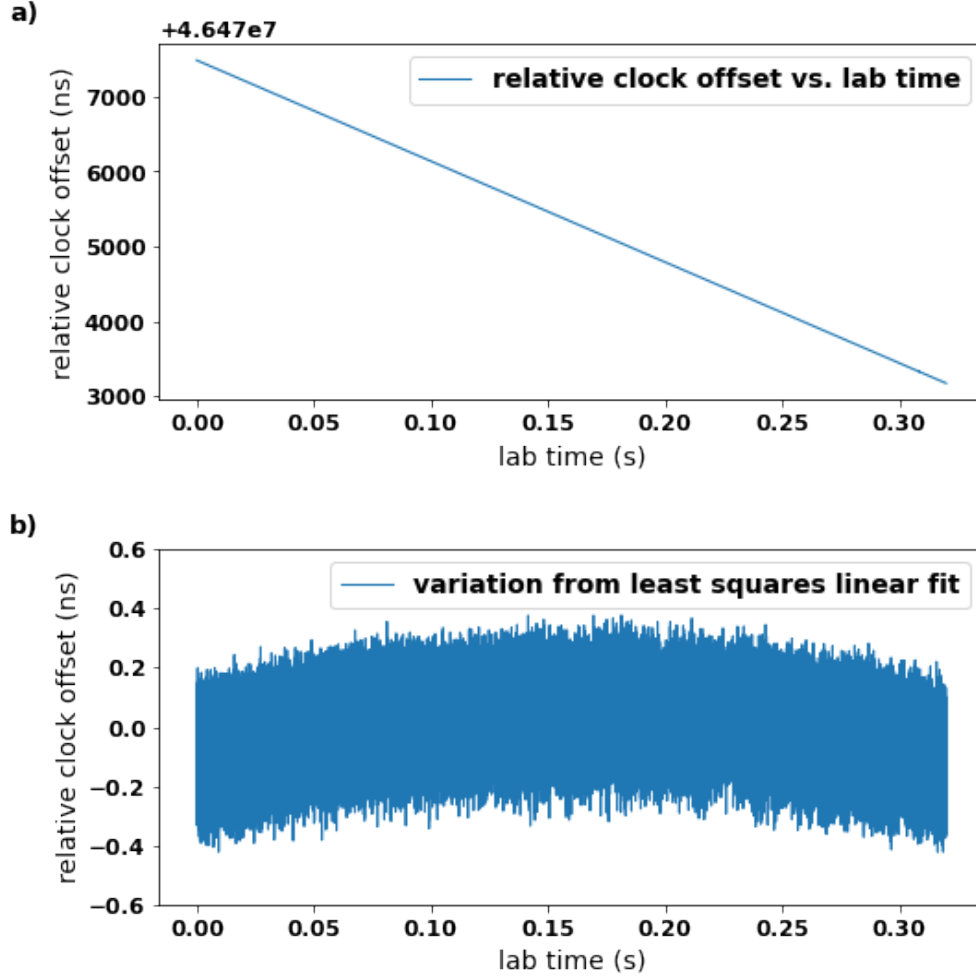


Figure 5.9: Plot of the relative clock offset between two FPGA phase lock loops (PLLs). We generate 12.5 MHz PLLs on two separate FPGAs and send the two signals to a time-tagger that measures the time differences between the two over time. There is a clear linear trend in a), indicating a small difference in the two PLL frequencies. b) shows the relative clock offset with a least squares linear fit subtracted off. Not only is the variation from the linear term small, but it is bounded within approximately 1 ns around the fit over short time scales. Over longer time scales, more deviation from the linear fit appears due to the non-linear time drift terms.

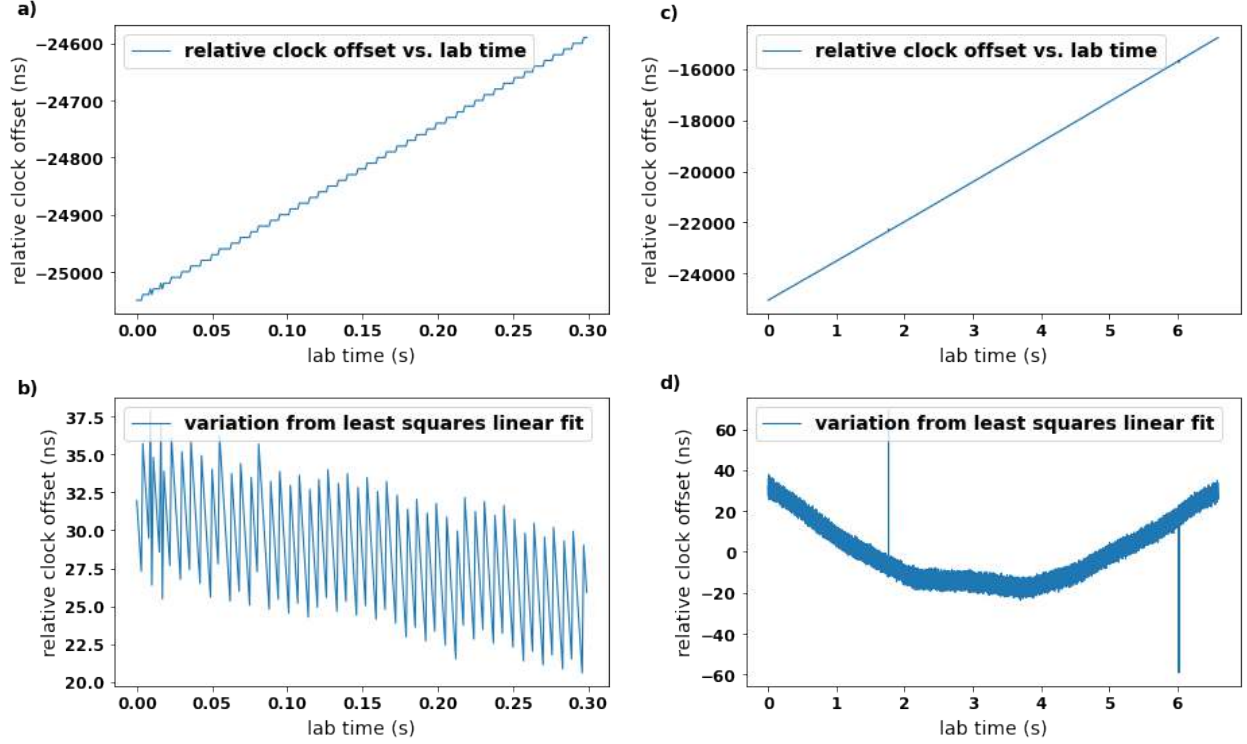


Figure 5.10: Plots of the recovered synchronization offset over time from an air-to-air QKD demonstration. a) and b) are plotted over 0.3 seconds to show the behavior on shorter time scales. Because the receiver samples at 100 MHz, the resolution of the offset recovery is limited to 10 ns. The linear part of the drift is the dominant effect. Over longer timescales, as shown in c) and d), the dominant drift term is still linear, but the non-linear drift appears in d) with greater curvature and variations from the linear fit.

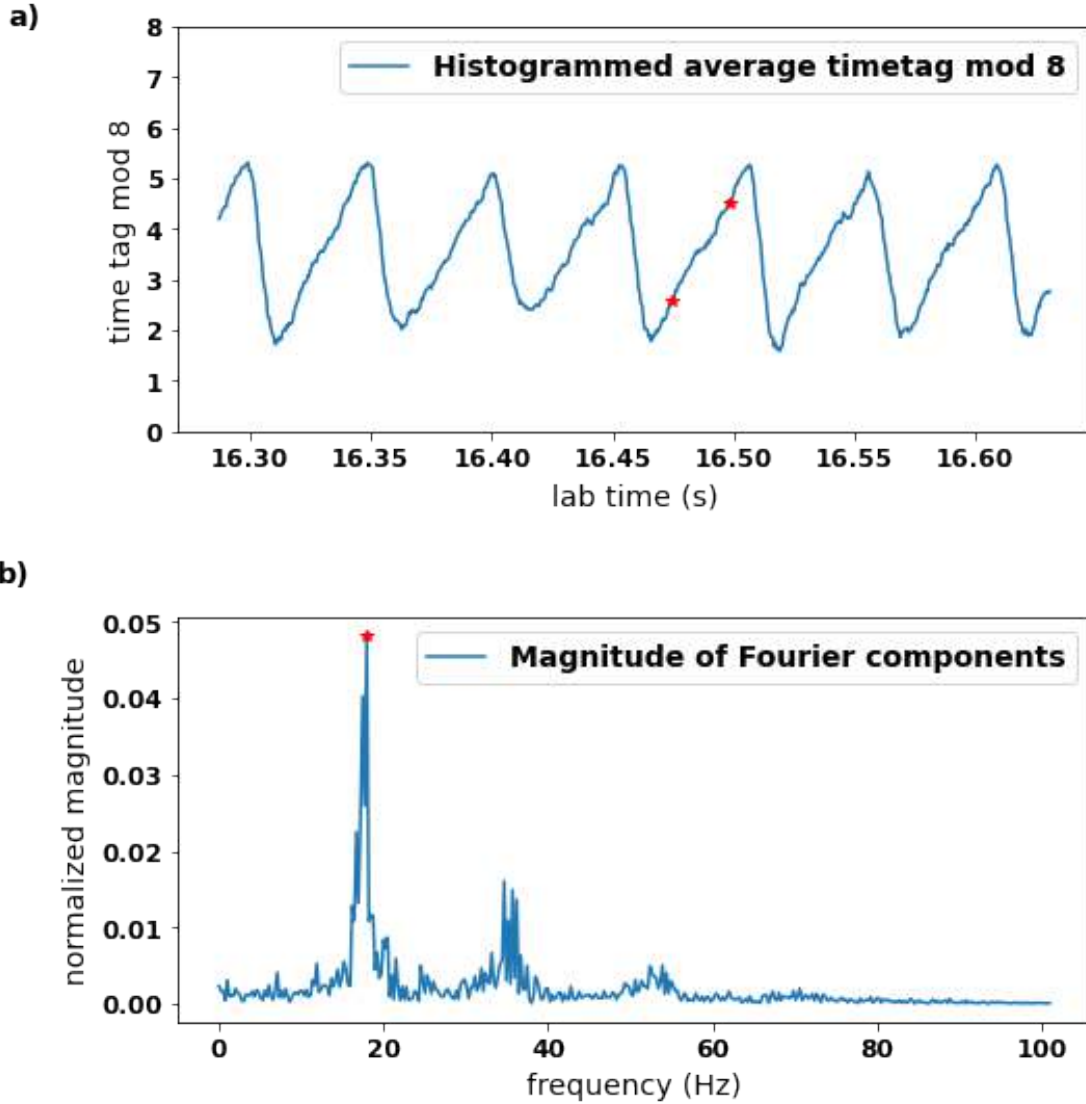


Figure 5.11: Methods for estimating the linear component of the clock drift. a) shows the drift over time of the mod 8 histogrammed time-tags for one channel of the receiver data. I estimate the clock drift by calculating the slope between the two red points. In b), I take the discrete Fourier transform of a larger segment of the mod 8 histogrammed time-tags (of which part a) is a subset), and identify the highest magnitude frequency component to estimate the rate of clock drift.

estimates the clock drift rate by taking the greatest magnitude frequency component of the discrete Fourier transform of the mod 8 time-tag drift. This tells me how quickly the clocks are falling 80 ns further out of synchronization, and it is more robust because it considers a larger segment of data. While these methods do not address the higher order clock drift terms, they allow me to remove the dominant term, leaving an easier task for the qubit-based synchronization algorithm that I apply later in the data processing.

5.4.2 Adapting the Algorithm

The fact that the offset drift is close to linear can be exploited in a few ways to improve processing speeds and diagnose errors. Attempting synchronization for the first processing batch of a blip must be very robust, because no prior information is known about the offset. I typically use a processing batch length $N = 10^5$, corresponding to 1 ms of collected data. Given the observed clock drift rates, the offset can be approximated as constant over this timescale (*i.e.*, it is very unlikely to see one sampling bin or 10 ns worth of clock drift in this time). However, the choice for the number of potential offsets M to be considered is more flexible and can in fact be updated dynamically. For the first processing batch, I choose $M = 2 \times 10^5$. This gives me a wide enough window to find the correct offset within the course synchronization produced by the stitching method. These choices result in a $N + M = 3 \times 10^5$ processing length for the FFT computations, which run efficiently due to the high factorizability of the length.

Once the offset has been found for the first processing batch, subsequent batches typically have the same or a slightly different offset. This allows me to reduce the number of candidate offsets dramatically. Here I find opportunity for significant computational advantage by computing the cross-correlation with a shifting method instead of with the FFT. The cross-

correlation can be computed directly by iterating over the M potential offsets and computing the dot product between the relevant sections of Alice's and Bob's data. The computational cost of the shifting method scales linearly with M , so in general, it is most competitive for small values of M . The computational cost of the FFT method scales better for higher values of M on the whole, though it is irregular because it depends on the factorizability of $N + M$. A comparison of the computation times for these methods for $N = 10^5$ and M ranging from 1 to 500 is shown in Figure 5.12. The FFT method used for the first processing batch (with $N = 10^5$ and $M = 2 \times 10^5$) takes ≈ 0.014 seconds per cross-correlation. When using the shifting method, $N = 10^5$ and $M = 75$ also yields ≈ 0.014 seconds.

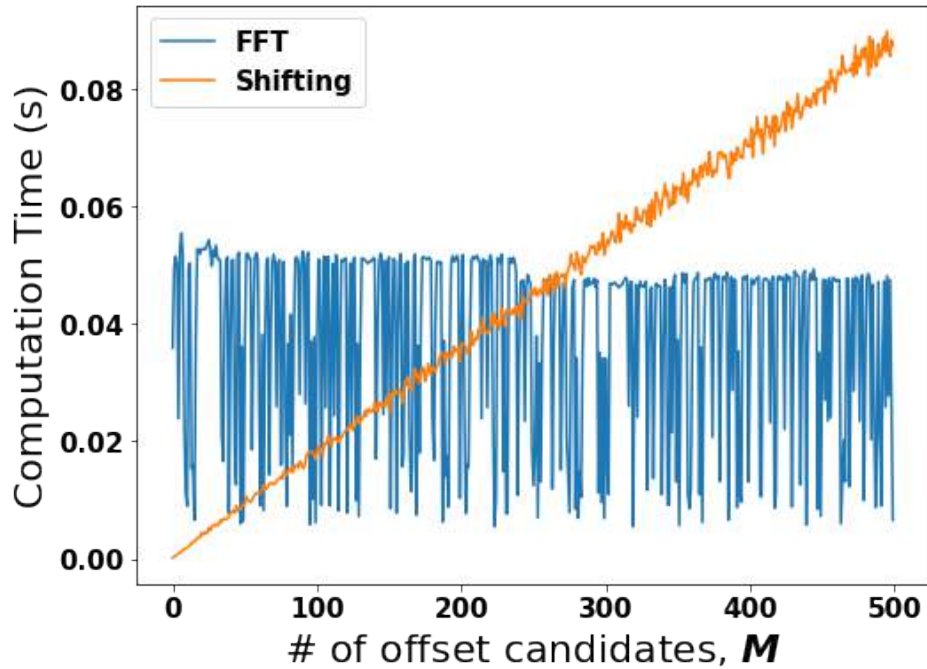


Figure 5.12: Comparison of the time required to compute a cross-correlation between Alice and Bob's data as a function of the number of offset candidates M for an FFT method vs. a shifting method. Here, I use the batch length $N = 10^5$.

Because the incoming signals from Alice come at every 8th sampling bin for Bob, the highest probability candidate offsets are typically found at multiples of 8 sampling bins sepa-

ration from the true offset. It is important when computing the synchronization probability to include viable alternative offset candidates. For example, if I were to choose $M = 8$ for a subsequent synchronization batch, the offset that is a multiple of 8 bins away from the true offset (of which there is only one) would easily stand out as the preferred one. Because the synchronization probability is re-normalized, the reported confidence would be inflated. Furthermore, with no viable alternatives, the algorithm would continue to choose that offset for subsequent batches. It is preferable for the algorithm to have some freedom to wander, because the first offset I find could be erroneous. Instead, I choose $M = 20$, which still provides improved computational speed, but ensures the presence of viable competing candidate offsets at ± 8 from the expected offset.

It is clear that the true offset shifts slowly over the course of the data set. If I observe that the recovered offset is wandering too quickly between batches, I conclude that the algorithm has lost the synchronization, and the full search with $M = 2 \times 10^5$ with the FFT method must be used to regain the correct offset. The criterion I set for this is if the offset wanders more than 12 bins over the course of two processing batches. This allows for the possibility of the algorithm briefly choosing a nearby candidate offset but returning on the following processing batch (which should not necessitate resuming a full search), as well as the gradual change in the offset resulting from the FPGA clock frequency differences.

5.4.3 Noise filtering

In addition to the post-processed time-gating outlined in Section 5.3.4, there are opportunities for significant noise filtering in the post-processing of the data. The most obvious of these is filtering out processing batches based on the calculated synchronization confidence. As previously mentioned, the synchronization algorithm outputs a probability, giving a reli-

able method for determining the confidence in having found the correct synchronization for that batch. If the wrong synchronization is found, the resulting processed data is completely randomized, and the received data is completely uncorrelated with the transmitted data. This can artificially inflate the error rates, because these errors are caused by an inability to correlate the data, rather than any actions of the eavesdropper or properties of the channel. Thus, I can noticeably improve the error rates by filtering out data with low synchronization confidence. Data regions with low synchronization confidence are often sparse signal regions with low throughput due to pointing-and-tracking issues. A consequence of this is that the regions I filter out often have few transmitted bits in the first place, which means little data is lost. I choose 95% as a reasonable value for this synchronization confidence filtering threshold.

The next filter I apply is based on an estimation of the signal-to-noise ratio in each processing batch. In the absence of an eavesdropper, with ideal optics sorting, the quantum bit error rate (QBER) caused by random noise is half of the noise fraction. Data regions with a QBER over 11% do not contribute to a secure key, but it is not allowed to filter based on the QBER directly because this would cause an underestimation of the extent of the Eavesdropper's activities and it cannot be calculated without publishing additional data that subsequently cannot be used for generating a secure key. However, I can estimate the noise fraction using Bob's data and Alice's published information. Alice must publish which times she sends the decoy and vacuum states as part of the decoy state analysis. Bob can estimate the noise fraction by computing the fraction of all his received events that arrive when vacuum is sent and comparing that to the fraction of the time vacuum is sent by Alice. In the extreme case when the transmission is zero, the fraction of all bits detected by Bob that arrive when vacuum is sent should be approximately equal to the fraction of Alice's states that are vacuum states. If I find using this method that a processing batch has a noise

fraction is greater than 0.22, I filter it out to improve the overall data quality and ultimate key length.

Chapter 6

Discussion and Results

Having introduced the system hardware, procedures, and data processing, I now present the results of the QKD data collection along with figures-of-merit. I highlight data sets that demonstrate the progression of difficulty in data collection from the laboratory to field experiments. These configurations of interest are **tabletop**, **ground-to-ground**, **air-to-air**, **air-to-car**, **car-to-car (low speed)**, and **car-to-car (high speed)**. The figures-of-merit and associated plots give insight into the particular challenges encountered in each configuration and the degree to which the system is able to address them. Table 6.1 gives a summary of the key metrics and figures-of-merit for the experiments in all the configurations discussed in this chapter.

Daniel Sanchez-Rosales, Andrew Conrad, Samantha Isaac, and I collaborated to collect these data sets. The plots and supporting figures-of-merit (*e.g.*, QBER, synchronization confidence, etc.) are outputs of my data analysis. The finite secure key rate, which is the primary figures-of-merit for these data sets, was calculated by Lars Kamin. In this chapter, “we” refers to the collective effort of Daniel Sanchez-Rosales, Andrew Conrad, Samantha Isaac, and I as we designed, implemented, and collected data with our mobile QKD system.

The key figure-of-merit in these experiments is the finite secure key rate. Our collabo-

	Collection Time (s)	Raw Key Rate (kbits/s)	Average QBER (%)	Finite Secure Key Rate (kbits/s)
Tabletop	93.7	501	3.06	41.3
Ground-to-Ground	209.1	165	2.83	6.05
Air-to-Air	166.4	247	2.21	7.98
Air-to-Car	129.8	113	3.31	1.52
Car-to-Car (5 mph)	93.7	235	2.57	19.6
Car-to-Car (70 mph)	144.2	9.26	3.74	0

Table 6.1: Summary table of the relevant figures-of-merit for QKD demonstrations in different mobile platform configurations.

rators at the University of Waterloo developed a security framework for the protocol and have calculated secure key rates for the collected data sets. This calculation is a complex optimization problem, but some of the factors that contribute to high finite secure key rate include large total number of signals sent, high channel transmission, low fluctuation of measurement statistics (including transmission), and low QBER. Our team made a large fraction of the design choices with these heuristics in mind, as we made efforts to maximize the finite secure key rate. The transmitter state sending fractions also hold implications for finite key generation. We used several different state sending fractions in these data sets as our understanding of the optimal fractions to use evolved between these experiments.

6.1 Tabletop QKD

The most basic rendition of the experiment is the tabletop version. While not a groundbreaking demonstration in its own right, it demonstrates the performance of the system design in a controlled environment with consistent high transmission between Alice and Bob, as the

locking does not have to rely on the pointing-and-tracking system. This provides a point of reference that contextualizes the figures-of-merit and the variability in performance found in the more complex configurations. This data set uses the following state sending fractions: signal R: 121/256, signal L: 121/256, signal H: 2/256, decoy R: 2/256, decoy L: 2/256, decoy H: 2/256, vacuum: 6/256. The mean photon number for the signal states exiting the transmitter is 0.427.

The first plot of interest (see Fig. 6.1) is the time-averaged counts in each channel over the course of the data collection with an integration time of 0.1 s. In this tabletop data, these counts are virtually constant except for the periodic dips, which are built into the transmitter data structure (see Section 5.1). The average values of these count rates demonstrate the total throughput of the system and its ability to generate raw key. The differences between the various count rates are a function of the relative sending frequency of the input states, the mean photon number balancing, the state and measurement fidelity, and any mismatch in detector efficiency. The data blips serve as the basic processing unit and course stitching tool (see Section 5.2), and accordingly, the processed data region begins and ends at a dip between blips. This data set includes 13 blips.

The second plot of interest (see Fig. 6.2) is the time-averaged synchronization confidence as a function of time with an integration time of 0.1 s. This plot only subtends the processed data region of the previous plot. Here, I mainly highlight the high quality of the data that allows such high average synchronization confidence, while on spottier data sets it highlights the robustness of the algorithm in achieving synchronization in spite of the quality of the data. Ultimately, this plot provides a visual representation of which regions on average pass the 95% threshold required for the data to be processed further. In this data set, only 0.078% of the processing batches fail to meet that threshold. Among the batches that met the synchronization threshold, none failed to meet the signal-to-noise criterion described in

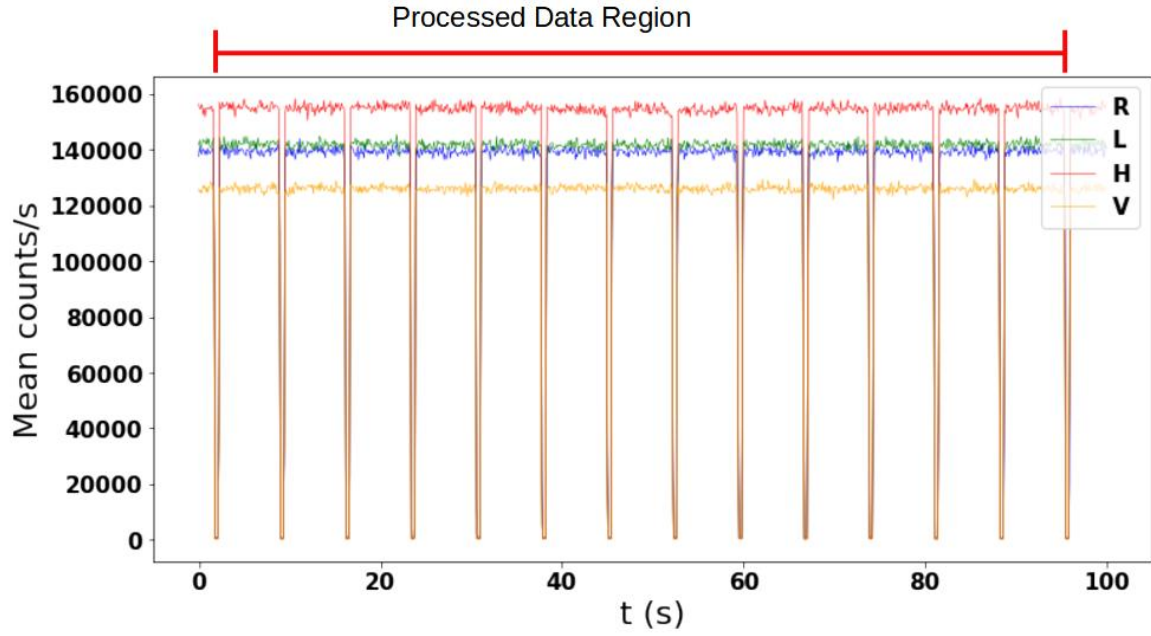


Figure 6.1: The mean counts per second for a tabletop QKD data set taken on August 8th, 2023 with a 0.1 s integration time. All four channels exhibit the expected 7.2 second duty cycle, including the 0.5 second period where the transmission stops and only background counts are observed. The background is especially low due to the controlled laboratory environment, and the overall throughput is consistent and high due to the stable alignment on an optical table. The processed data region spans 13 blips.

Section 5.4.3.

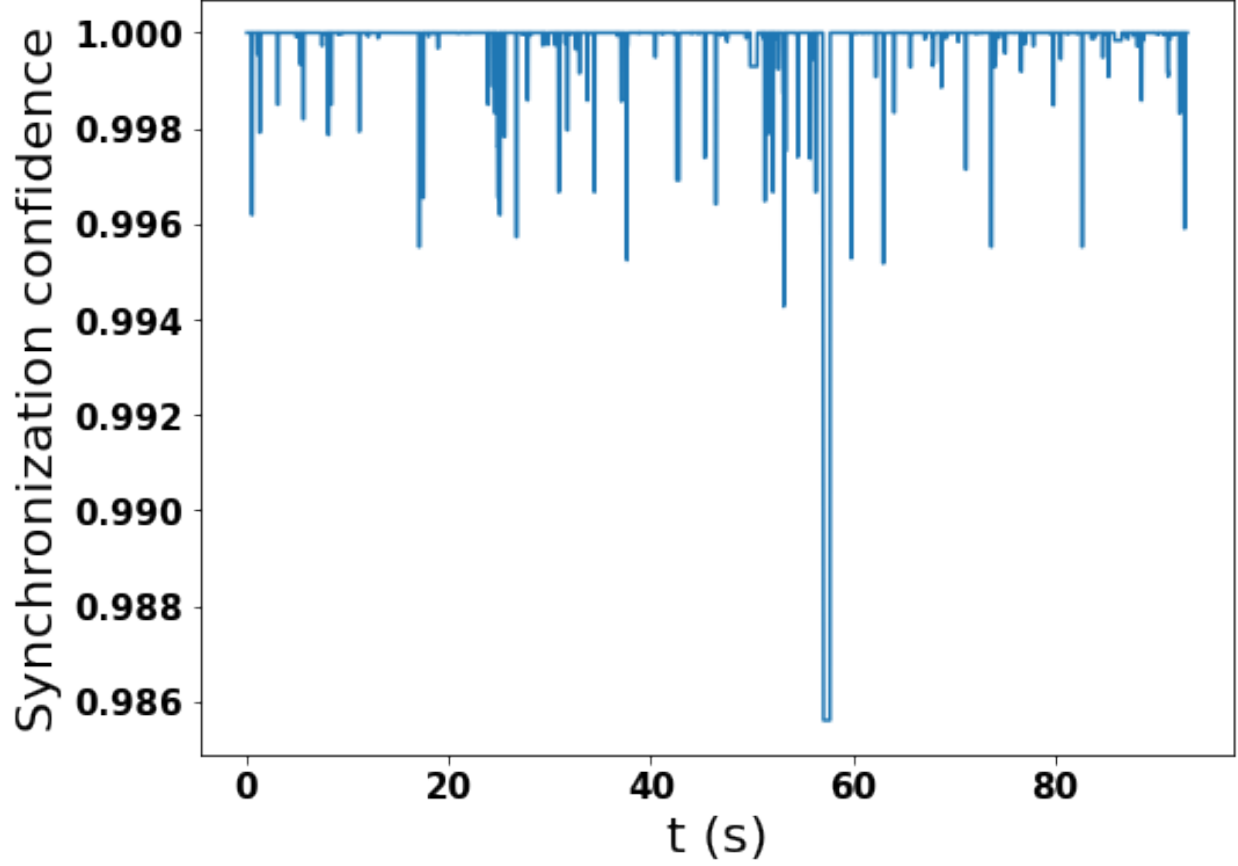


Figure 6.2: The time-averaged synchronization confidence per processing batch for a tabletop QKD data set taken on August 8th, 2023 with a 0.1 s integration time. Although the average does not fall below the 95% synchronization rejection threshold, this does not guarantee that no individual processing batches are rejected.

The third plot of interest (see Fig. 6.3) is the QBER as a function of time. This plot shows the variability of the error rates over the course of the data set. While the QBERs do not change systematically over time in the tabletop dataset, they do fluctuate due to shot noise in the measurement. The H QBER fluctuations are greater in magnitude because there are fewer H states sent in this dataset compared to L and R. The average value of the QBERs is determined by the state fidelity, measurement fidelity, and the background noise. These

QBERs are a key metric in determining how much secure key rate can be generated, and they fall well below the 11% threshold below which secure key can be generated asymptotically.

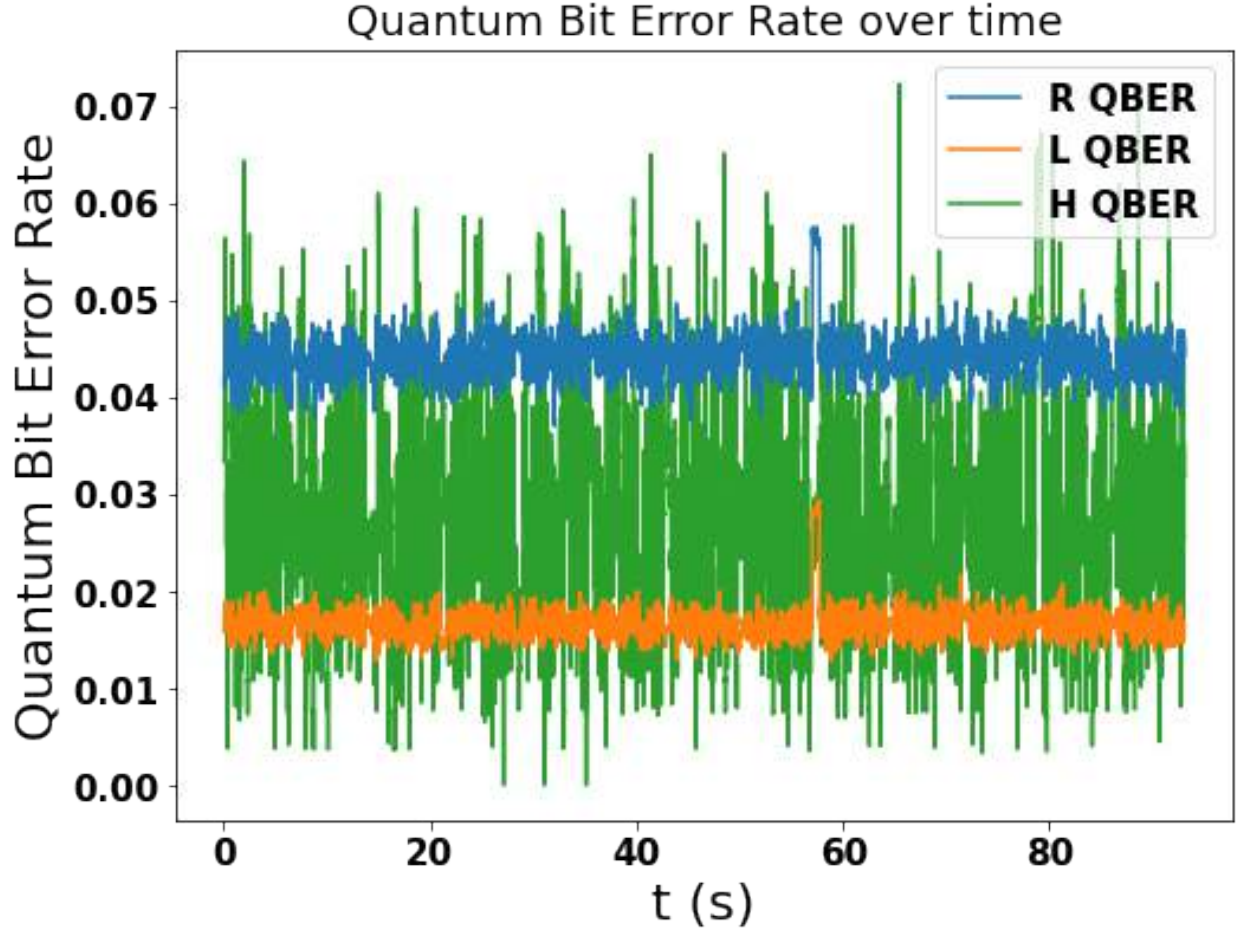


Figure 6.3: The time-averaged QBER over the data collection for a tabletop QKD data set taken on August 8th, 2023 with a 0.1 s integration time. These values fall well below the 11% threshold below which asymptotic secure key can be generated.

Ultimately, we are able to achieve a finite secure key rate of 41.3 kbits/s with the tabletop QKD demonstration. This data set has a relatively short collection time, resulting in a lower number of transmitted states. This limits the finite secure key rate to some extent, but overall this finite secure key rate sets a benchmark for that which can be achieved in less

ideal configurations.

6.2 Ground-to-Ground QKD

The ground-to-ground QKD experiment at 10 m separation is the next proving ground of the system. This configuration demonstrates the viability of the system with the pointing-and-tracking system engaged in an outdoor setting, but without dealing with the extreme vibrations that make locking so challenging in the mobile QKD configurations. This provides the next point of reference to contextualize the plots and figures-of-merit, giving a floor for the performance of the full system sans vibrations. This data set uses the following state sending fractions: signal R: 51/256, signal L: 51/256, signal H: 78/256, decoy R: 14/256, decoy L: 14/256, decoy H: 21/256, vacuum: 27/256. The mean photon number for the signal states is 0.44.

The plot of the time-averaged counts (with a 0.1 s integration time) shown in Figure 6.4 exhibits more variability in transmission, amid the 7.2 s duty cycle, than its tabletop counterpart due to the innate variation in the pointing-and-tracking control architecture. Nevertheless, the total throughput is still relatively stable and high compared to the mobile QKD demonstrations that follow. The background is higher than in the tabletop experiment due to increased background light outdoors and the presence of beacon light. As before, the differences between the various count rates are a function of the relative sending frequency of the input states, the mean photon number balancing, the state and measurement fidelity, and any mismatch in detector efficiency. The processed data region here is 29 blips, limited only by the amount of data saved by the transmitter.

The time-averaged synchronization over the data set (with an integration time of 0.1 s) seen in Figure 6.5 shows how well the synchronization algorithm performs over different

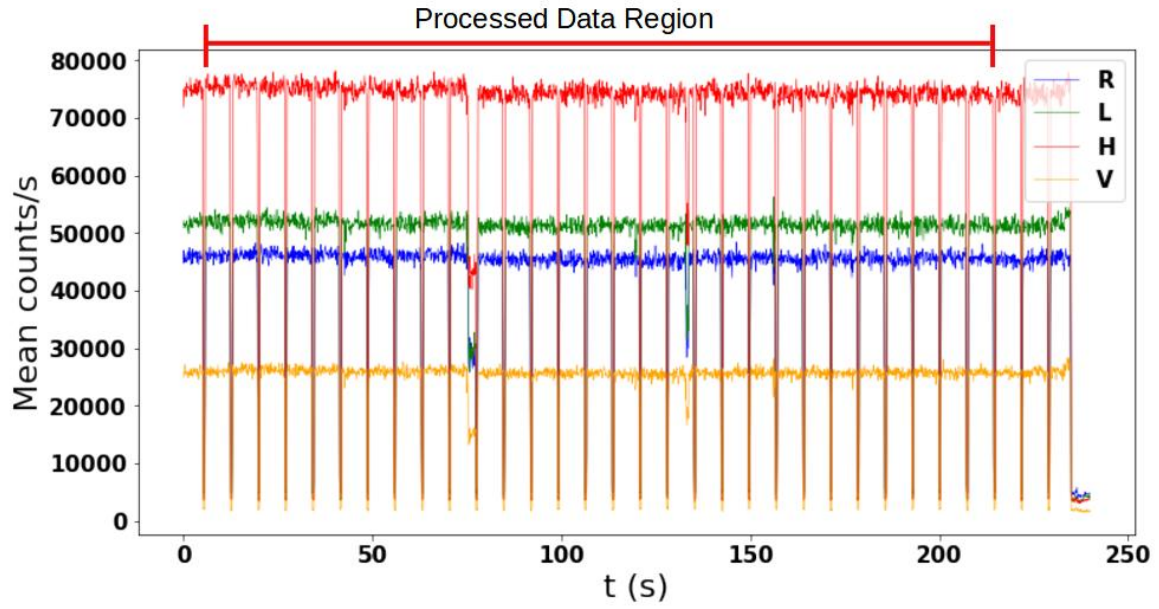


Figure 6.4: The mean counts per second for a ground-to-ground QKD data set taken on November 14th, 2023 with a 0.1 s integration time. In addition to the expected 7.2 second duty cycle, there is minor variability in the transmission due to fluctuations from the control architecture. The background is higher than the tabletop, and more in line with the full field tests because it is outdoors and the pointing-and-tracking beacons are engaged. The processed data region spans 29 blips.

sections of this dataset. In this data set there is high average synchronization confidence for the entire duration, and only 0.376% of the batches fail to meet the 95% synchronization confidence threshold. Among the batches that met the synchronization threshold, only 0.11% failed to meet the signal-to-noise criterion.

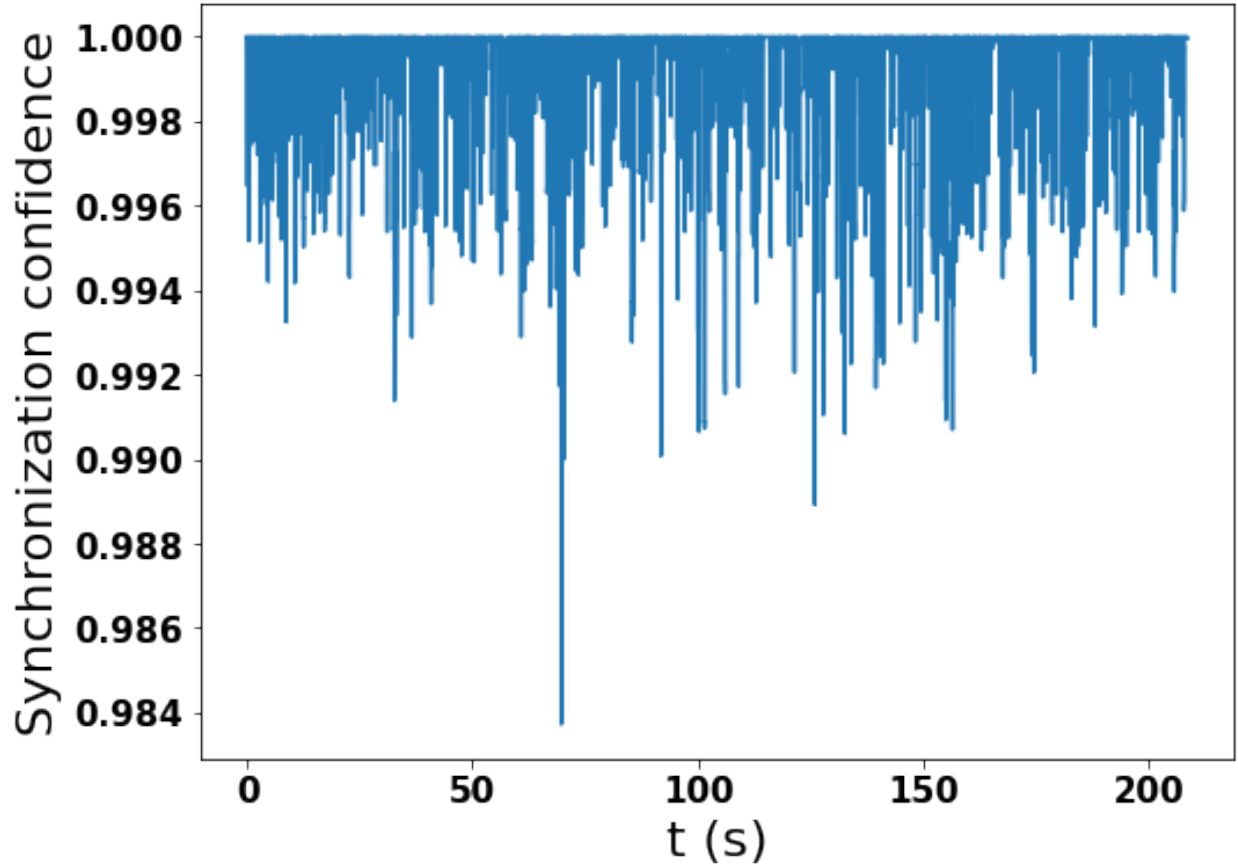


Figure 6.5: The time-averaged synchronization confidence per processing batch for a ground-to-ground QKD data set taken on November 14th, 2023 with a 0.1 s integration time. Although the average does not fall below the 95% synchronization rejection threshold, this does not guarantee that no individual processing batches are rejected.

The plot of QBER over time (see Fig. 6.6) shows the variability of the error rates over the course of the data set. The fluctuations shown here are caused by shot noise in the measurement as well as variability in the signal-to-noise ratio cause by the pointing-

and-tracking system. The shot noise in the H QBER measurement is less compared with the tabletop data set because there are many more H states sent with these state sending fractions. The QBERs are a key metric in determining how much secure key rate can be generated, and they fall well below the 11% threshold below which secure key can be generated asymptotically.

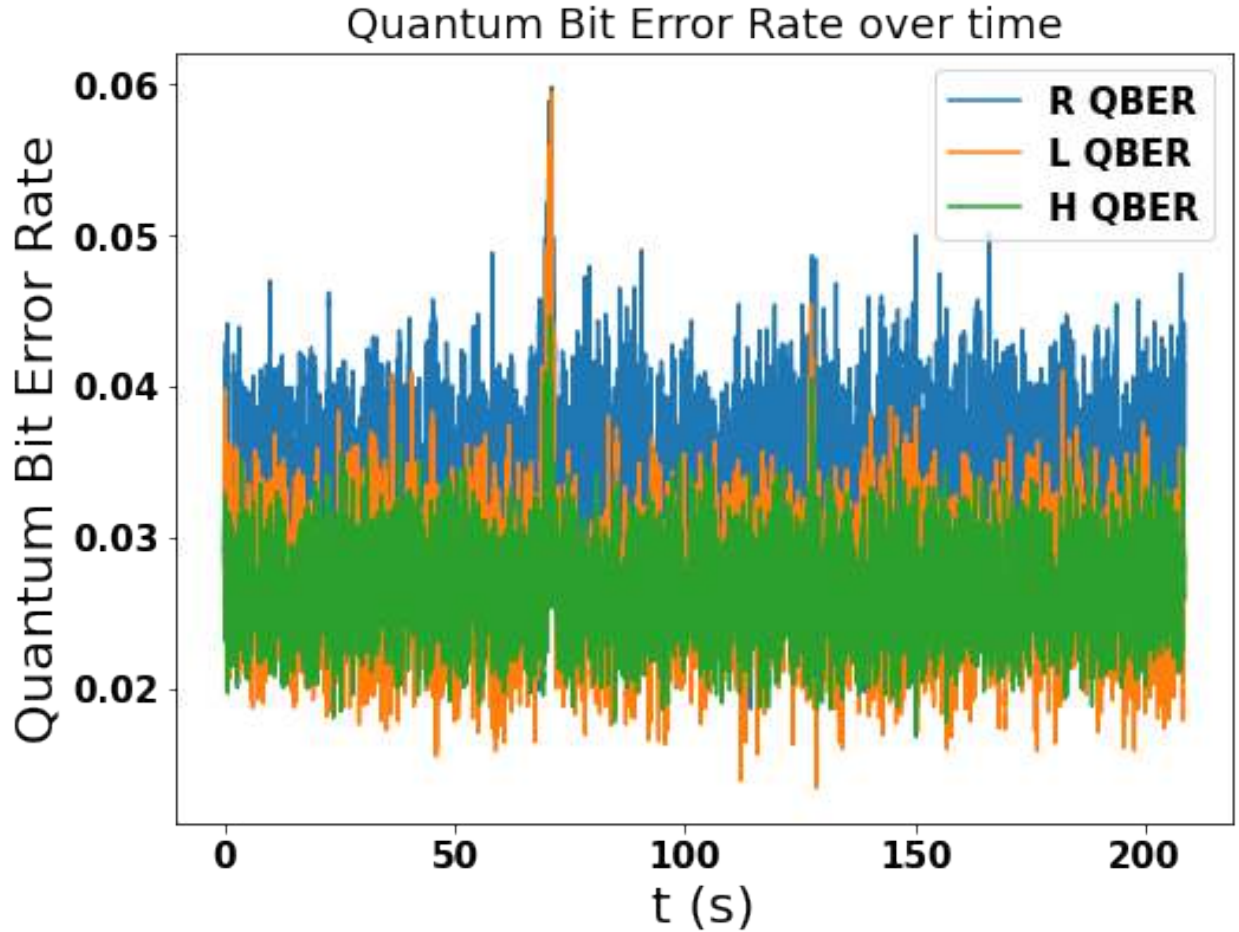


Figure 6.6: The time-averaged QBER over the data collection for a ground-to-ground QKD data set taken on November 14th, 2023 with a 0.1 s integration time. These values fall well below the 11% threshold below which asymptotic secure key can be generated.

Ultimately, we are able to achieve a finite secure key rate of 6.05 kbits/s with the ground-

to-ground QKD demonstration. This data set has a relatively long collection time, resulting in a higher number of transmitted states. This boosts the finite secure key rate to some extent, but does not exceed the key rate of the more ideal tabletop experiment. This provides a soft upper limit for the finite secure key that can be achieved in the more challenging mobile QKD experiments, while still incorporating the effects of the pointing-and-tracking system.

6.3 Air-to-Air QKD

The air-to-air QKD experiment at 10 m separation is the first of the four mobile QKD configurations, and most of our team’s design work emphasizing low SWaP was with this configuration in mind. A full system demonstrating air-to-air quantum communication paves the way toward mobile quantum networks. This dataset contends with the full array of field testing challenges, including non-ideal or variable temperature, humidity, wind, ambient light, and in-flight vibration. The best air-to-air data our team collected used an early version of the transmitter and receiver FPGA code with the following state sending fractions: signal R: 4/16, signal L: 4/16, signal H: 4/16, decoy R: 1/16, decoy L: 1/16, decoy H: 1/16, vacuum: 1/16. The mean photon number for the signal states is 0.776.

The plot of the time-averaged counts (with a 0.1 s integration time) shown in Figure 6.7 exhibits a higher level of variability in transmission, amid the 5.37 s duty cycle, than its ground-to-ground counterpart due to the vibrations and motion associated with outdoor drone flight. The transmission in this dataset is best described as approximately constant with short dropout regions where the pointing-and-tracking system momentarily loses coupling between the drones before returning to the same level. This leaves a significant fraction of “good” data regions that have high consistent throughput and that can most effectively generate secure key. The background is similar to that of the ground-to-ground dataset,

showing that the noise filtering is robust in the full system demonstration. As before, the differences between the various count rates are a function of the relative sending frequency of the input states, the mean photon number balancing, the state and measurement fidelity, and any mismatch in detector efficiency. The processed data region here is 31 blips, limited by the amount of data saved by the transmitter.

The time-averaged synchronization over the data set (with an integration time of 0.1 s) seen in Figure 6.8 shows how well the synchronization algorithm performs over different sections of this dataset. The synchronization confidence remains high in the regions with high transmission, but drops off in the regions with low transmission. This is because the qubit-based synchronization algorithm depends on the data itself and low transmission areas present fewer differences between the true synchronization point and the other candidates. Only 11.2% of the batches fail to meet the 95% synchronization confidence threshold, and among the batches that met the synchronization threshold, only 5.36% failed to meet the signal-to-noise criterion.

The plot of QBER over time (see Fig. 6.9) shows the variability of the error rates over the course of the data set. The fluctuations shown here are caused by shot noise in the measurement as well as variability in the signal-to-noise ratio cause by drone motion and vibrations countered by the pointing-and-tracking system. Accordingly, the spikes in the QBER largely correspond to the dips in transmission that affect the signal-to-noise ratio. The average QBER values are low compared to the 11% threshold for asymptotic secure key, which indicates the capacity of this dataset to generate a high finite secure key rate.

Ultimately, we are able to achieve a finite secure key rate of 7.98 kbits/s with the air-to-air QKD demonstration. This is in some ways the most challenging demonstration, given the difficulties posed by the environmental factors and the drone motion and vibration affecting both the transmitter and receiver. These factors make it all the more impressive that this

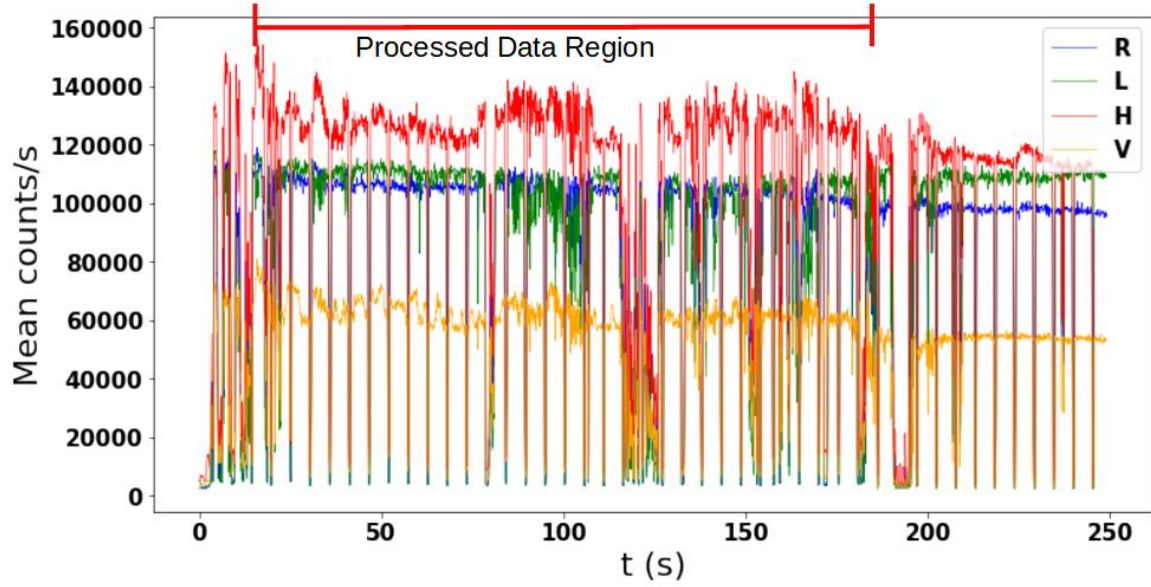


Figure 6.7: The mean counts per second for an air-to-air QKD data set taken on November 2nd, 2022 with a 0.1 s integration time. In addition to the expected 5.37 second duty cycle used on that version of the transmitter FPGA code, there is significant variability in the transmission as the control architecture contends with the vibrations and motion endemic to outdoor drone flight. The background is similar to that of the ground-to-ground dataset, showing that the noise filtering is robust in the full system demonstration. The processed data region spans 31 (5.37 s long) blips.

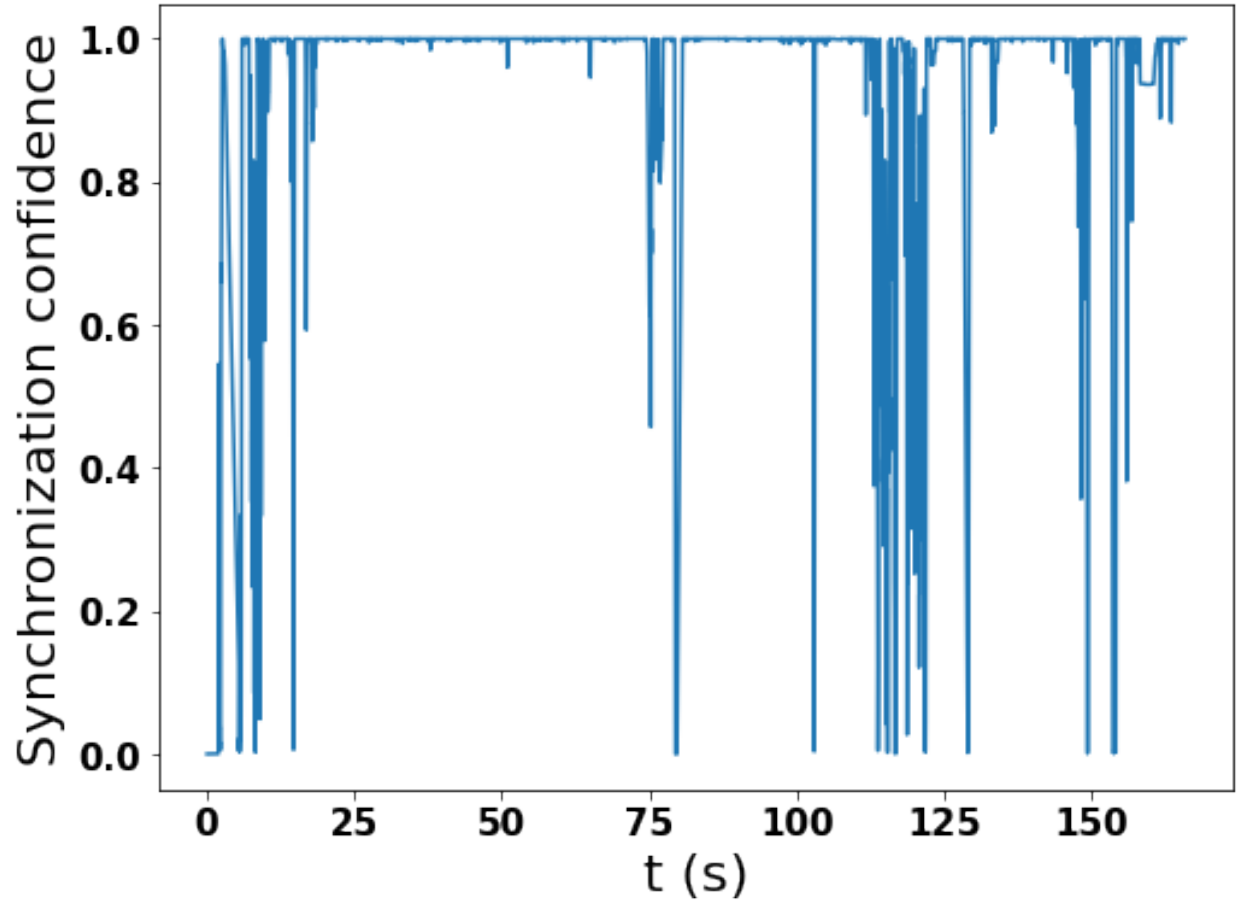


Figure 6.8: The time-averaged synchronization confidence per processing batch for an air-to-air QKD data set taken on November 2nd, 2022 with a 0.1 s integration time. The confidence drops particularly in the regions where the transmission drops.

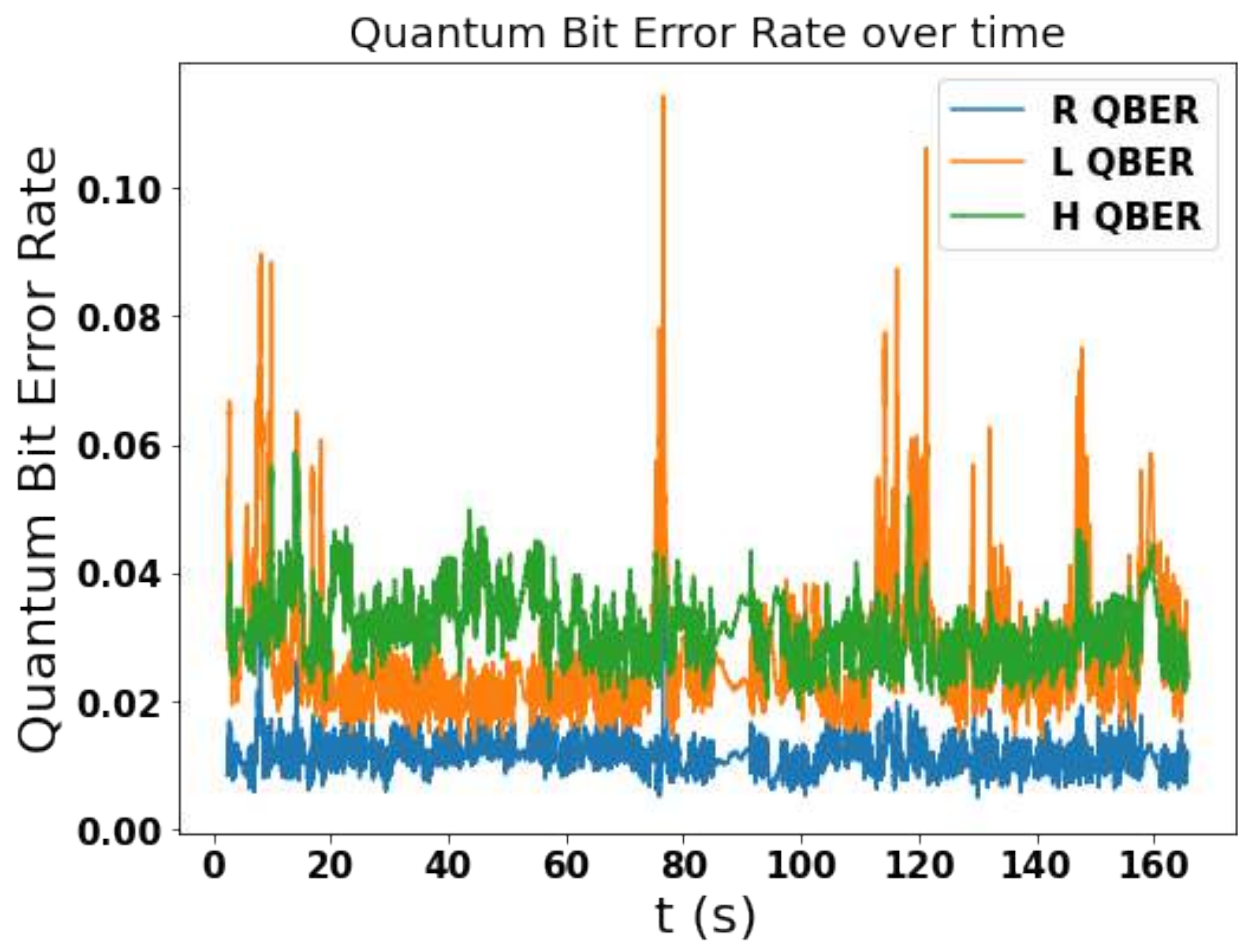


Figure 6.9: The time-averaged QBER over the data collection for an air-to-air QKD data set taken on November 2nd, 2022 with a 0.1 s integration time. These values are low compared to the 11% threshold below which asymptotic secure key can be generated.

configuration achieved higher key rate than the relatively simple ground-to-ground configuration (though it largely owes this advantage to the high mean photon number achieved for this data set). To my knowledge, this is the first known demonstration of a fully airborne quantum communication system.

6.4 Air-to-Car QKD

The air-to-car QKD experiment at 10 mph speeds and 10 m separation is the second of the four mobile QKD configurations. This demonstration paves the way toward integration between airborne and vehicle quantum nodes, which could be an asset to future mobile quantum networks. This dataset contends with most of the same testing challenges as the air-to-air demonstration, including non-ideal or variable temperature, humidity, wind, ambient light, and in-flight vibration. However, the air-to-car only has to deal with flight instability on the transmitter, while the receiver resides on the relatively more stable vehicle platform. This dataset uses the following state sending fractions: signal R: 51/256, signal L: 51/256, signal H: 78/256, decoy R: 14/256, decoy L: 14/256, decoy H: 21/256, vacuum: 27/256. The mean photon number for the signal states is 0.44.

The plot of the time-averaged counts (with a 0.1 s integration time) shown in Figure 6.10 exhibits a higher level of variability in transmission, amid the 7.2 s duty cycle, than its ground-to-ground counterpart due to the vibrations and motion associated with outdoor drone flight, vehicle vibration, and the 10 mph motion of both systems. In this demonstration, the pointing-and-tracking system lost locking just before data collection began and did not regain locking until around 100 seconds into the experiment. The transmission in this dataset after the locking re-engaged is best described as approximately constant with short dropout regions where the pointing-and-tracking system momentarily loses coupling

between the transmitter and receiver before returning to the same level. This leaves a significant fraction of “good” data regions that have high consistent throughput and that can most effectively generate secure key. The background is somewhat higher than that of the ground-to-ground and air-to-air datasets because of vehicle lights. However, these effects are not overwhelming, showing that the noise filtering is robust in the full system demonstration. As before, the differences between the various count rates are a function of the relative sending frequency of the input states, the mean photon number balancing, the state and measurement fidelity, and any mismatch in detector efficiency. The processed data region here is 18 blips, limited by the amount of data saved by the transmitter and the delay before the locking became fully engaged.

The time-averaged synchronization over the data set (with an integration time of 0.1 s) seen in Figure 6.11 shows how well the synchronization algorithm performs over different sections of this dataset. The synchronization confidence remains high in the regions with high transmission, but drops off in the regions with low transmission. This is because the qubit-based synchronization algorithm depends on the data itself and low transmission areas present fewer differences between the true synchronization point and the other candidates. Only 5.25% of the batches fail to meet the 95% synchronization confidence threshold, and among the batches that met the synchronization threshold, only 2.73% failed to meet the signal-to-noise criterion.

The plot of QBER over time (see Fig. 6.12) shows the variability of the error rates over the course of the data set. The fluctuations shown here are caused by shot noise in the measurement as well as variability in the signal-to-noise ratio caused by drone motion and vibrations, the car vibrations, and the motion of both platforms. Accordingly, the spikes in the QBER largely correspond to the dips in transmission that affect the signal-to-noise ratio, but the wide variance in these values is also driven by greater short-term variation in the

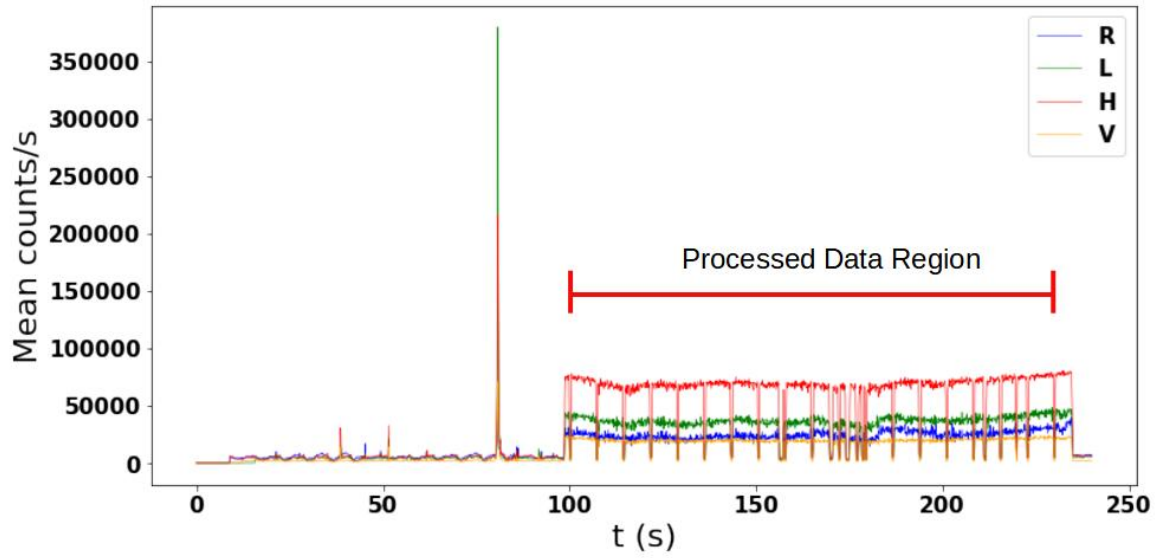


Figure 6.10: The mean counts per second for an air-to-car QKD data set taken on November 14th, 2023 with a 0.1 s integration time. In addition to the expected 7.2 second duty cycle, there is significant variability in the transmission as the control architecture contends with the vibrations and motion endemic to outdoor drone flight and the vehicle vibration. The background is slightly higher than the air-to-air data due to the presence of car lights, but not enough to significantly affect the signal-to-noise ratio. The processed data region spans 18 blips.

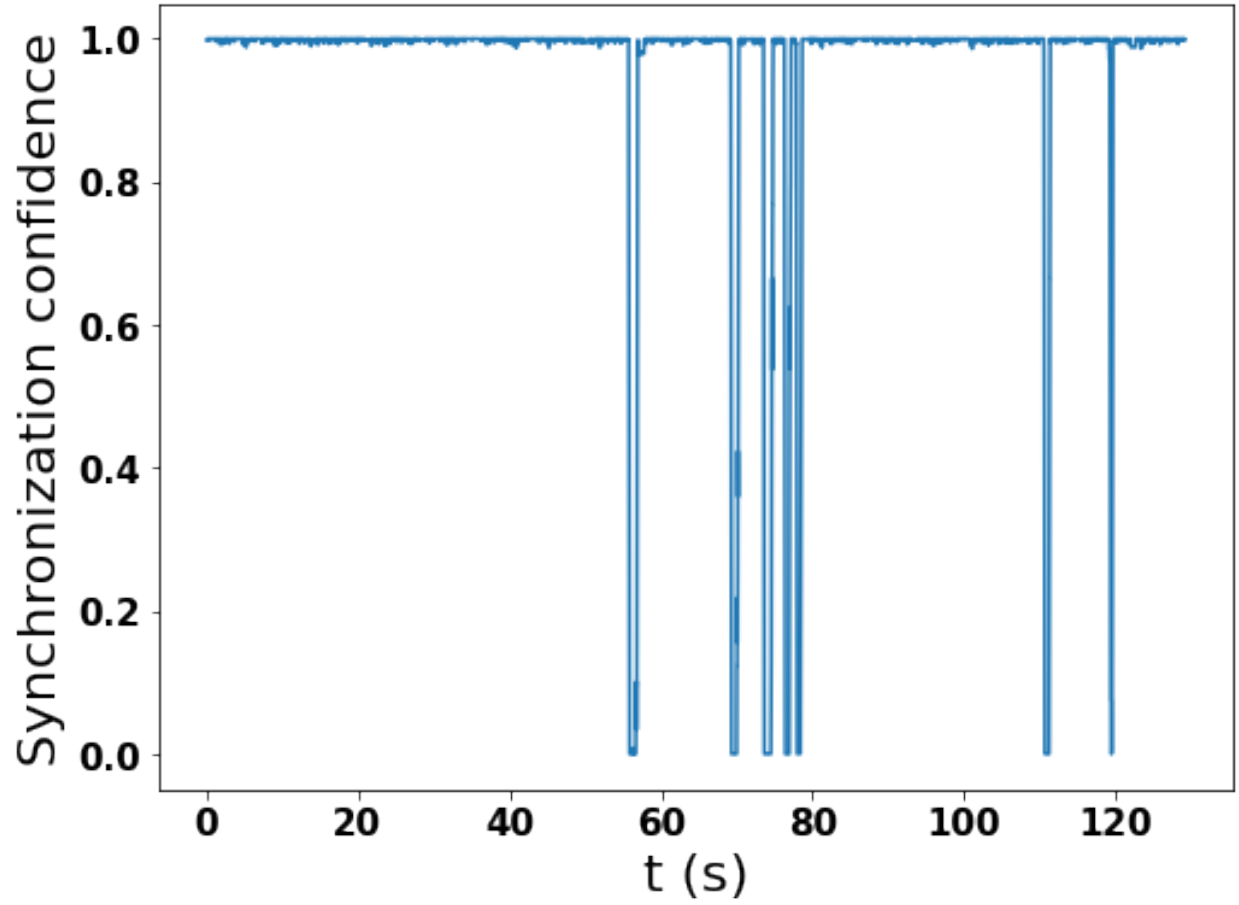


Figure 6.11: The time-averaged synchronization confidence per processing batch for an air-to-car QKD data set taken on November 14th, 2023 with a 0.1 s integration time. The confidence drops particularly in the regions where the transmission drops.

locking in this configuration. The average QBER values are lower than the 11% threshold for asymptotic secure key, which allows the generation of a finite secure key rate.

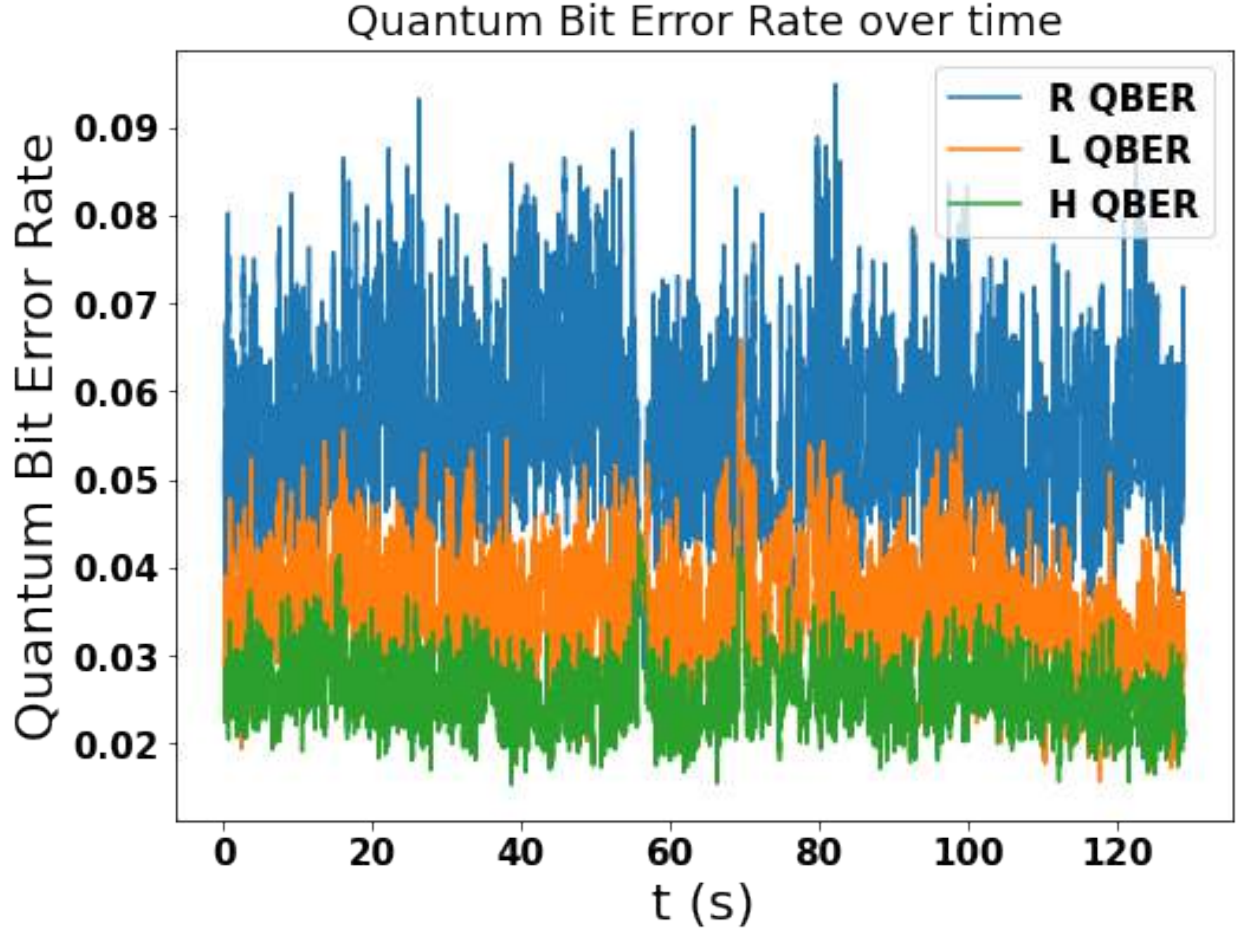


Figure 6.12: The time-averaged QBER over the data collection for a air-to-car QKD data set taken on November 14th, 2023 with a 0.1 s integration time. These values do not exceed the 11% threshold below which asymptotic secure key can be generated.

Ultimately, we are able to achieve a finite secure key rate of 1.52 kbits/s with the air-to-car QKD demonstration. The lower finite secure key rate compared to the air-to-air data set is largely driven by the lower collection time (and thus fewer transmitted states), the higher average QBERs, and the lower mean photon number. While the system is not as well

optimized in this configuration, the success of this experiment showcases the modularity and adaptability of the apparatus in different configurations. To my knowledge, this is the first known demonstration of an air-to-car QKD experiment.

6.5 Car-to-Car QKD (Low Speed)

The car-to-car QKD experiment at 5 mph speeds between two cars on a closed test track is the third of the four mobile QKD configurations. This demonstration showcases the capability of the system to provide reliable connections between mobile vehicles. Moving the system to vehicles for both users shows that the system is highly modular and drone flight is in no way essential to its operation. While this dataset does not include an airborne user, it still contends with some of the same testing challenges as the air-to-air and air-to-car demonstration, including non-ideal or variable temperature, humidity, wind, ambient light, and platform vibration. This dataset uses the following state sending fractions: signal R: 51/256, signal L: 51/256, signal H: 78/256, decoy R: 14/256, decoy L: 14/256, decoy H: 21/256, vacuum: 27/256. The mean photon number for the signal states is 0.368.

The plot of the time-averaged counts (with a 0.1 s integration time) shown in Figure 6.13 exhibits an similar but slightly higher level of variability in transmission compared with its ground-to-ground counterpart, due to the vibrations and motion associated with vehicle vibration at 5 mph speeds. In this demonstration, the pointing-and-tracking largely maintained locking throughout the data set. The cars are in motion from 30 seconds to 125 seconds into the data set, limited primarily by the length of the test track. The transmission in this dataset is best described as slightly variable with a few brief sections of reduced transmission. The data set is almost entirely comprised of “good” data regions that have high consistent throughput and would most effectively generate secure key. The background

is somewhat higher than that of the ground-to-ground and air-to-air datasets because of vehicle lights. As before, the differences between the various count rates are a function of the relative sending frequency of the input states, the mean photon number balancing, the state and measurement fidelity, and any mismatch in detector efficiency. The overall transmission is fairly high, due to the stable transmission and healthy mean photon number. The processed data region here is 13 blips.

The time-averaged synchronization over the data set (with an integration time of 0.1 s) seen in Figure 6.14 shows how well the synchronization algorithm performs over different sections of this dataset. The synchronization confidence is uniformly high in this dataset, with no major drop outs. The qubit-based synchronization algorithm depends on the data itself and the consistently high transmission in this data set provide sufficient information for high synchronization confidence throughout. A mere 0.28% of the batches fail to meet the 95% synchronization confidence threshold, and among the batches that met the synchronization threshold, none failed to meet the signal-to-noise criterion.

The plot of QBER over time (see Fig. 6.15) shows the variability of the error rates over the course of the data set. The fluctuations shown here are caused by shot noise in the measurement as well as variability in the signal-to-noise ratio cause by car motion and vibrations at 5 mph. The average QBER values are much lower than the 11% threshold for asymptotic secure key, with only minor variability over the course of the data collection.

Ultimately, we are able to achieve a finite secure key rate of 19.6 kbits/s with the 5 mph car-to-car QKD demonstration. This is mostly driven by the reasonably high mean photon number, the strong and stable transmission, and relatively low average QBER values. While this data set has a relatively short collection time compared to the other mobile configurations, the quality and consistency of the data set allow us the achieve the highest finite secure key rate out of any of our mobile demonstrations. This experiment showcases

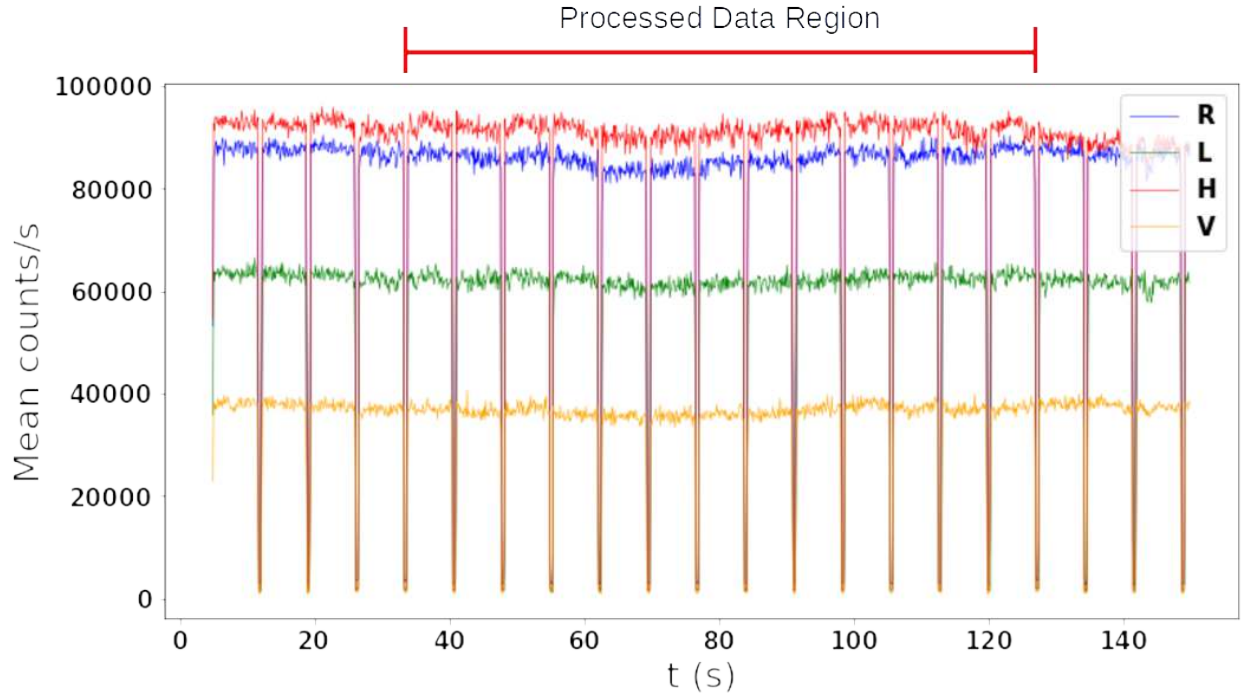


Figure 6.13: The mean counts per second for a car-to-car QKD data set taken on March 27th, 2024 with a 0.1 s integration time. In addition to the expected 7.2 second duty cycle, there is minor variability in the transmission as the control architecture contends with the vibrations driving at 5 mph on a closed test track. The overall throughput is fairly high. The background is higher than the air-to-air data due to the presence of car lights. The processed data region spans 13 blips.

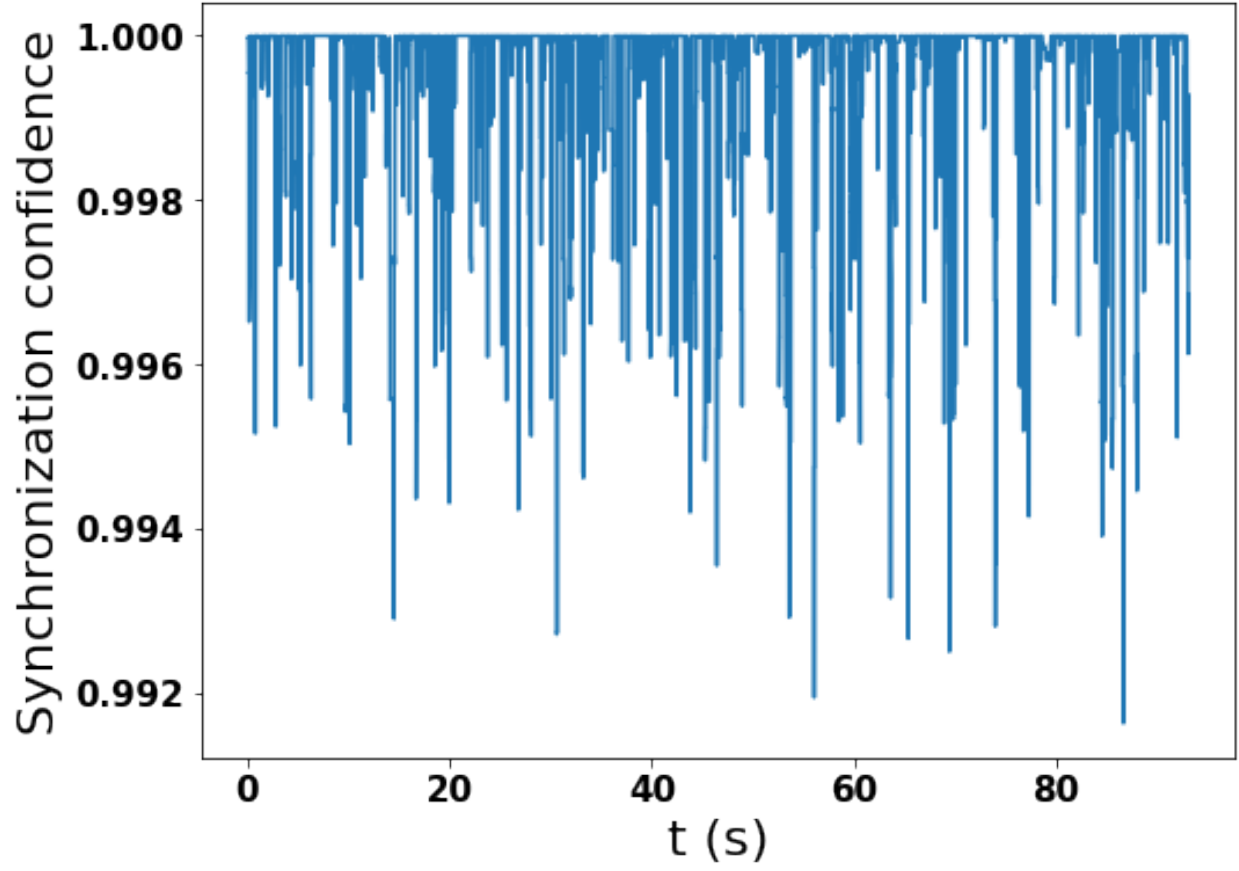


Figure 6.14: The time-averaged synchronization confidence per processing batch for a car-to-car QKD data set taken on March 27th, 2024 with a 0.1 s integration time. The confidence is consistently high throughout the data collection.

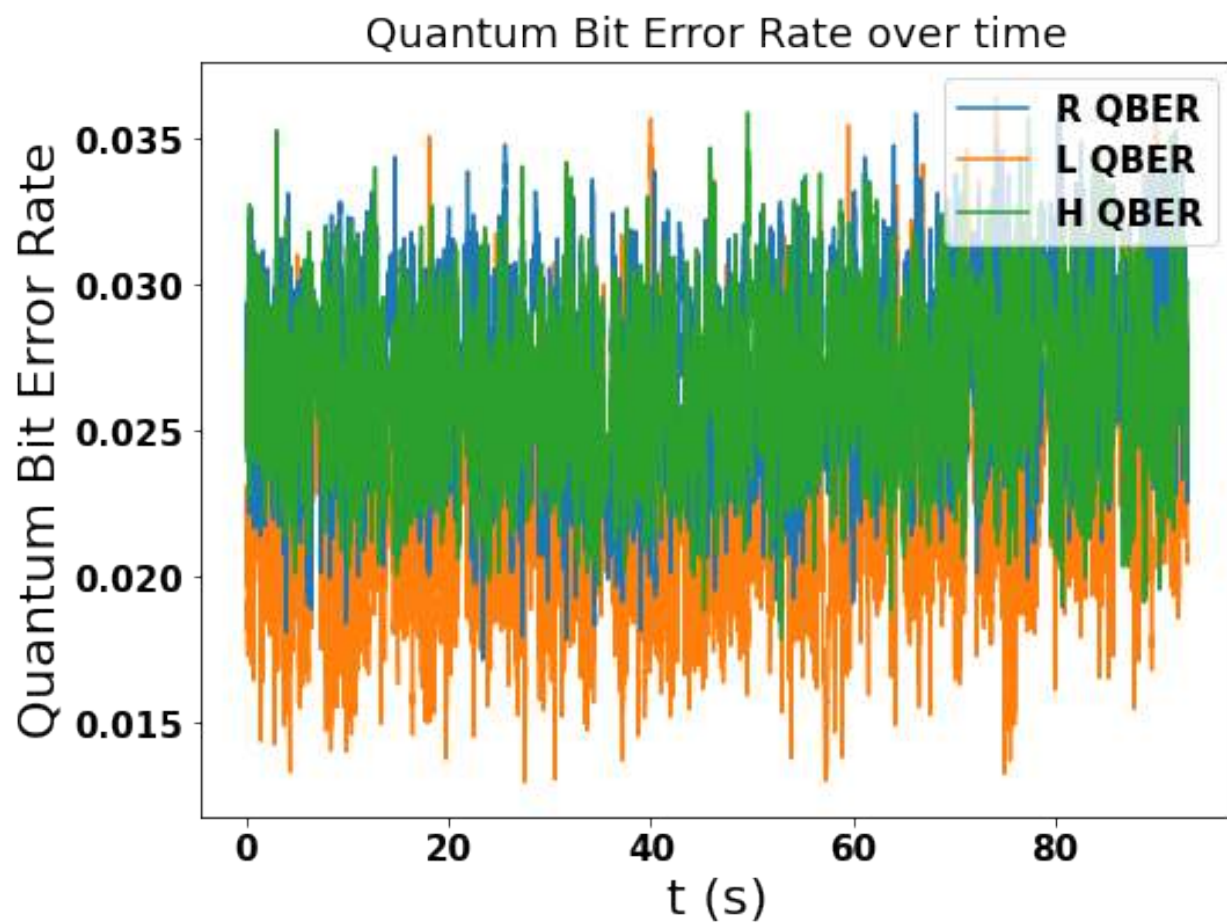


Figure 6.15: The time-averaged QBER over the data collection for a car-to-car QKD data set taken on March 27th, 2024 with a 0.1 s integration time. The QBERs do not exceed the 11% threshold (below which asymptotic secure key can be generated).

the modularity of the system further and demonstrates high performance even when neither user is airborne. To my knowledge, this is the first known demonstration of a car-to-car QKD experiment.

6.6 Car-to-Car QKD (High Speed)

The car-to-car QKD experiment at 70 mph speeds between two highway lanes is the fourth of the four mobile QKD configurations. This demonstration showcases the capability of the system to provide connections between mobile vehicles in a real world environment. While this dataset does not include an airborne user, it still contends with some of the same testing challenges as the air-to-air and air-to-car demonstration, including non-ideal or variable temperature, humidity, wind, ambient light, and platform vibration. The other significant challenge is not being able to engage the inner loop to due safety considerations on a public road. With only the outer loop engaged, the transmission is significantly lower and less consistent. The experiment in this configuration is still able to perform some quantum communication, but it has not yet achieved any finite secure key. This dataset uses the following state sending fractions: signal R: 51/256, signal L: 51/256, signal H: 78/256, decoy R: 14/256, decoy L: 14/256, decoy H: 21/256, vacuum: 27/256. During the collection of this dataset, the mean photon number for the signal states was tuned to 0.58, but the receiver data indicates that the mean photon number for the L channel decreased significantly prior to the data collection.

The plot of the time-averaged counts (with a 0.1 s integration time) shown in Figure 6.16 exhibits an extreme level of variability in transmission compared with its ground-to-ground counterpart, to the point where the 7.2 s duty cycle is barely discernible, due to the vibrations and motion associated with vehicle vibration at 70 mph speeds. In this

demonstration, the pointing-and-tracking struggled to maintain locking throughout the data set. With careful examination, Andrew was able to identify a set of blip edges so that the data processing could be performed. The transmission in this dataset is best described as extremely variable with brief sections throughout where the typical data structure can be observed. This leaves very few “good” data regions that have high consistent throughput and would most effectively generate secure key. The background is somewhat higher than that of the datasets in other configurations because of vehicle lights and especially headlights from other vehicles. As before, the differences between the various count rates are a function of the relative sending frequency of the input states, the mean photon number balancing, the state and measurement fidelity, and any mismatch in detector efficiency. The processed data region here is 20 blips, limited by the amount of data saved by the transmitter and the difficulty in identifying a blip edge to initiate data processing.

The time-averaged synchronization over the data set (with an integration time of 0.1 s) seen in Figure 6.17 shows how well the synchronization algorithm performs over different sections of this dataset. The synchronization confidence is typically low in this dataset, with occasional spikes corresponding brief moments of higher transmission. This is because the qubit-based synchronization algorithm depends on the data itself and low transmission areas present fewer differences between the true synchronization point and the other candidates. A significant 61.1% of the batches fail to meet the 95% synchronization confidence threshold, and among the batches that met the synchronization threshold, 20.1% failed to meet the signal-to-noise criterion.

The plot of QBER over time (see Fig. 6.12) shows the variability of the error rates over the course of the data set. The fluctuations shown here are caused by shot noise in the measurement as well as variability in the signal-to-noise ratio cause by car motion and vibrations at 70 mph. The extreme variability of the transmission means that large spikes

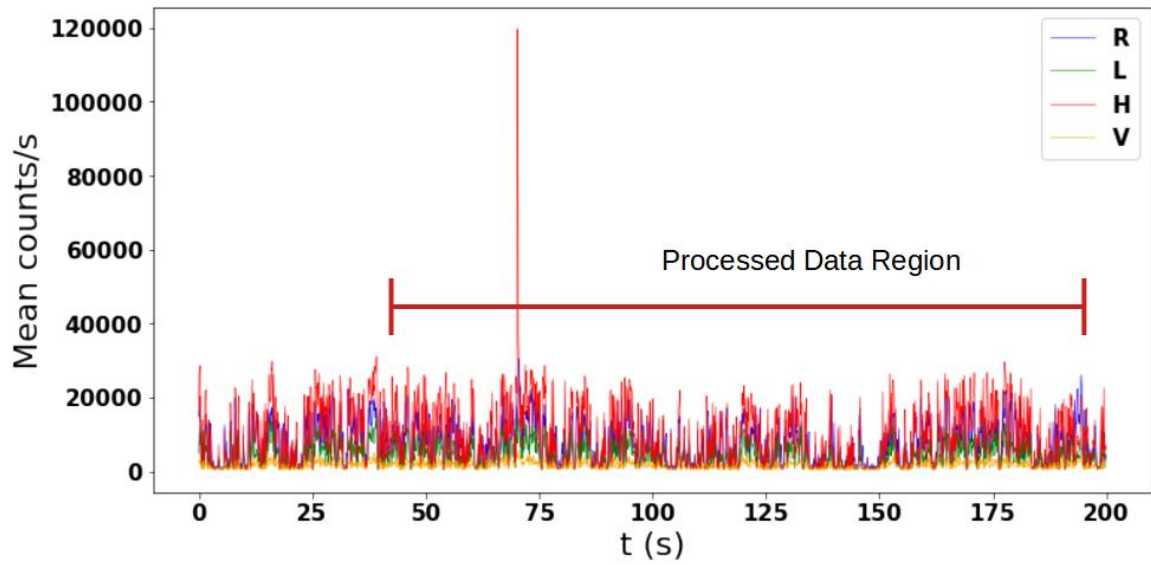


Figure 6.16: The mean counts per second for a highway car-to-car QKD data set taken on November 16th, 2023 with a 0.1 s integration time. In addition to the expected 7.2 second duty cycle, there is extreme variability in the transmission as the control architecture (without the inner loop) contends with the vibrations driving at 70 mph on a highway. The background is higher than the air-to-air data due to the presence of car lights. The processed data region spans 20 blips.

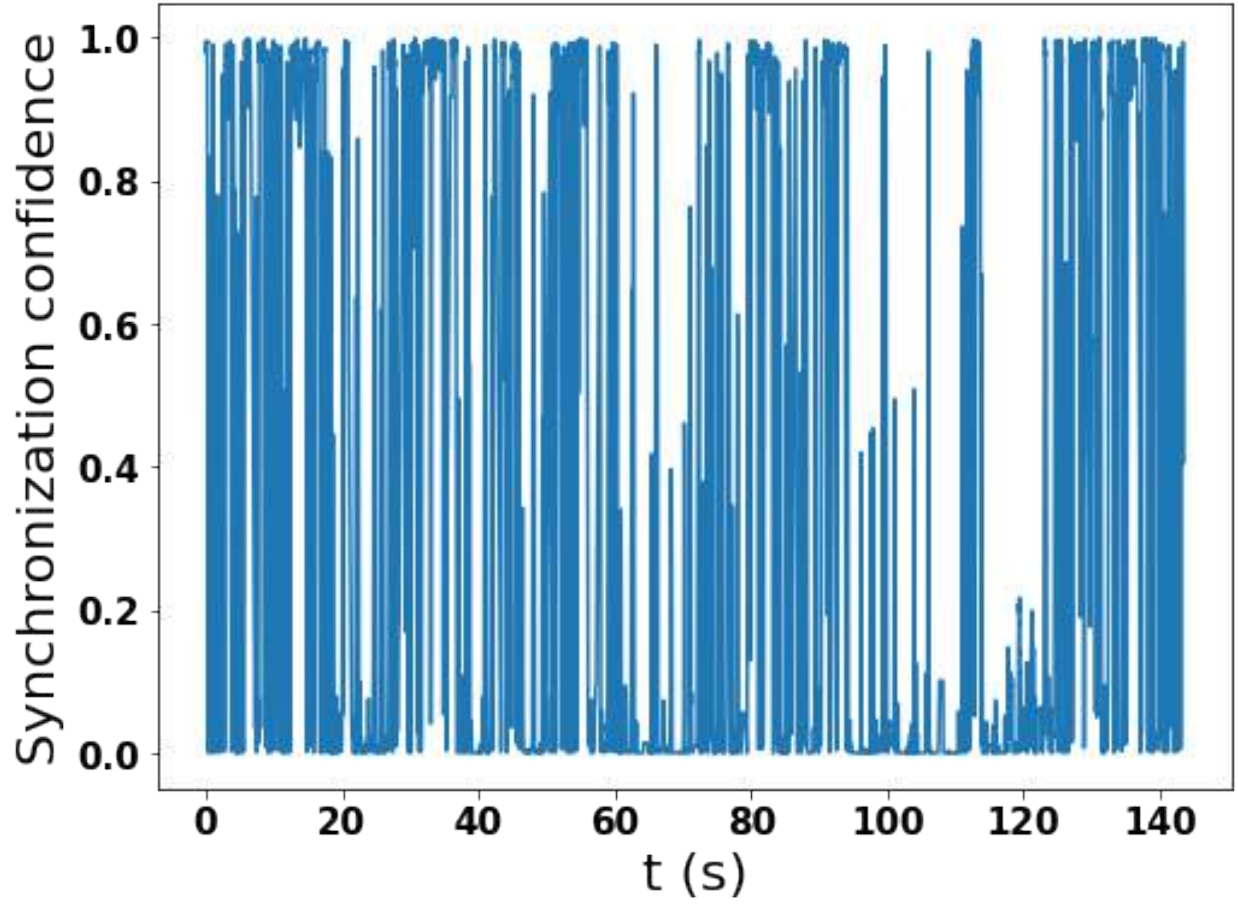


Figure 6.17: The time-averaged synchronization confidence per processing batch for a highway car-to-car QKD data set taken on November 16th, 2023 with a 0.1 s integration time. The confidence fluctuates wildly across the dataset as the system struggles to maintain high transmission locking and collect enough data to successfully synchronize.

in the QBER values are the norm rather than the exception. While the average QBER values for R and H are lower than the 11% threshold for asymptotic secure key, the L QBER fluctuates wildly and frequently exceeds this threshold. This high L QBER is driven primarily an unexpectedly low mean photon number in the L channel.

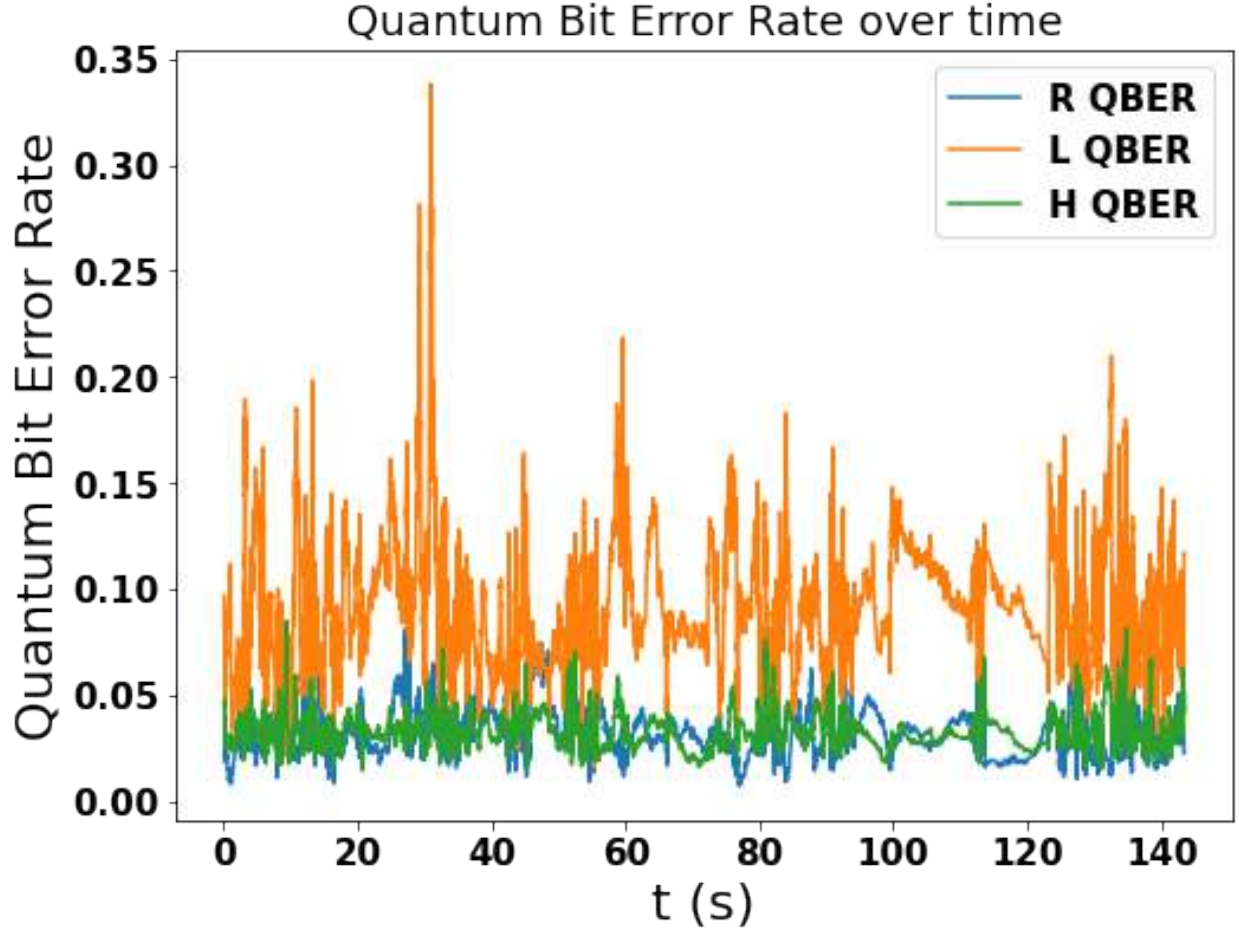


Figure 6.18: The time-averaged QBER over the data collection for a highway car-to-car QKD data set taken on November 16th, 2023 with a 0.1 s integration time. The R and H QBERs do not exceed the 11% threshold (below which asymptotic secure key can be generated), but the L QBER fluctuates wildly and frequently exceeds this threshold.

Ultimately, we are unable to achieve a finite secure key rate with the highway car-to-car QKD demonstration. This is mostly driven by the high L QBER, the low L channel mean

photon number, the low overall transmission, and the high variability of the transmission. The system is not as well optimized in this configuration, particularly because the inner loop system could not be deployed, but the system’s ability to transmit some quantum states (if not a full secure key) lays the groundwork for improvements to the system that will allow secure key in the future. To my knowledge, this is the first known demonstration of highway car-to-car quantum transmission.

6.7 Comparison to State-of-the-Art

To summarize, this experimental design demonstrates a 3 state prepare-and-measure QKD protocol with decoy states wherein the system employs an FPGA-driven attenuated LED source over a free-space channel in the low SWaP environment of airborne and other mobile platforms. To contextualize these results, I compare the figures-of-merit to other QKD demonstrations that share some of these characteristics.

Fiber-based QKD demonstrations have achieved some of the highest secure key rates and at some of the greatest distances out of all QKD implementations [15, 14]. Fiber QKD implementations incorporating LEDs and decoy states have achieved finite secure key rates of 10.9 kbits/s over a 1 km fiber [47]. A key advantage of the LED as the quantum source is the SWaP advantage, which has been leveraged in handheld QKD systems to achieve finite secure key rates up to 30 kbits/s over a 0.5 m separation [48].

Airborne quantum platforms have been implemented by other groups seeking to exploit the last-mile advantages of reconfigurable platforms and the high transmission allowed by free-space systems. Previous experiments have demonstrated satellite-to-ground QKD (with secure key rate of 1.38×10^{-1} bits/s) [18], aircraft-to-ground QKD (with secure key rate of 3.47 kbits/s) [49], and drone-to-ground QKD (with secure key rate of 8 kbits/s) as well as

entanglement distribution [50, 51].

Our team’s experiments demonstrate the viability of new and challenging configurations, yet still achieve comparable secure key rates to these. This cements our team’s work as an important proof of principle experiment towards reconfigurable quantum networks, while also setting the bar for performance in future quantum systems on mobile platforms.

6.8 Future Work

Future phases of this project will primarily focus on integrating an entanglement source on the transmitter with the goal of showing Bell state violations on mobile platforms, and to continue progress towards realizing a mobile quantum network.

We will continue to optimize other aspects of the system that improve its capacity for quantum communication. We hope to continue efforts to implement an external synchronization source that will be more robust in low-transmission environments. We also hope to continue improvements to the pointing-and-tracking system to increase transmission and stability and ultimately perform successful locking at distances up to 1 km.

6.9 Conclusion

In total, we have designed and implemented a complete QKD system on mobile platforms in an outdoor environment, including full data analysis and key generation. We have proved out this system in air-to-air, air-to-car, and car-to-car configurations and reported finite secure key rates comparable to other experiments in the literature. Additionally, we report the leaked mutual information metrics that quantify the security of our system against side-channel attacks. This system represents a strong proof-of-principle experiment for an

end-to-end fully mobile QKD system, and lays the groundwork for future mobile quantum communication and quantum networking.

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